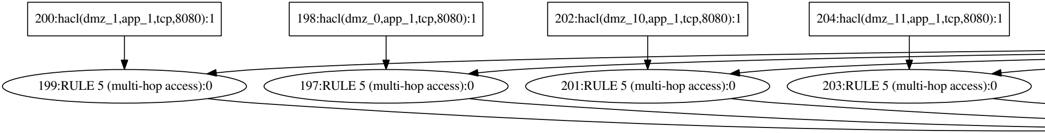
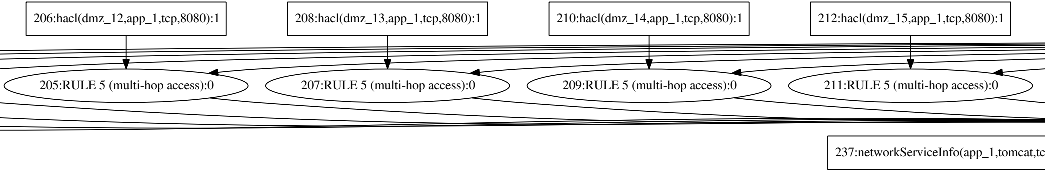
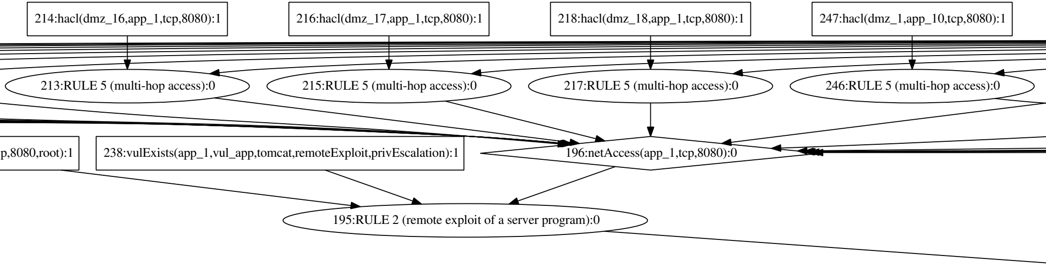
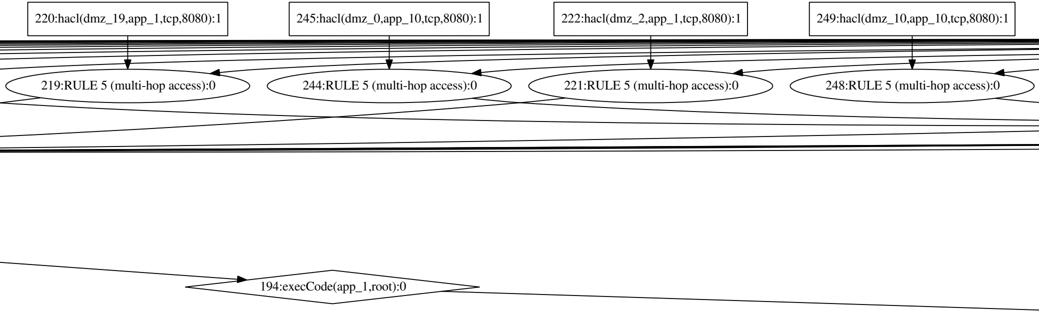
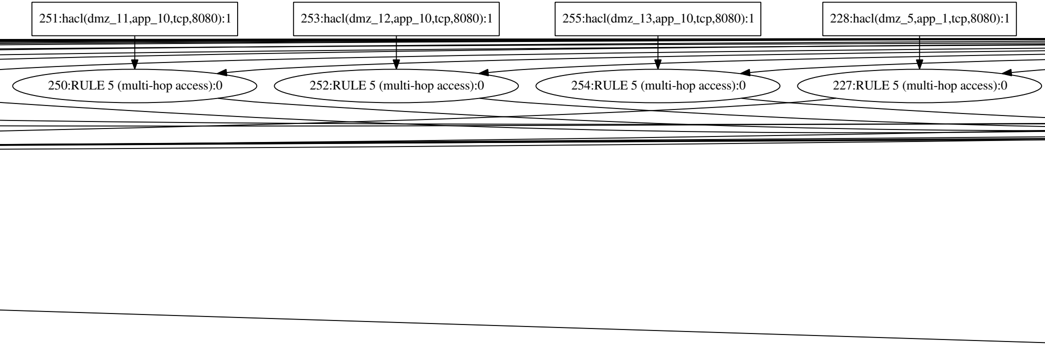
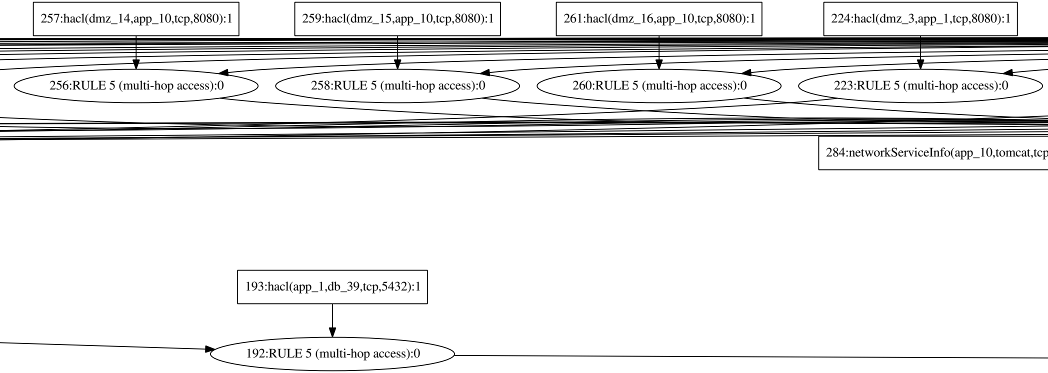

200:hacl(dmz_1,app_1,tcp,8080):1
198:hacl(dmz_0,app_1,tcp,8080):1
202:hacl(dmz_10,app_1,tcp,8080):1
204:hacl(dmz_11,app_1,tcp,8080):1
199:RULE 5 (multi-hop access):0
197:RULE 5 (multi-hop access):0
201:RULE 5 (multi-hop access):0
203:RULE 5 (multi-hop access):0

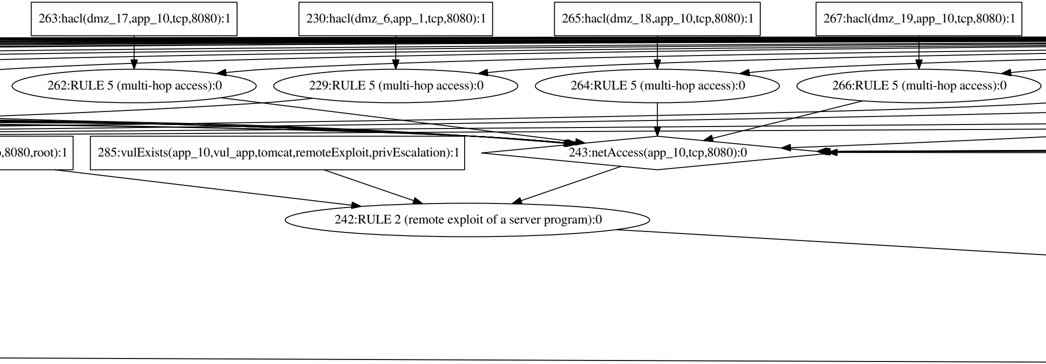

206:hacl(dmz_12,app_1,tcp,8080):1
208:hacl(dmz_13,app_1,tcp,8080):1
210:hacl(dmz_14,app_1,tcp,8080):1
212:hacl(dmz_15,app_1,tcp,8080):1
205:RULE 5 (multi-hop access):0
207:RULE 5 (multi-hop access):0
209:RULE 5 (multi-hop access):0
211:RULE 5 (multi-hop access):0
237:networkServiceInfo(app_1,tomcat,tc

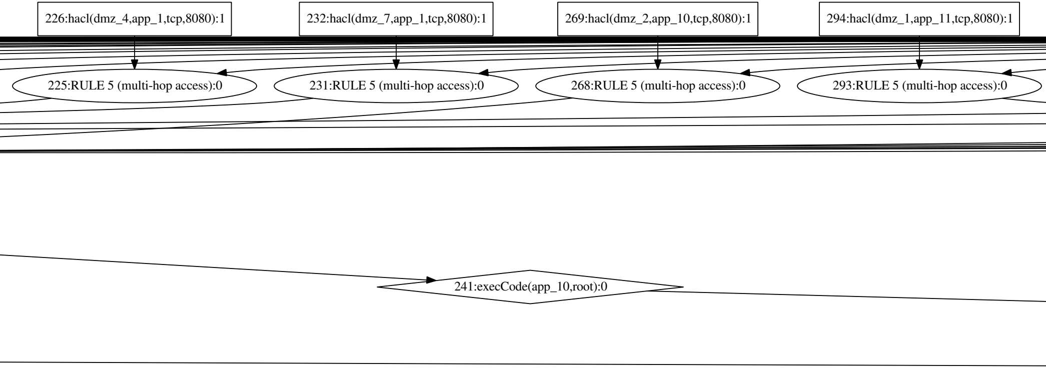

214:hacl(dmz_16,app_1,tcp,8080):1
216:hacl(dmz_17,app_1,tcp,8080):1
218:hacl(dmz_18,app_1,tcp,8080):1
247:hacl(dmz_1,app_10,tcp,8080):1
213:RULE 5 (multi-hop access):0
215:RULE 5 (multi-hop access):0
217:RULE 5 (multi-hop access):0
246:RULE 5 (multi-hop access):0
p,8080,root):1
238:vulExists(app_1,vul_app,tomcat,remoteExploit,privEscalation):1
196:netAccess(app_1,tcp,8080):0
195:RULE 2 (remote exploit of a server program):0

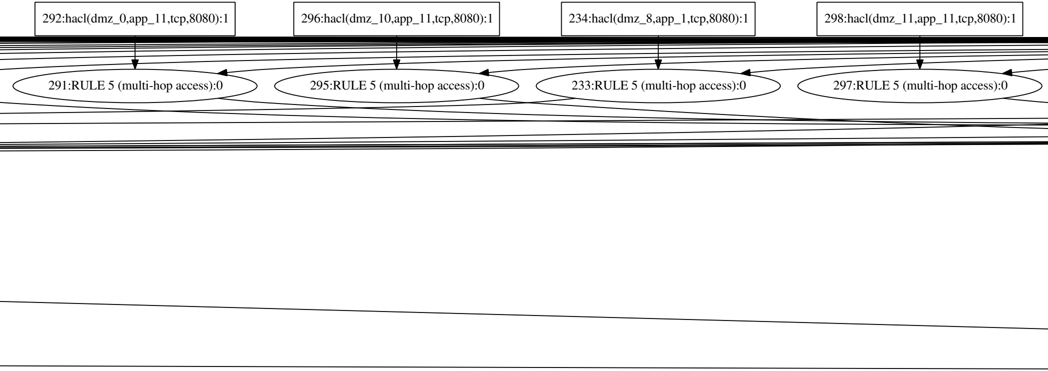

220:hacl(dmz_19,app_1,tcp,8080):1
245:hacl(dmz_0,app_10,tcp,8080):1
222:hacl(dmz_2,app_1,tcp,8080):1
249:hacl(dmz_10,app_10,tcp,8080):1
219:RULE 5 (multi-hop access):0
244:RULE 5 (multi-hop access):0
221:RULE 5 (multi-hop access):0
248:RULE 5 (multi-hop access):0
194:execCode(app_1,root):0

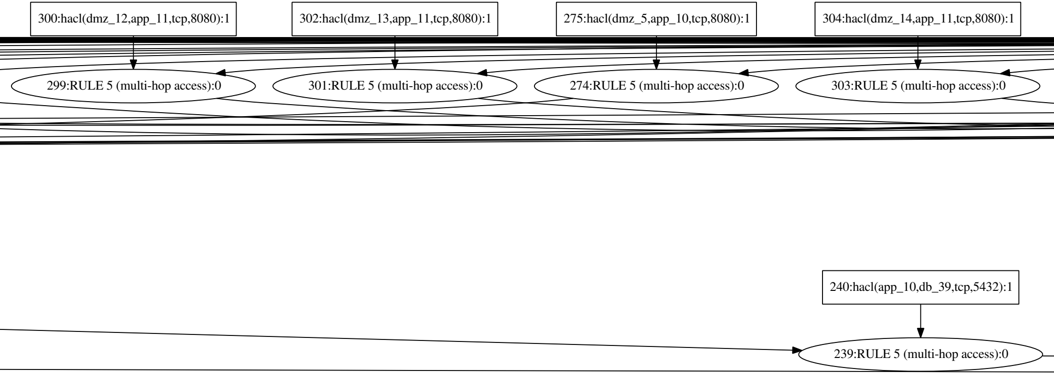

251:hacl(dmz_11,app_10,tcp,8080):1
253:hacl(dmz_12,app_10,tcp,8080):1
255:hacl(dmz_13,app_10,tcp,8080):1
228:hacl(dmz_5,app_1,tcp,8080):1
250:RULE 5 (multi-hop access):0
252:RULE 5 (multi-hop access):0
254:RULE 5 (multi-hop access):0
227:RULE 5 (multi-hop access):0

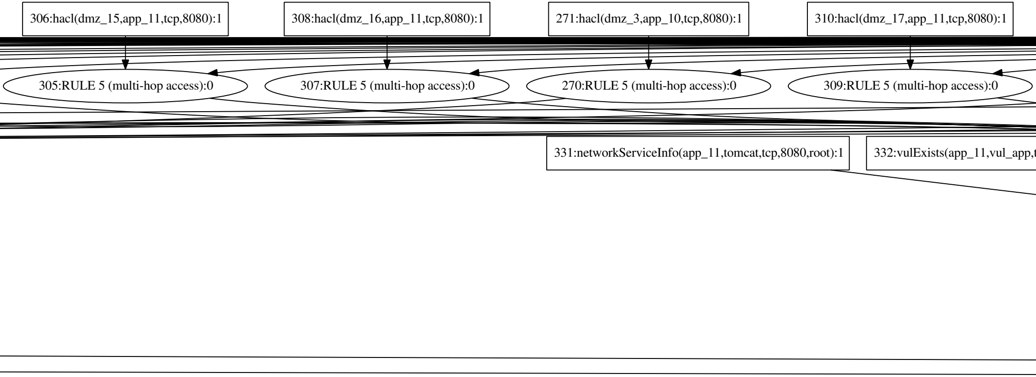

257:hacl(dmz_14,app_10,tcp,8080):1
259:hacl(dmz_15,app_10,tcp,8080):1
261:hacl(dmz_16,app_10,tcp,8080):1
224:hacl(dmz_3,app_1,tcp,8080):1
256:RULE 5 (multi-hop access):0
258:RULE 5 (multi-hop access):0
260:RULE 5 (multi-hop access):0
223:RULE 5 (multi-hop access):0
284:networkServiceInfo(app_10,tomcat,tcp
193:hacl(app_1,db_39,tcp,5432):1
192:RULE 5 (multi-hop access):0

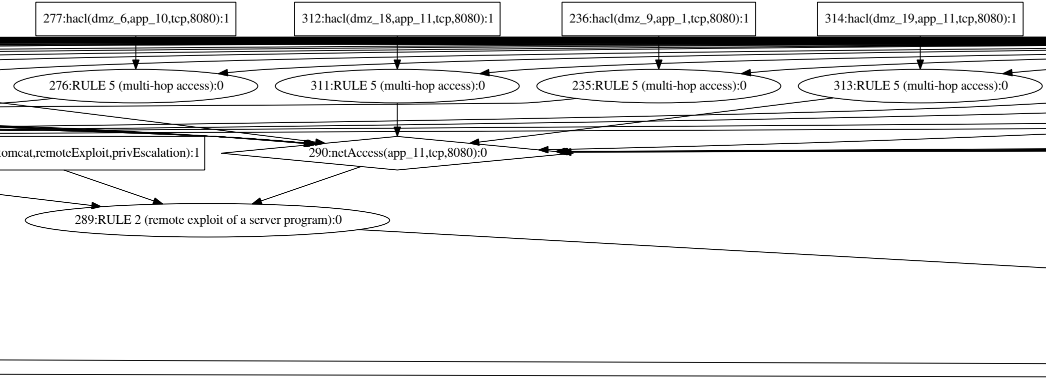

263:hacl(dmz_17,app_10,tcp,8080):1
230:hacl(dmz_6,app_1,tcp,8080):1
265:hacl(dmz_18,app_10,tcp,8080):1
267:hacl(dmz_19,app_10,tcp,8080):1
262:RULE 5 (multi-hop access):0
229:RULE 5 (multi-hop access):0
264:RULE 5 (multi-hop access):0
266:RULE 5 (multi-hop access):0
,8080,root):1
285:vulExists(app_10,vul_app,tomcat,remoteExploit,privEscalation):1
243:netAccess(app_10,tcp,8080):0
242:RULE 2 (remote exploit of a server program):0

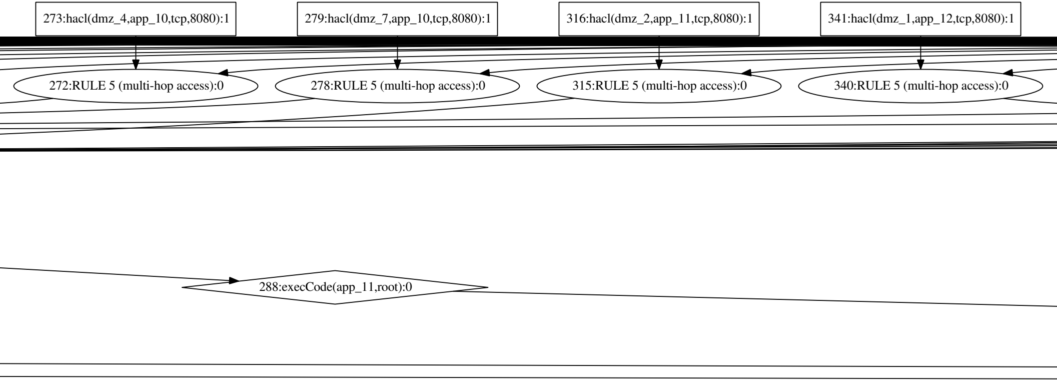

226:hacl(dmz_4,app_1,tcp,8080):1
232:hacl(dmz_7,app_1,tcp,8080):1
269:hacl(dmz_2,app_10,tcp,8080):1
294:hacl(dmz_1,app_11,tcp,8080):1
225:RULE 5 (multi-hop access):0
231:RULE 5 (multi-hop access):0
268:RULE 5 (multi-hop access):0
293:RULE 5 (multi-hop access):0
241:execCode(app_10,root):0

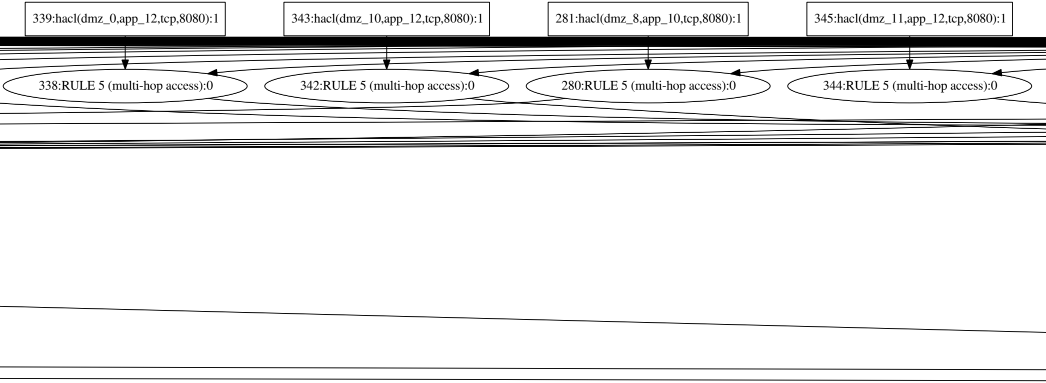

292:hacl(dmz_0,app_11,tcp,8080):1
296:hacl(dmz_10,app_11,tcp,8080):1
234:hacl(dmz_8,app_1,tcp,8080):1
298:hacl(dmz_11,app_11,tcp,8080):1
291:RULE 5 (multi-hop access):0
295:RULE 5 (multi-hop access):0
233:RULE 5 (multi-hop access):0
297:RULE 5 (multi-hop access):0

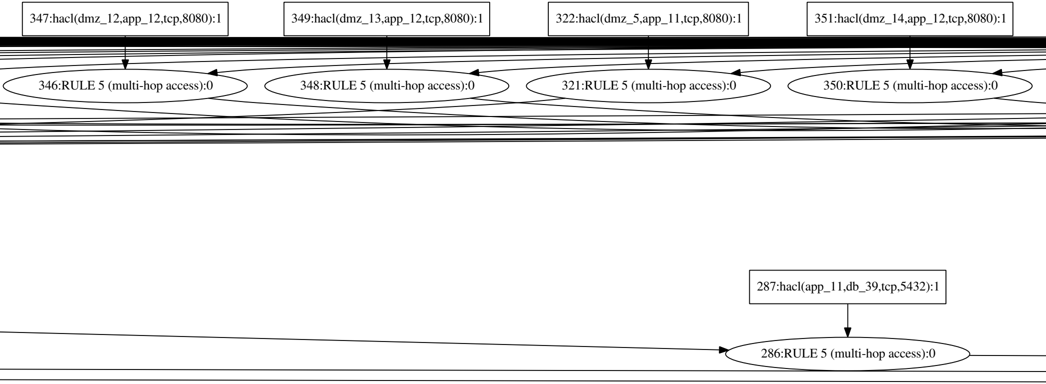

300:hacl(dmz_12,app_11,tcp,8080):1
302:hacl(dmz_13,app_11,tcp,8080):1
275:hacl(dmz_5,app_10,tcp,8080):1
304:hacl(dmz_14,app_11,tcp,8080):1
299:RULE 5 (multi-hop access):0
301:RULE 5 (multi-hop access):0
274:RULE 5 (multi-hop access):0
303:RULE 5 (multi-hop access):0
240:hacl(app_10,db_39,tcp,5432):1
239:RULE 5 (multi-hop access):0

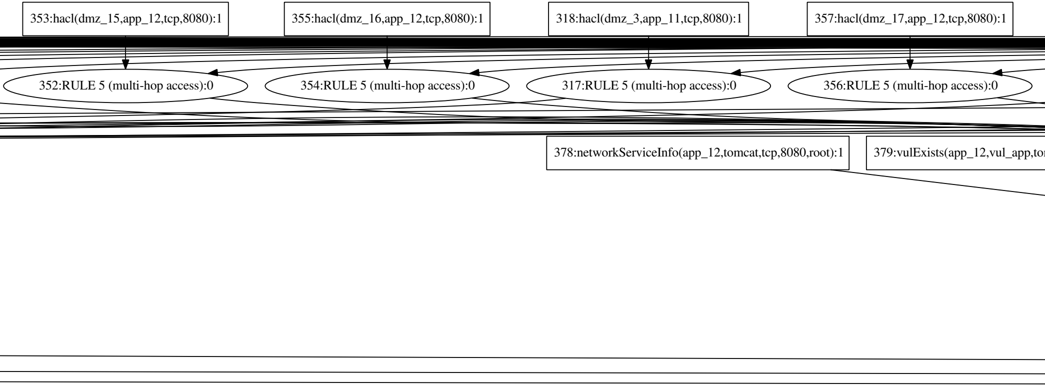

306:hacl(dmz_15,app_11,tcp,8080):1
308:hacl(dmz_16,app_11,tcp,8080):1
271:hacl(dmz_3,app_10,tcp,8080):1
310:hacl(dmz_17,app_11,tcp,8080):1
305:RULE 5 (multi-hop access):0
307:RULE 5 (multi-hop access):0
270:RULE 5 (multi-hop access):0
309:RULE 5 (multi-hop access):0
331:networkServiceInfo(app_11,tomcat,tcp,8080,root):1
332:vulExists(app_11,vul_app,t


277:hacl(dmz_6,app_10,tcp,8080):1
312:hacl(dmz_18,app_11,tcp,8080):1
236:hacl(dmz_9,app_1,tcp,8080):1
314:hacl(dmz_19,app_11,tcp,8080):1
276:RULE 5 (multi-hop access):0
311:RULE 5 (multi-hop access):0
235:RULE 5 (multi-hop access):0
313:RULE 5 (multi-hop access):0
omcat,remoteExploit,privEscalation):1
290:netAccess(app_11,tcp,8080):0
289:RULE 2 (remote exploit of a server program):0

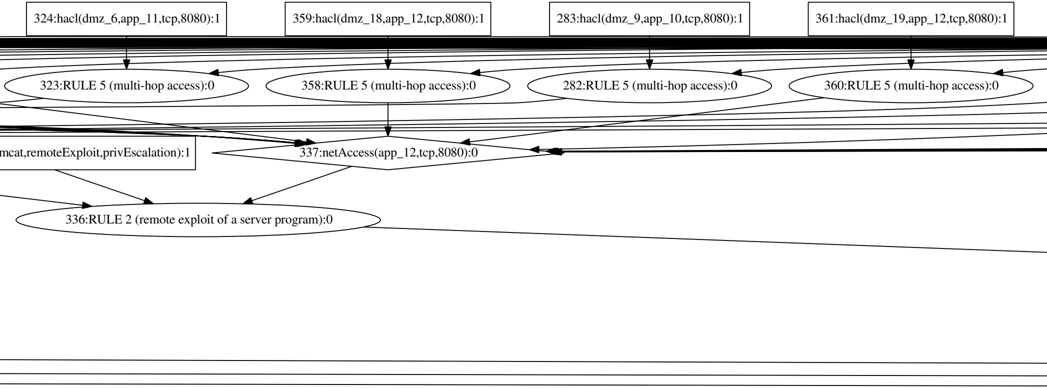

273:hacl(dmz_4,app_10,tcp,8080):1
279:hacl(dmz_7,app_10,tcp,8080):1
316:hacl(dmz_2,app_11,tcp,8080):1
341:hacl(dmz_1,app_12,tcp,8080):1
272:RULE 5 (multi-hop access):0
278:RULE 5 (multi-hop access):0
315:RULE 5 (multi-hop access):0
340:RULE 5 (multi-hop access):0
288:execCode(app_11,root):0

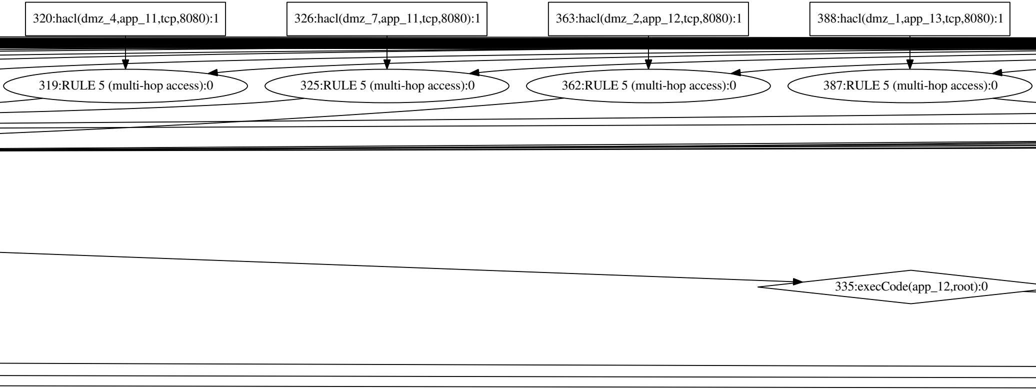

339:hacl(dmz_0,app_12,tcp,8080):1
343:hacl(dmz_10,app_12,tcp,8080):1
281:hacl(dmz_8,app_10,tcp,8080):1
345:hacl(dmz_11,app_12,tcp,8080):1
338:RULE 5 (multi-hop access):0
342:RULE 5 (multi-hop access):0
280:RULE 5 (multi-hop access):0
344:RULE 5 (multi-hop access):0

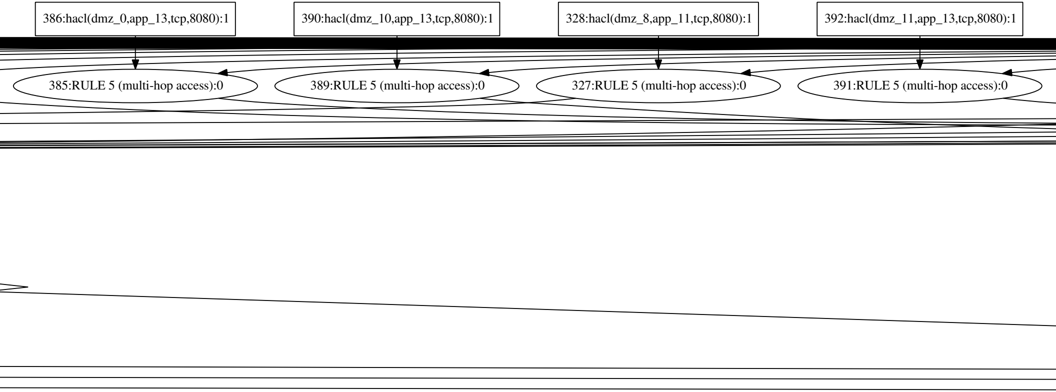

347:hacl(dmz_12,app_12,tcp,8080):1
349:hacl(dmz_13,app_12,tcp,8080):1
322:hacl(dmz_5,app_11,tcp,8080):1
351:hacl(dmz_14,app_12,tcp,8080):1
346:RULE 5 (multi-hop access):0
348:RULE 5 (multi-hop access):0
321:RULE 5 (multi-hop access):0
350:RULE 5 (multi-hop access):0
287:hacl(app_11,db_39,tcp,5432):1
286:RULE 5 (multi-hop access):0

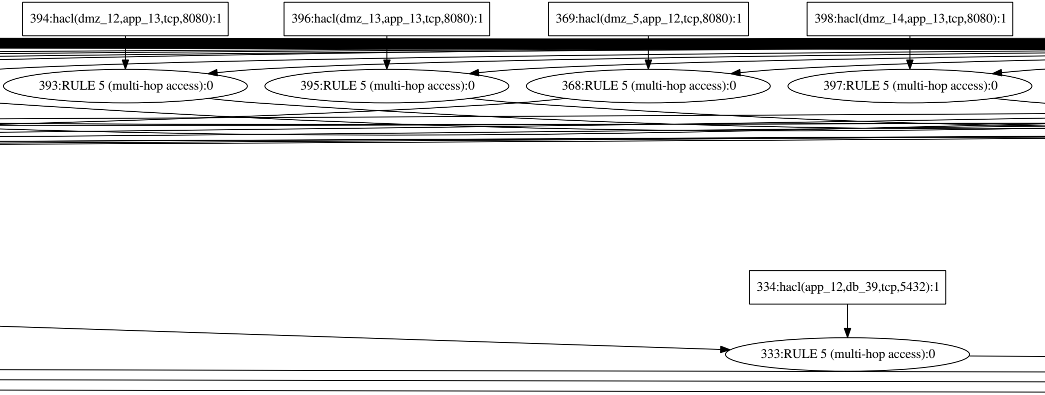


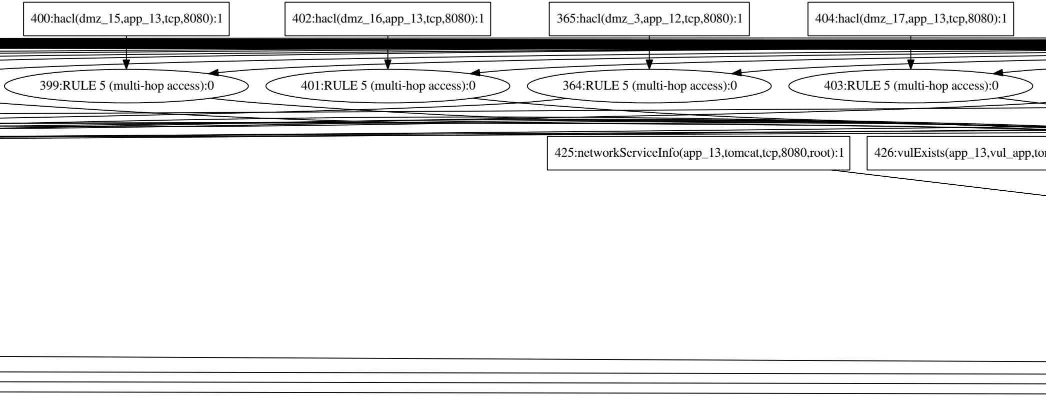
353:hacl(dmz_15,app_12,tcp,8080):1
355:hacl(dmz_16,app_12,tcp,8080):1
318:hacl(dmz_3,app_11,tcp,8080):1
357:hacl(dmz_17,app_12,tcp,8080):1
352:RULE 5 (multi-hop access):0
354:RULE 5 (multi-hop access):0
317:RULE 5 (multi-hop access):0
356:RULE 5 (multi-hop access):0
378:networkServiceInfo(app_12,tomcat,tcp,8080,root):1
379:vulExists(app_12,vul_app,to

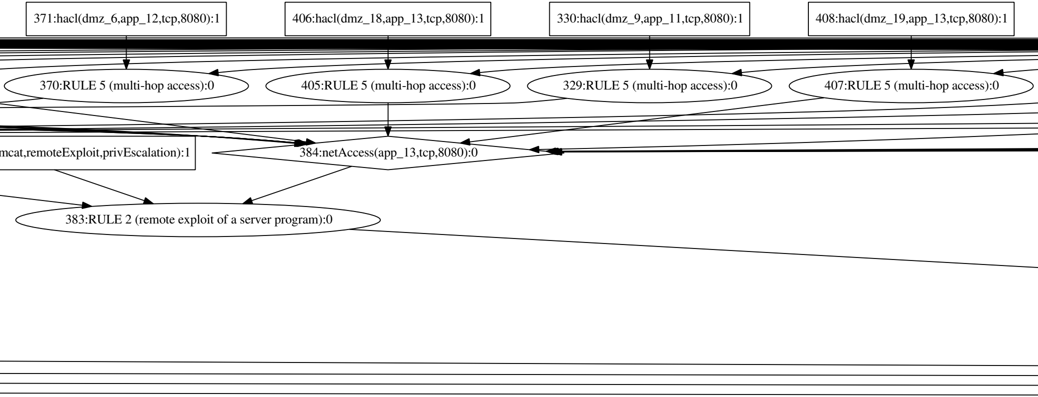

324:hacl(dmz_6,app_11,tcp,8080):1
359:hacl(dmz_18,app_12,tcp,8080):1
283:hacl(dmz_9,app_10,tcp,8080):1
361:hacl(dmz_19,app_12,tcp,8080):1
323:RULE 5 (multi-hop access):0
358:RULE 5 (multi-hop access):0
282:RULE 5 (multi-hop access):0
360:RULE 5 (multi-hop access):0
mcat,remoteExploit,privEscalation):1
337:netAccess(app_12,tcp,8080):0
336:RULE 2 (remote exploit of a server program):0

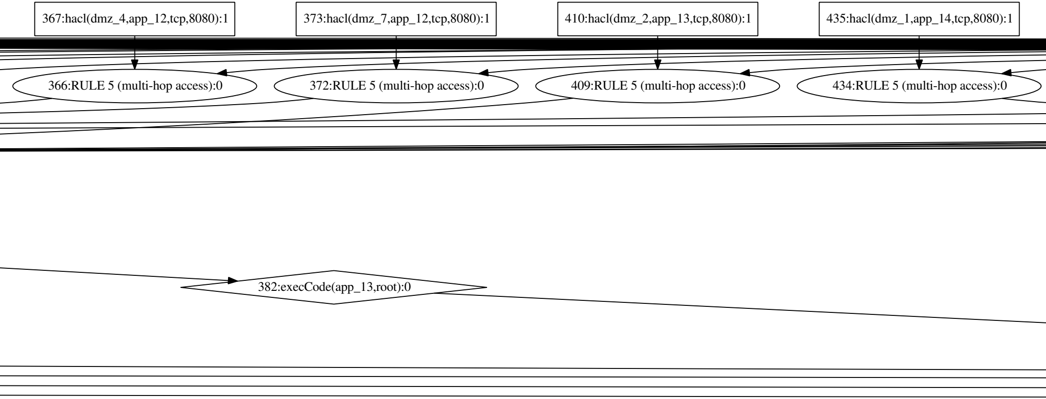

320:hacl(dmz_4,app_11,tcp,8080):1
326:hacl(dmz_7,app_11,tcp,8080):1
363:hacl(dmz_2,app_12,tcp,8080):1
388:hacl(dmz_1,app_13,tcp,8080):1
319:RULE 5 (multi-hop access):0
325:RULE 5 (multi-hop access):0
362:RULE 5 (multi-hop access):0
387:RULE 5 (multi-hop access):0
335:execCode(app_12,root):0

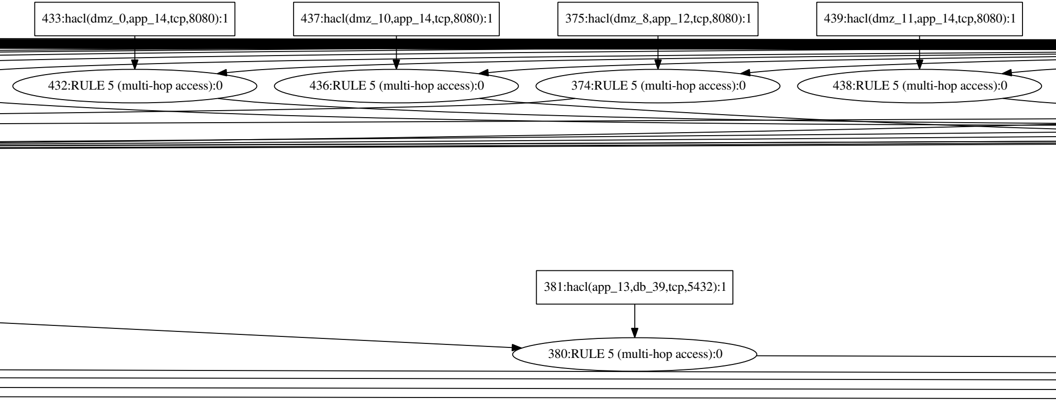

386:hacl(dmz_0,app_13,tcp,8080):1
390:hacl(dmz_10,app_13,tcp,8080):1
328:hacl(dmz_8,app_11,tcp,8080):1
392:hacl(dmz_11,app_13,tcp,8080):1
385:RULE 5 (multi-hop access):0
389:RULE 5 (multi-hop access):0
327:RULE 5 (multi-hop access):0
391:RULE 5 (multi-hop access):0

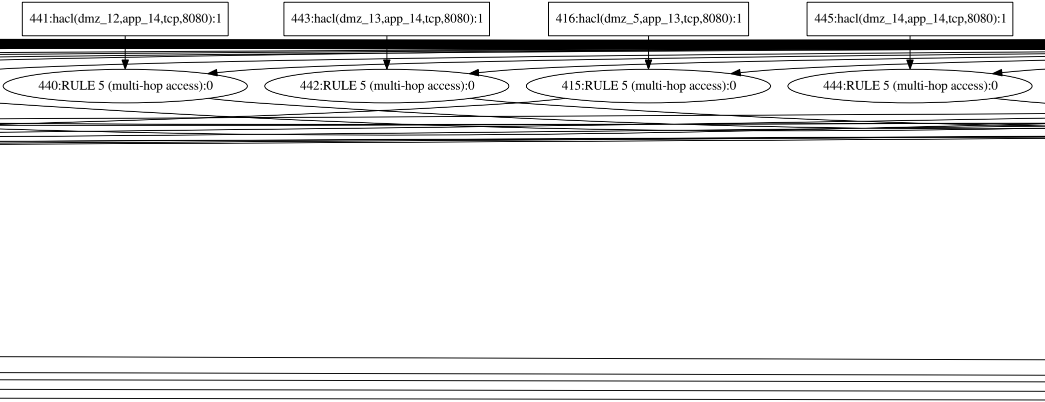

394:hacl(dmz_12,app_13,tcp,8080):1
396:hacl(dmz_13,app_13,tcp,8080):1
369:hacl(dmz_5,app_12,tcp,8080):1
398:hacl(dmz_14,app_13,tcp,8080):1
393:RULE 5 (multi-hop access):0
395:RULE 5 (multi-hop access):0
368:RULE 5 (multi-hop access):0
397:RULE 5 (multi-hop access):0
334:hacl(app_12,db_39,tcp,5432):1
333:RULE 5 (multi-hop access):0

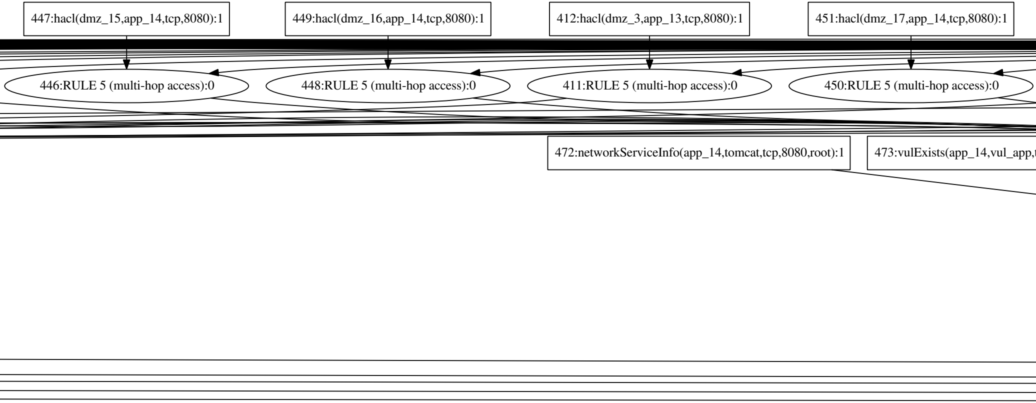

400:hacl(dmz_15,app_13,tcp,8080):1
402:hacl(dmz_16,app_13,tcp,8080):1
365:hacl(dmz_3,app_12,tcp,8080):1
404:hacl(dmz_17,app_13,tcp,8080):1
399:RULE 5 (multi-hop access):0
401:RULE 5 (multi-hop access):0
364:RULE 5 (multi-hop access):0
403:RULE 5 (multi-hop access):0
425:networkServiceInfo(app_13,tomcat,tcp,8080,root):1
426:vulExists(app_13,vul_app,to

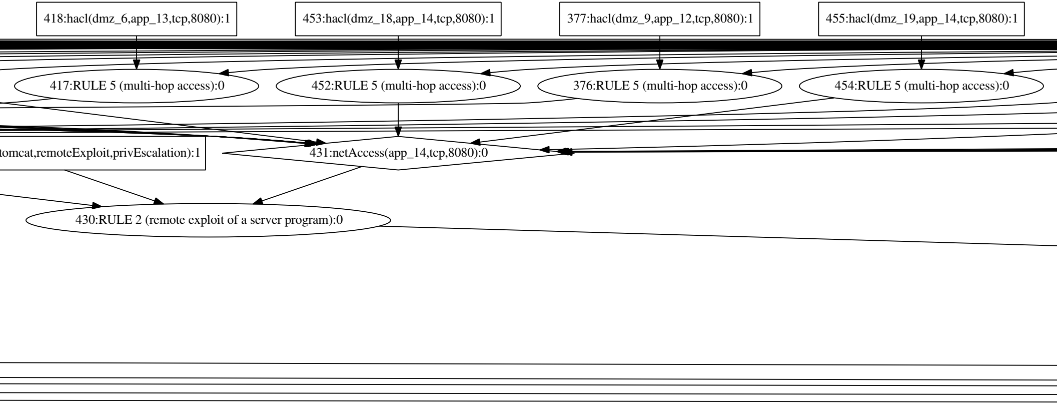

371:hacl(dmz_6,app_12,tcp,8080):1
406:hacl(dmz_18,app_13,tcp,8080):1
330:hacl(dmz_9,app_11,tcp,8080):1
408:hacl(dmz_19,app_13,tcp,8080):1
370:RULE 5 (multi-hop access):0
405:RULE 5 (multi-hop access):0
329:RULE 5 (multi-hop access):0
407:RULE 5 (multi-hop access):0
mcat,remoteExploit,privEscalation):1
384:netAccess(app_13,tcp,8080):0
383:RULE 2 (remote exploit of a server program):0

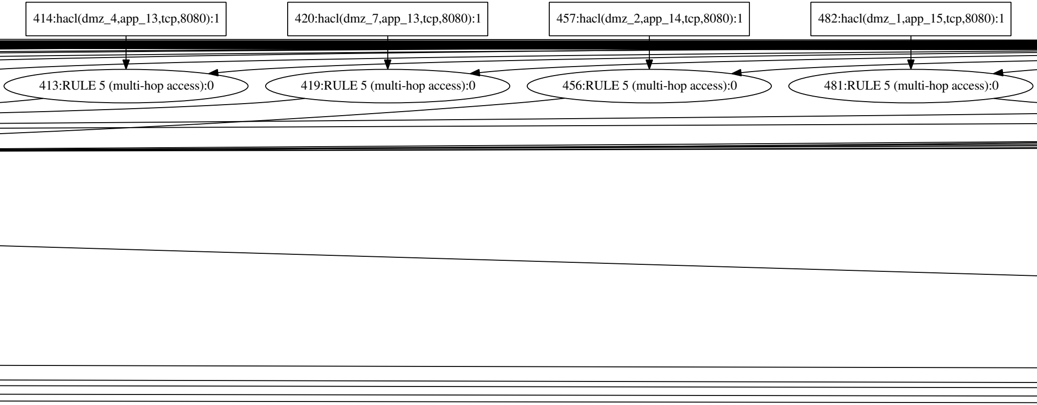

367:hacl(dmz_4,app_12,tcp,8080):1
373:hacl(dmz_7,app_12,tcp,8080):1
410:hacl(dmz_2,app_13,tcp,8080):1
435:hacl(dmz_1,app_14,tcp,8080):1
366:RULE 5 (multi-hop access):0
372:RULE 5 (multi-hop access):0
409:RULE 5 (multi-hop access):0
434:RULE 5 (multi-hop access):0
382:execCode(app_13,root):0

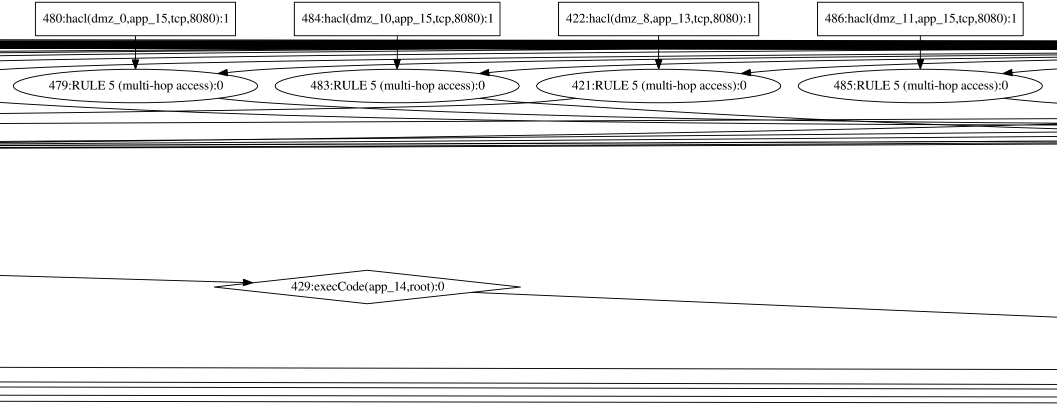

433:hacl(dmz_0,app_14,tcp,8080):1
437:hacl(dmz_10,app_14,tcp,8080):1
375:hacl(dmz_8,app_12,tcp,8080):1
439:hacl(dmz_11,app_14,tcp,8080):1
432:RULE 5 (multi-hop access):0
436:RULE 5 (multi-hop access):0
374:RULE 5 (multi-hop access):0
438:RULE 5 (multi-hop access):0
381:hacl(app_13,db_39,tcp,5432):1
380:RULE 5 (multi-hop access):0

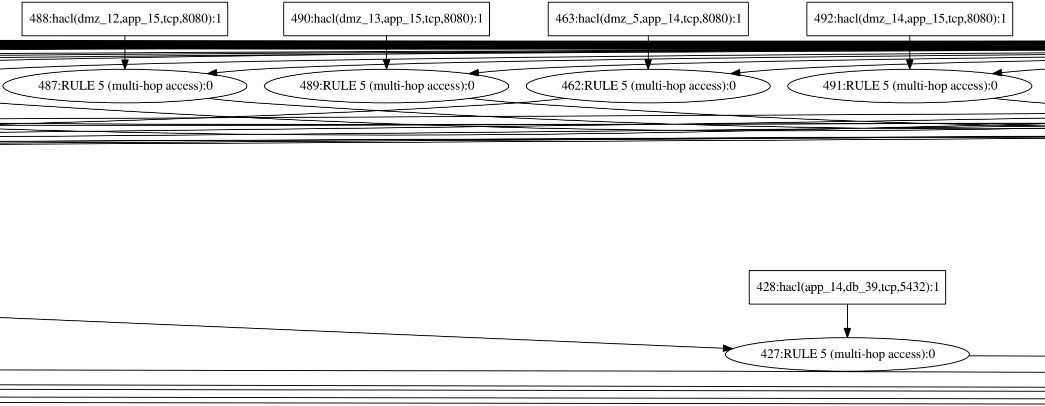

441:hacl(dmz_12,app_14,tcp,8080):1
443:hacl(dmz_13,app_14,tcp,8080):1
416:hacl(dmz_5,app_13,tcp,8080):1
445:hacl(dmz_14,app_14,tcp,8080):1
440:RULE 5 (multi-hop access):0
442:RULE 5 (multi-hop access):0
415:RULE 5 (multi-hop access):0
444:RULE 5 (multi-hop access):0

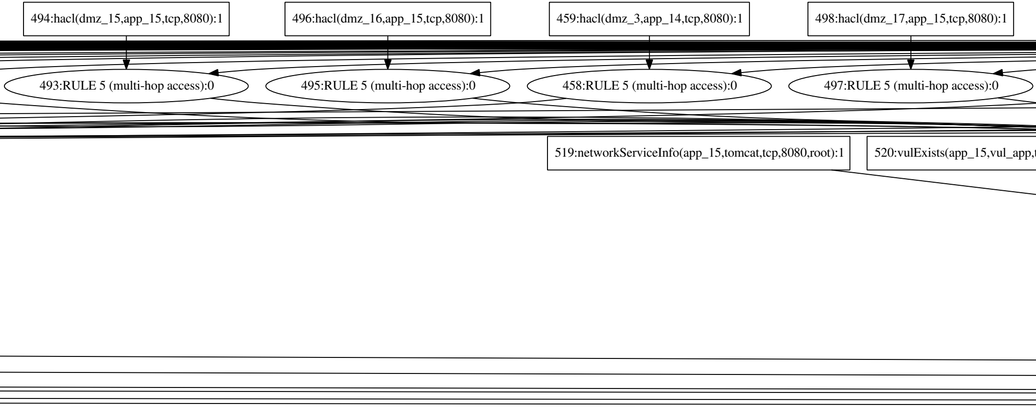

447:hacl(dmz_15,app_14,tcp,8080):1
449:hacl(dmz_16,app_14,tcp,8080):1
412:hacl(dmz_3,app_13,tcp,8080):1
451:hacl(dmz_17,app_14,tcp,8080):1
446:RULE 5 (multi-hop access):0
448:RULE 5 (multi-hop access):0
411:RULE 5 (multi-hop access):0
450:RULE 5 (multi-hop access):0
472:networkServiceInfo(app_14,tomcat,tcp,8080,root):1
473:vulExists(app_14,vul_app,

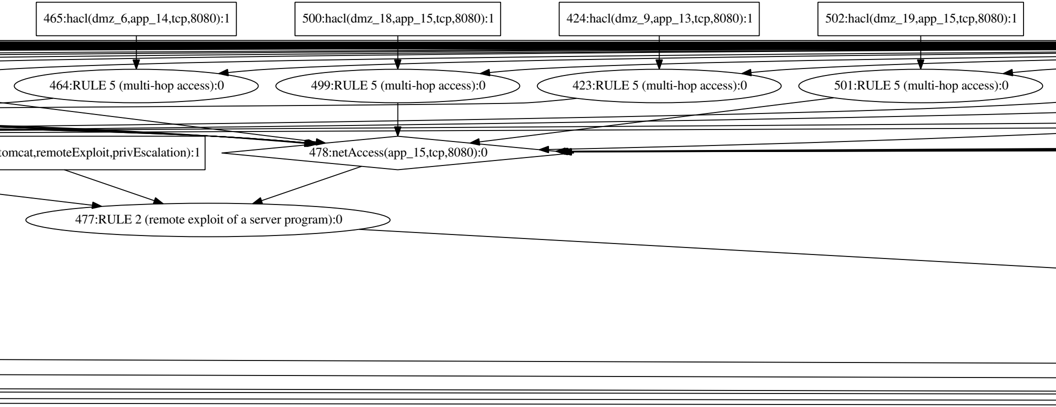

418:hacl(dmz_6,app_13,tcp,8080):1
453:hacl(dmz_18,app_14,tcp,8080):1
377:hacl(dmz_9,app_12,tcp,8080):1
455:hacl(dmz_19,app_14,tcp,8080):1
417:RULE 5 (multi-hop access):0
452:RULE 5 (multi-hop access):0
376:RULE 5 (multi-hop access):0
454:RULE 5 (multi-hop access):0
tomcat,remoteExploit,privEscalation):1
431:netAccess(app_14,tcp,8080):0
430:RULE 2 (remote exploit of a server program):0

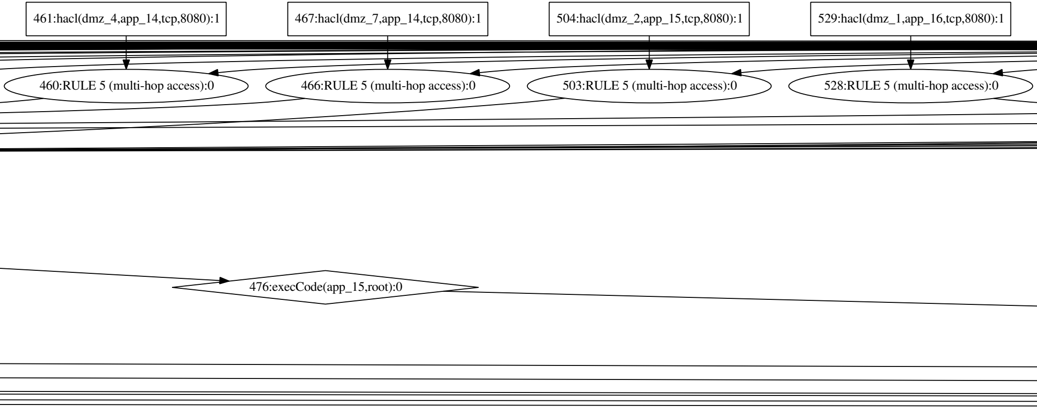

414:hacl(dmz_4,app_13,tcp,8080):1
420:hacl(dmz_7,app_13,tcp,8080):1
457:hacl(dmz_2,app_14,tcp,8080):1
482:hacl(dmz_1,app_15,tcp,8080):1
413:RULE 5 (multi-hop access):0
419:RULE 5 (multi-hop access):0
456:RULE 5 (multi-hop access):0
481:RULE 5 (multi-hop access):0

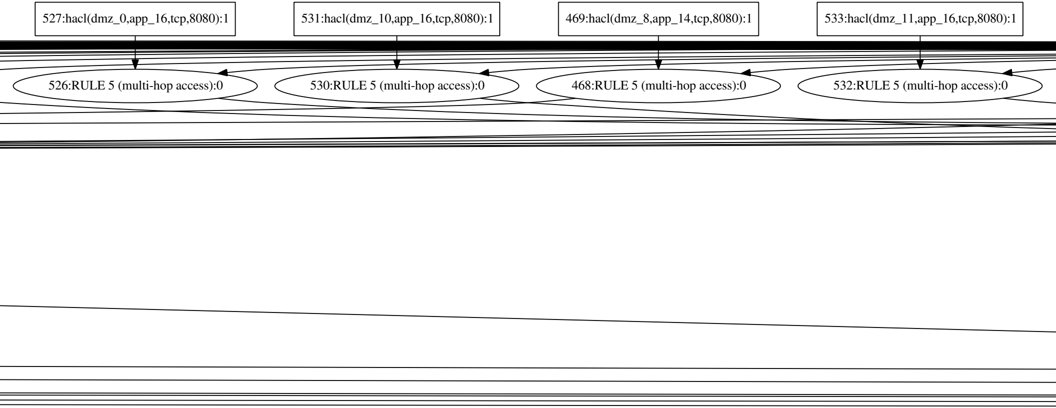

480:hacl(dmz_0,app_15,tcp,8080):1
484:hacl(dmz_10,app_15,tcp,8080):1
422:hacl(dmz_8,app_13,tcp,8080):1
486:hacl(dmz_11,app_15,tcp,8080):1
479:RULE 5 (multi-hop access):0
483:RULE 5 (multi-hop access):0
421:RULE 5 (multi-hop access):0
485:RULE 5 (multi-hop access):0
429:execCode(app_14,root):0

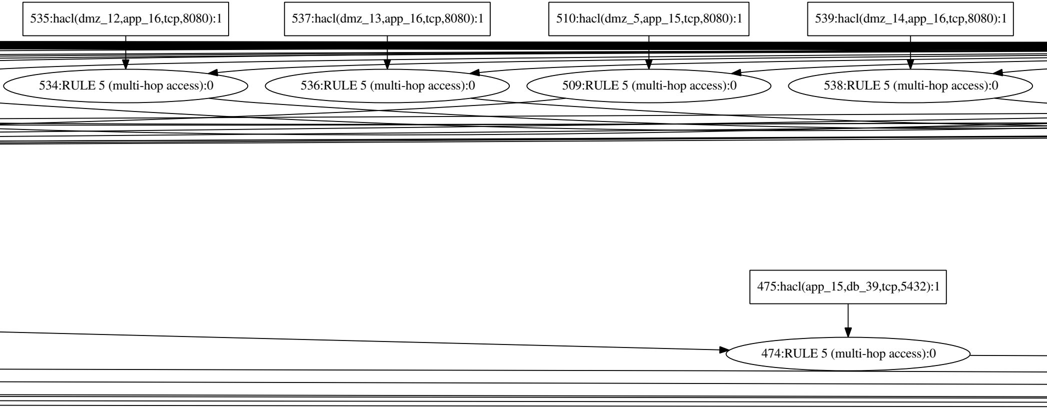

488:hacl(dmz_12,app_15,tcp,8080):1
490:hacl(dmz_13,app_15,tcp,8080):1
463:hacl(dmz_5,app_14,tcp,8080):1
492:hacl(dmz_14,app_15,tcp,8080):1
487:RULE 5 (multi-hop access):0
489:RULE 5 (multi-hop access):0
462:RULE 5 (multi-hop access):0
491:RULE 5 (multi-hop access):0
428:hacl(app_14,db_39,tcp,5432):1
427:RULE 5 (multi-hop access):0

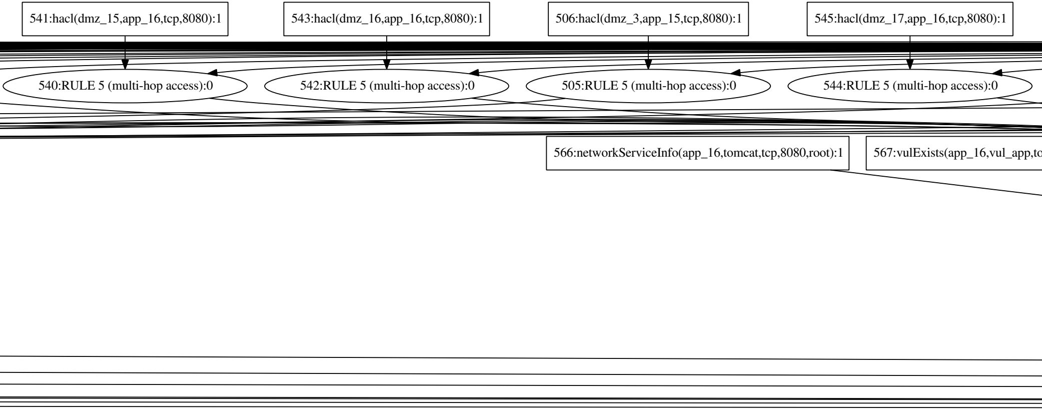

494:hacl(dmz_15,app_15,tcp,8080):1
496:hacl(dmz_16,app_15,tcp,8080):1
459:hacl(dmz_3,app_14,tcp,8080):1
498:hacl(dmz_17,app_15,tcp,8080):1
493:RULE 5 (multi-hop access):0
495:RULE 5 (multi-hop access):0
458:RULE 5 (multi-hop access):0
497:RULE 5 (multi-hop access):0
519:networkServiceInfo(app_15,tomcat,tcp,8080,root):1
520:vulExists(app_15,vul_app,

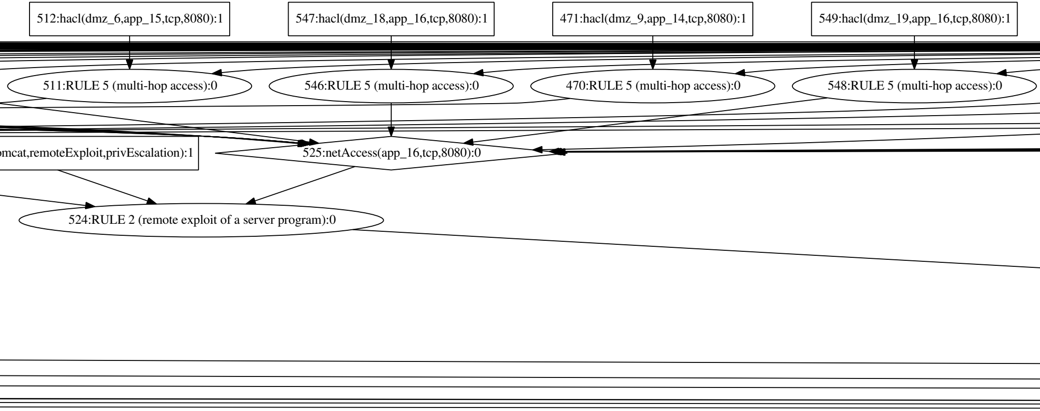

465:hacl(dmz_6,app_14,tcp,8080):1
500:hacl(dmz_18,app_15,tcp,8080):1
424:hacl(dmz_9,app_13,tcp,8080):1
502:hacl(dmz_19,app_15,tcp,8080):1
464:RULE 5 (multi-hop access):0
499:RULE 5 (multi-hop access):0
423:RULE 5 (multi-hop access):0
501:RULE 5 (multi-hop access):0
tomcat,remoteExploit,privEscalation):1
478:netAccess(app_15,tcp,8080):0
477:RULE 2 (remote exploit of a server program):0

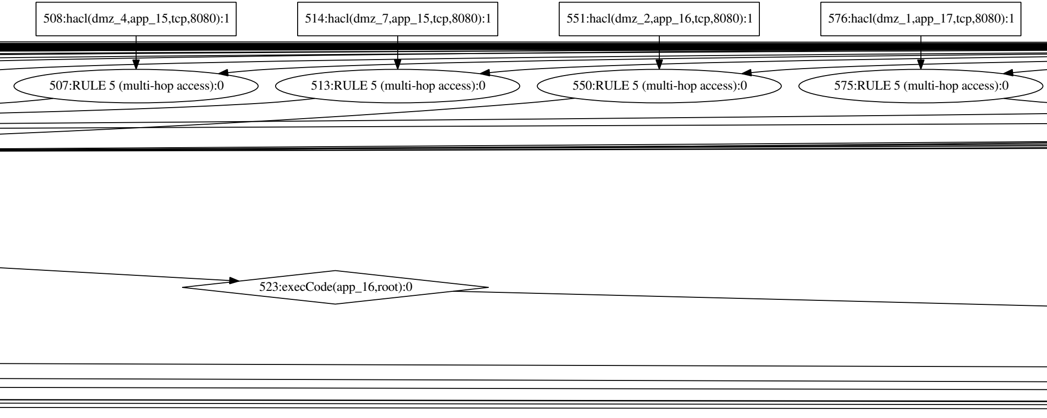

461:hacl(dmz_4,app_14,tcp,8080):1
467:hacl(dmz_7,app_14,tcp,8080):1
504:hacl(dmz_2,app_15,tcp,8080):1
529:hacl(dmz_1,app_16,tcp,8080):1
460:RULE 5 (multi-hop access):0
466:RULE 5 (multi-hop access):0
503:RULE 5 (multi-hop access):0
528:RULE 5 (multi-hop access):0
476:execCode(app_15,root):0

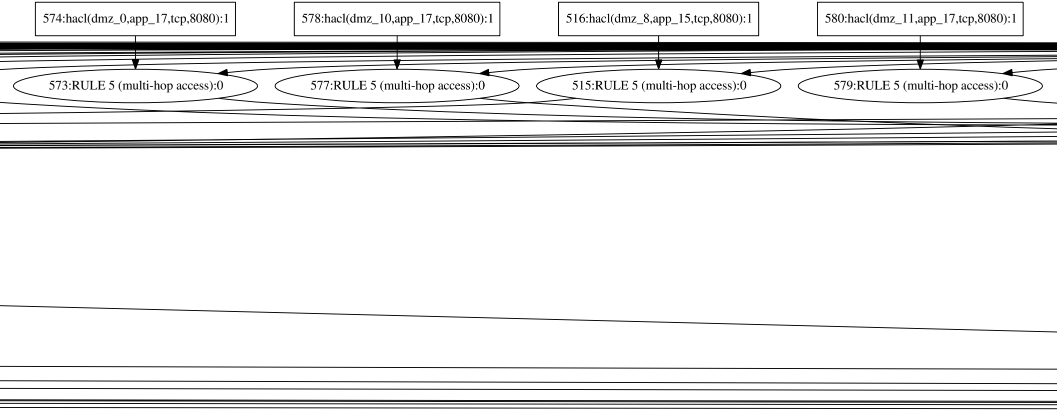

527:hacl(dmz_0,app_16,tcp,8080):1
531:hacl(dmz_10,app_16,tcp,8080):1
469:hacl(dmz_8,app_14,tcp,8080):1
533:hacl(dmz_11,app_16,tcp,8080):1
526:RULE 5 (multi-hop access):0
530:RULE 5 (multi-hop access):0
468:RULE 5 (multi-hop access):0
532:RULE 5 (multi-hop access):0

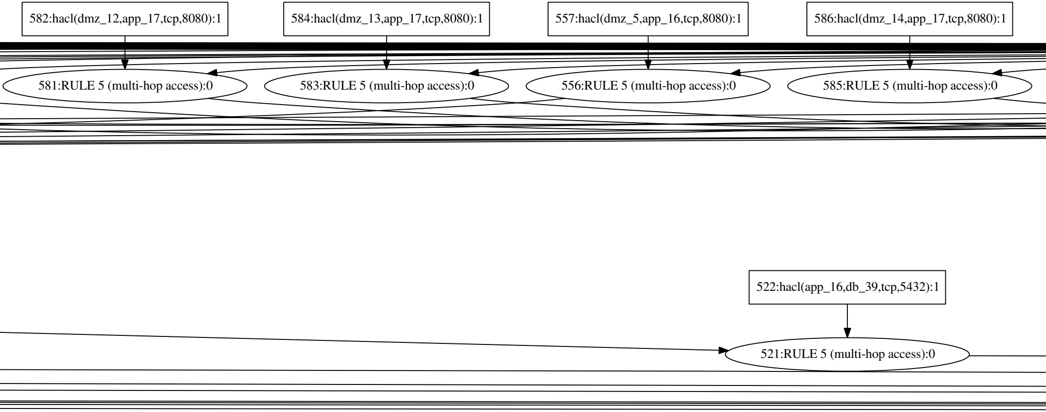

535:hacl(dmz_12,app_16,tcp,8080):1
537:hacl(dmz_13,app_16,tcp,8080):1
510:hacl(dmz_5,app_15,tcp,8080):1
539:hacl(dmz_14,app_16,tcp,8080):1
534:RULE 5 (multi-hop access):0
536:RULE 5 (multi-hop access):0
509:RULE 5 (multi-hop access):0
538:RULE 5 (multi-hop access):0
475:hacl(app_15,db_39,tcp,5432):1
474:RULE 5 (multi-hop access):0

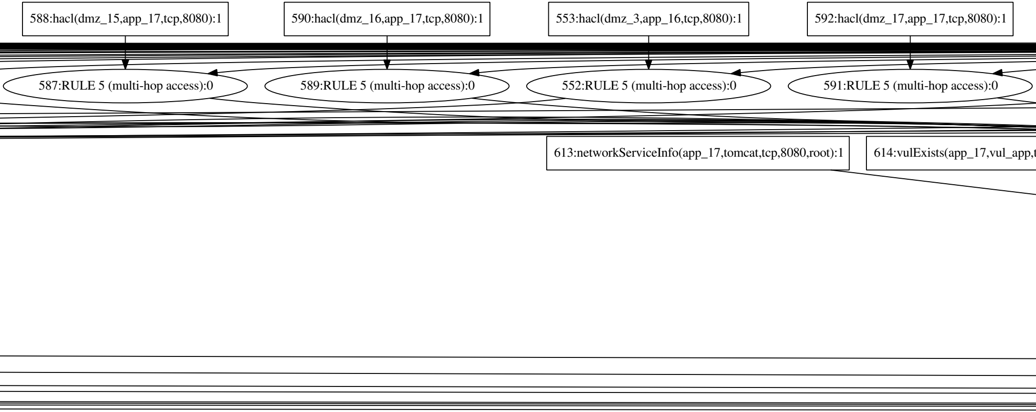

541:hacl(dmz_15,app_16,tcp,8080):1
543:hacl(dmz_16,app_16,tcp,8080):1
506:hacl(dmz_3,app_15,tcp,8080):1
545:hacl(dmz_17,app_16,tcp,8080):1
540:RULE 5 (multi-hop access):0
542:RULE 5 (multi-hop access):0
505:RULE 5 (multi-hop access):0
544:RULE 5 (multi-hop access):0
566:networkServiceInfo(app_16,tomcat,tcp,8080,root):1
567:vulExists(app_16,vul_app,to

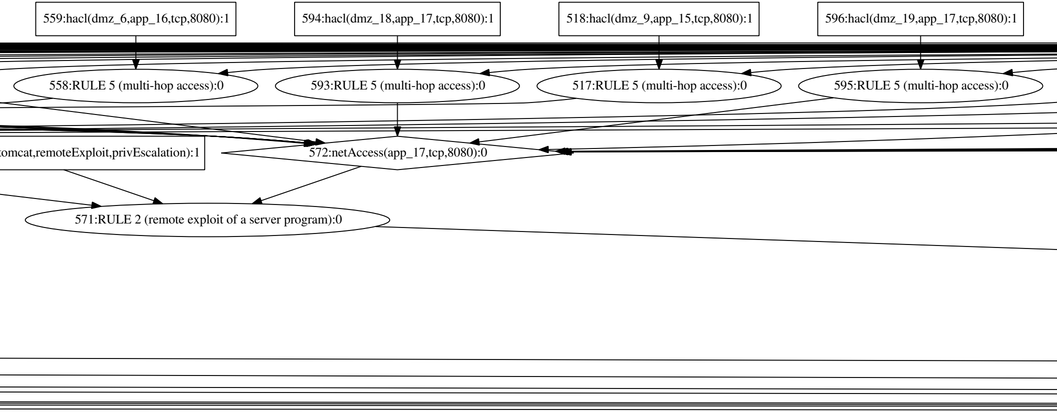

512:hacl(dmz_6,app_15,tcp,8080):1
547:hacl(dmz_18,app_16,tcp,8080):1
471:hacl(dmz_9,app_14,tcp,8080):1
549:hacl(dmz_19,app_16,tcp,8080):1
511:RULE 5 (multi-hop access):0
546:RULE 5 (multi-hop access):0
470:RULE 5 (multi-hop access):0
548:RULE 5 (multi-hop access):0
omcat,remoteExploit,privEscalation):1
525:netAccess(app_16,tcp,8080):0
524:RULE 2 (remote exploit of a server program):0

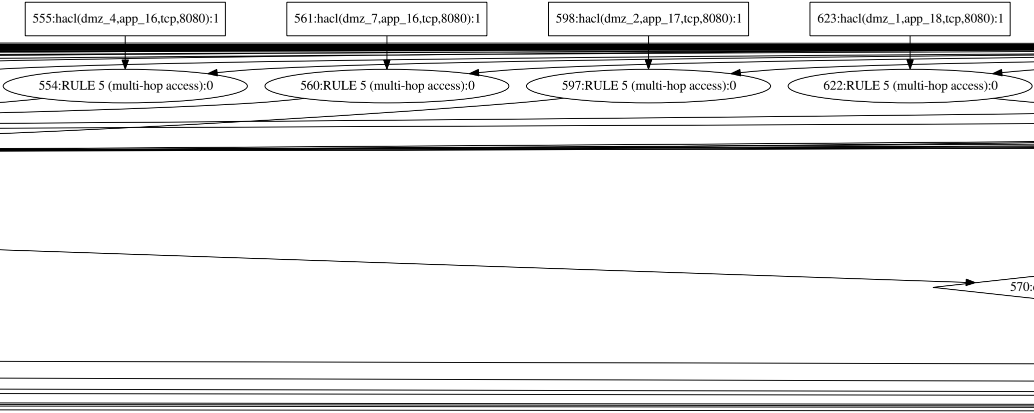

508:hacl(dmz_4,app_15,tcp,8080):1
514:hacl(dmz_7,app_15,tcp,8080):1
551:hacl(dmz_2,app_16,tcp,8080):1
576:hacl(dmz_1,app_17,tcp,8080):1
507:RULE 5 (multi-hop access):0
513:RULE 5 (multi-hop access):0
550:RULE 5 (multi-hop access):0
575:RULE 5 (multi-hop access):0
523:execCode(app_16,root):0

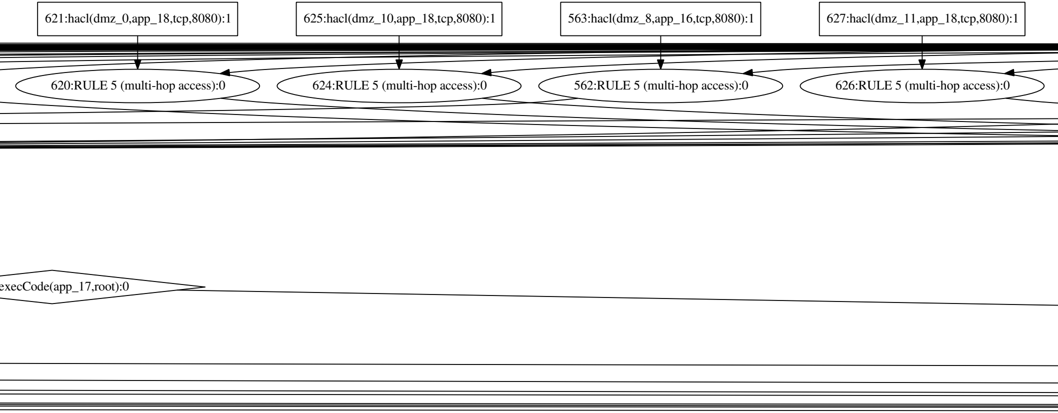

574:hacl(dmz_0,app_17,tcp,8080):1
578:hacl(dmz_10,app_17,tcp,8080):1
516:hacl(dmz_8,app_15,tcp,8080):1
580:hacl(dmz_11,app_17,tcp,8080):1
573:RULE 5 (multi-hop access):0
577:RULE 5 (multi-hop access):0
515:RULE 5 (multi-hop access):0
579:RULE 5 (multi-hop access):0

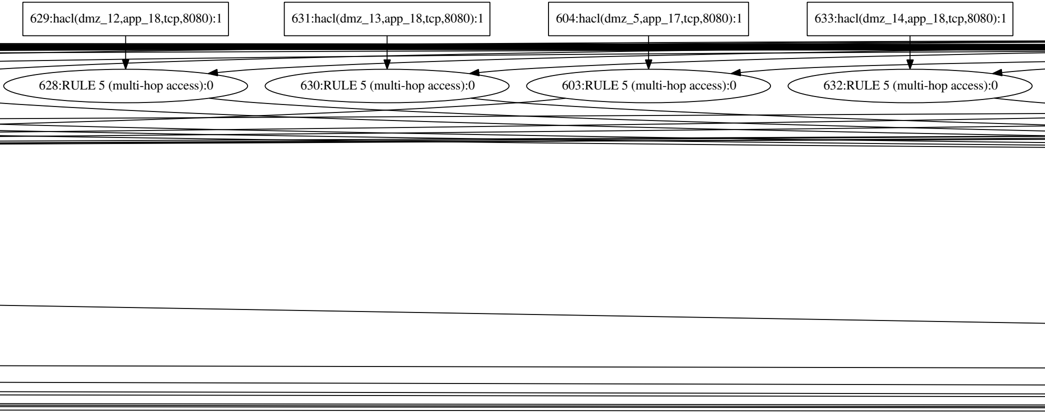

582:hacl(dmz_12,app_17,tcp,8080):1
584:hacl(dmz_13,app_17,tcp,8080):1
557:hacl(dmz_5,app_16,tcp,8080):1
586:hacl(dmz_14,app_17,tcp,8080):1
581:RULE 5 (multi-hop access):0
583:RULE 5 (multi-hop access):0
556:RULE 5 (multi-hop access):0
585:RULE 5 (multi-hop access):0
522:hacl(app_16,db_39,tcp,5432):1
521:RULE 5 (multi-hop access):0

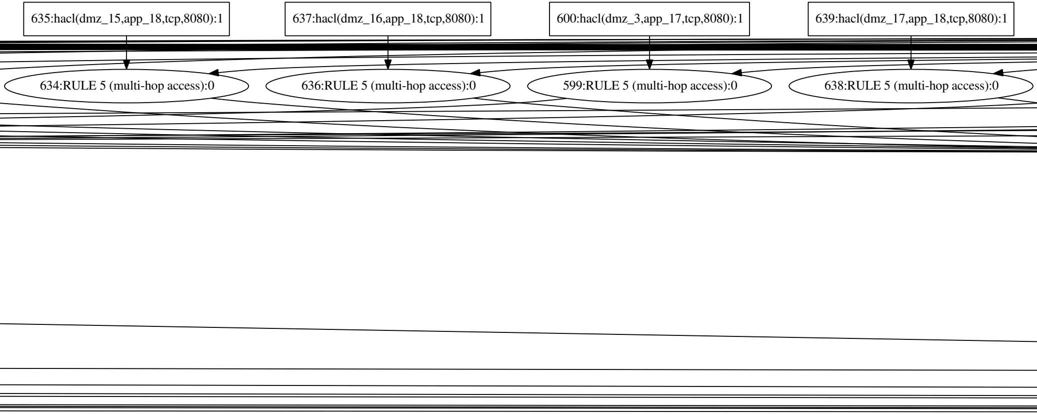

588:hacl(dmz_15,app_17,tcp,8080):1
590:hacl(dmz_16,app_17,tcp,8080):1
553:hacl(dmz_3,app_16,tcp,8080):1
592:hacl(dmz_17,app_17,tcp,8080):1
587:RULE 5 (multi-hop access):0
589:RULE 5 (multi-hop access):0
552:RULE 5 (multi-hop access):0
591:RULE 5 (multi-hop access):0
613:networkServiceInfo(app_17,tomcat,tcp,8080,root):1
614:vulExists(app_17,vul_app,t

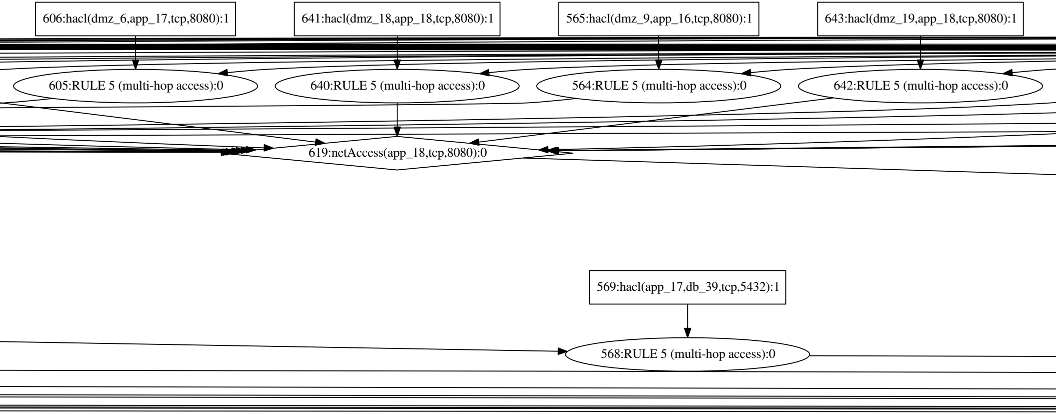

559:hacl(dmz_6,app_16,tcp,8080):1
594:hacl(dmz_18,app_17,tcp,8080):1
518:hacl(dmz_9,app_15,tcp,8080):1
596:hacl(dmz_19,app_17,tcp,8080):1
558:RULE 5 (multi-hop access):0
593:RULE 5 (multi-hop access):0
517:RULE 5 (multi-hop access):0
595:RULE 5 (multi-hop access):0
tomcat,remoteExploit,privEscalation):1
572:netAccess(app_17,tcp,8080):0
571:RULE 2 (remote exploit of a server program):0

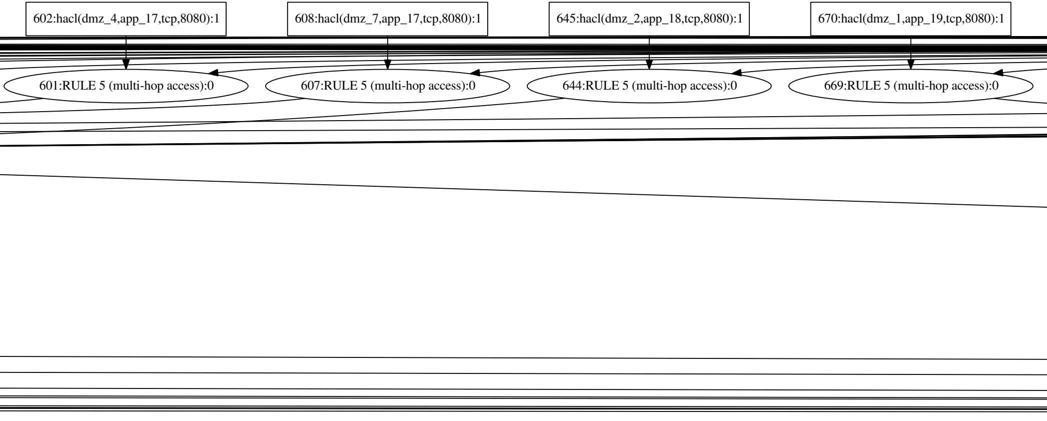

555:hacl(dmz_4,app_16,tcp,8080):1
561:hacl(dmz_7,app_16,tcp,8080):1
598:hacl(dmz_2,app_17,tcp,8080):1
623:hacl(dmz_1,app_18,tcp,8080):1
554:RULE 5 (multi-hop access):0
560:RULE 5 (multi-hop access):0
597:RULE 5 (multi-hop access):0
622:RULE 5 (multi-hop access):0
570:

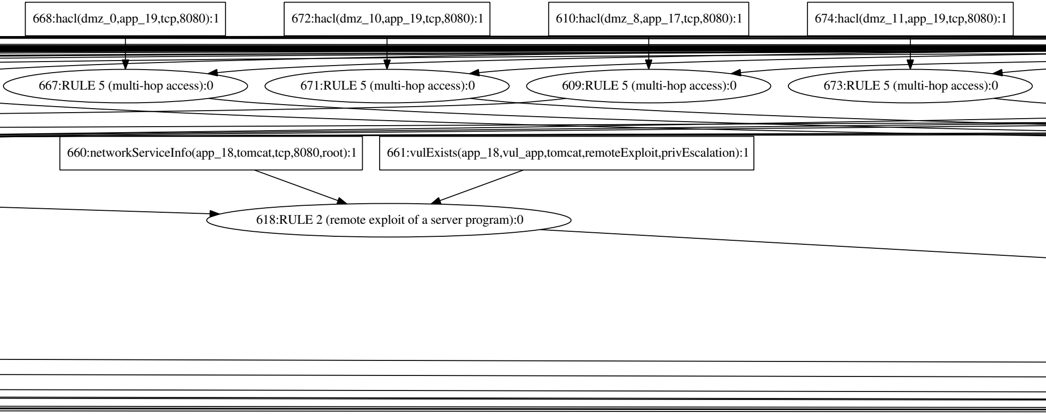

621:hacl(dmz_0,app_18,tcp,8080):1
625:hacl(dmz_10,app_18,tcp,8080):1
563:hacl(dmz_8,app_16,tcp,8080):1
627:hacl(dmz_11,app_18,tcp,8080):1
620:RULE 5 (multi-hop access):0
624:RULE 5 (multi-hop access):0
562:RULE 5 (multi-hop access):0
626:RULE 5 (multi-hop access):0
execCode(app_17,root):0

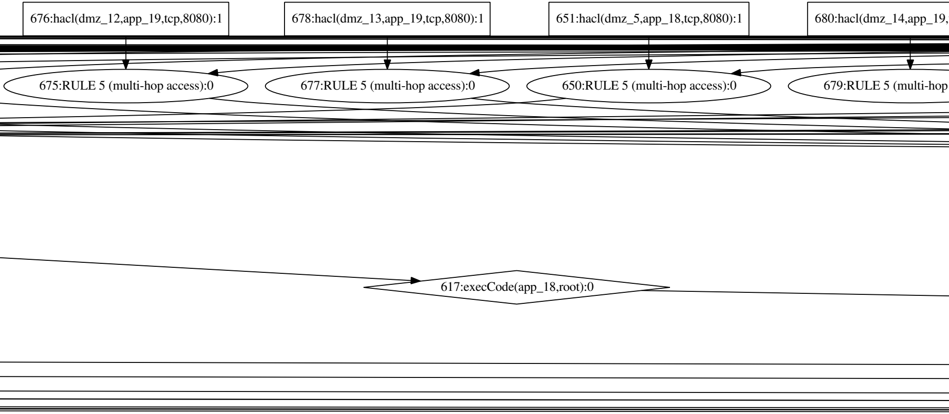

629:hacl(dmz_12,app_18,tcp,8080):1
631:hacl(dmz_13,app_18,tcp,8080):1
604:hacl(dmz_5,app_17,tcp,8080):1
633:hacl(dmz_14,app_18,tcp,8080):1
628:RULE 5 (multi-hop access):0
630:RULE 5 (multi-hop access):0
603:RULE 5 (multi-hop access):0
632:RULE 5 (multi-hop access):0

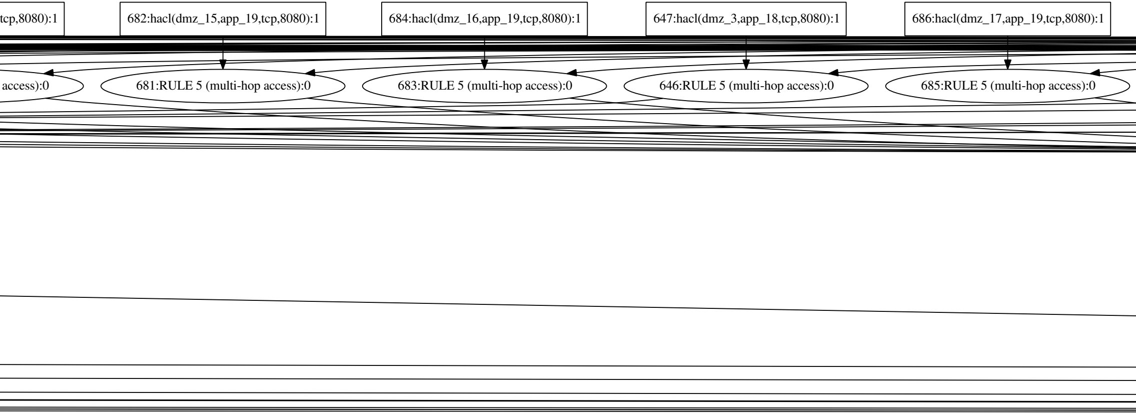

635:hacl(dmz_15,app_18,tcp,8080):1
637:hacl(dmz_16,app_18,tcp,8080):1
600:hacl(dmz_3,app_17,tcp,8080):1
639:hacl(dmz_17,app_18,tcp,8080):1
634:RULE 5 (multi-hop access):0
636:RULE 5 (multi-hop access):0
599:RULE 5 (multi-hop access):0
638:RULE 5 (multi-hop access):0

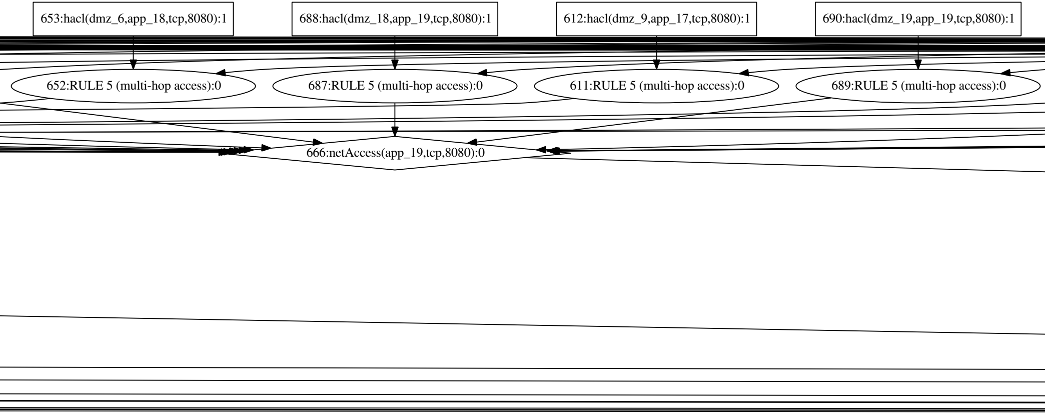

606:hacl(dmz_6,app_17,tcp,8080):1
641:hacl(dmz_18,app_18,tcp,8080):1
565:hacl(dmz_9,app_16,tcp,8080):1
643:hacl(dmz_19,app_18,tcp,8080):1
605:RULE 5 (multi-hop access):0
640:RULE 5 (multi-hop access):0
564:RULE 5 (multi-hop access):0
642:RULE 5 (multi-hop access):0
619:netAccess(app_18,tcp,8080):0
569:hacl(app_17,db_39,tcp,5432):1
568:RULE 5 (multi-hop access):0

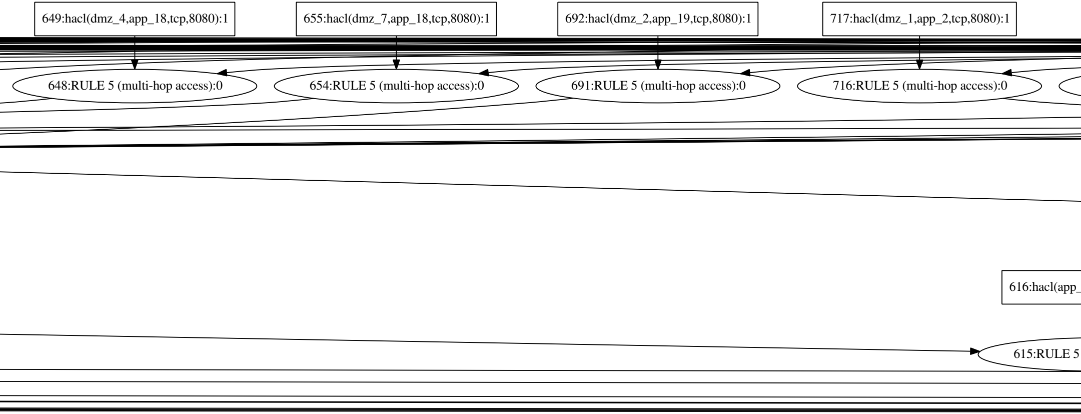

602:hacl(dmz_4,app_17,tcp,8080):1
608:hacl(dmz_7,app_17,tcp,8080):1
645:hacl(dmz_2,app_18,tcp,8080):1
670:hacl(dmz_1,app_19,tcp,8080):1
601:RULE 5 (multi-hop access):0
607:RULE 5 (multi-hop access):0
644:RULE 5 (multi-hop access):0
669:RULE 5 (multi-hop access):0

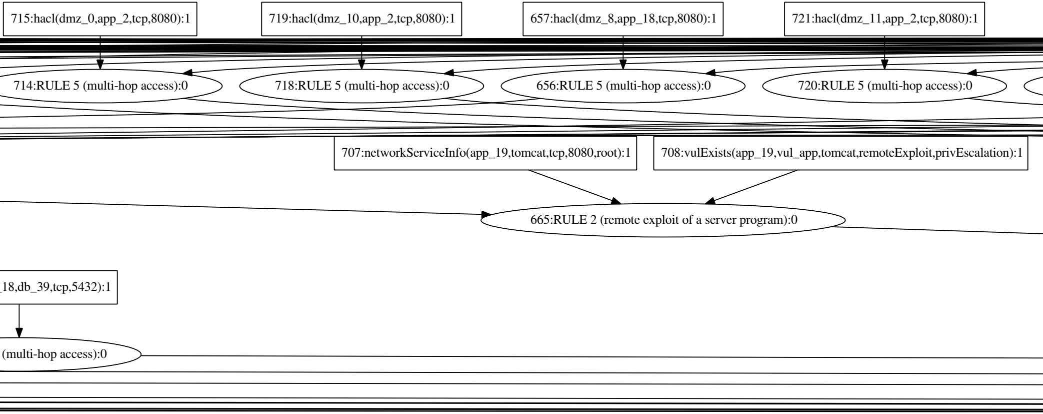

668:hacl(dmz_0,app_19,tcp,8080):1
672:hacl(dmz_10,app_19,tcp,8080):1
610:hacl(dmz_8,app_17,tcp,8080):1
674:hacl(dmz_11,app_19,tcp,8080):1
667:RULE 5 (multi-hop access):0
671:RULE 5 (multi-hop access):0
609:RULE 5 (multi-hop access):0
673:RULE 5 (multi-hop access):0
660:networkServiceInfo(app_18,tomcat,tcp,8080,root):1
661:vulExists(app_18,vul_app,tomcat,remoteExploit,privEscalation):1
618:RULE 2 (remote exploit of a server program):0

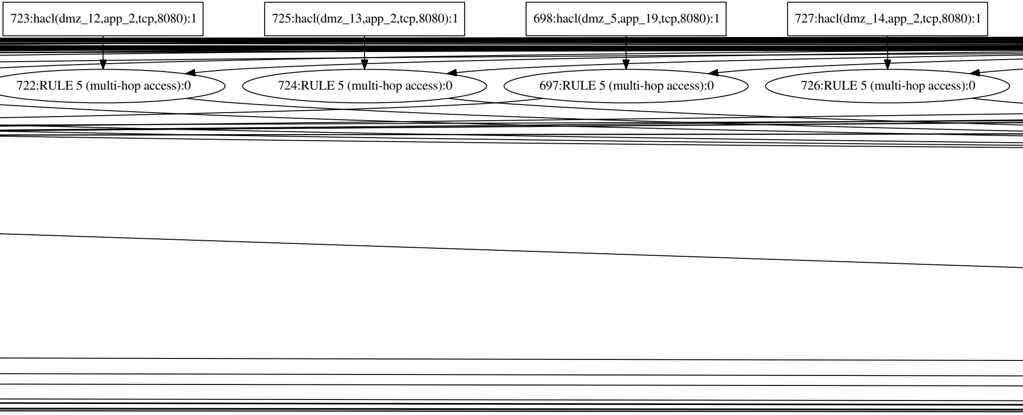

676:hacl(dmz_12,app_19,tcp,8080):1
678:hacl(dmz_13,app_19,tcp,8080):1
651:hacl(dmz_5,app_18,tcp,8080):1
680:hacl(dmz_14,app_19,
675:RULE 5 (multi-hop access):0
677:RULE 5 (multi-hop access):0
650:RULE 5 (multi-hop access):0
679:RULE 5 (multi-hop
617:execCode(app_18,root):0

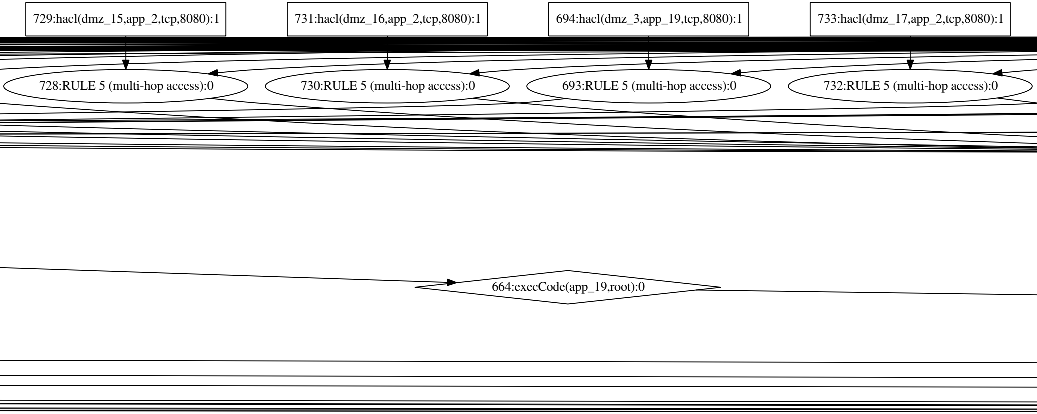

tcp,8080):1
682:hacl(dmz_15,app_19,tcp,8080):1
684:hacl(dmz_16,app_19,tcp,8080):1
647:hacl(dmz_3,app_18,tcp,8080):1
686:hacl(dmz_17,app_19,tcp,8080):1
access):0
681:RULE 5 (multi-hop access):0
683:RULE 5 (multi-hop access):0
646:RULE 5 (multi-hop access):0
685:RULE 5 (multi-hop access):0

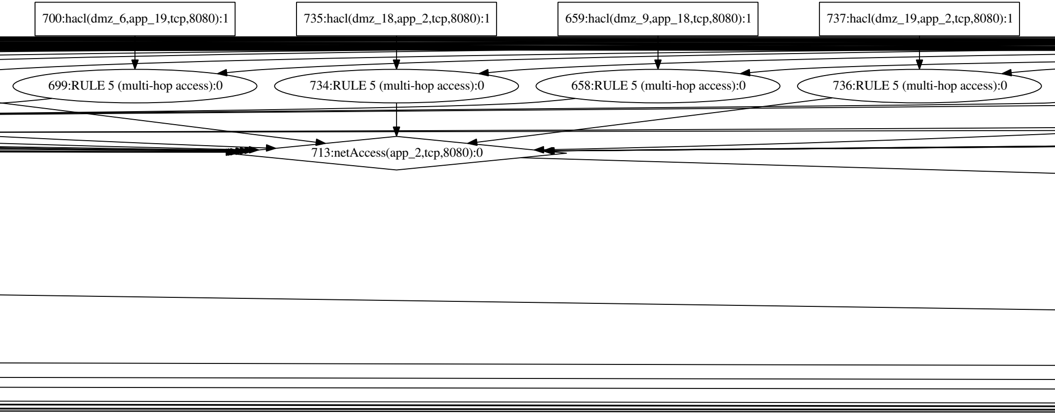

653:hacl(dmz_6,app_18,tcp,8080):1
688:hacl(dmz_18,app_19,tcp,8080):1
612:hacl(dmz_9,app_17,tcp,8080):1
690:hacl(dmz_19,app_19,tcp,8080):1
652:RULE 5 (multi-hop access):0
687:RULE 5 (multi-hop access):0
611:RULE 5 (multi-hop access):0
689:RULE 5 (multi-hop access):0
666:netAccess(app_19,tcp,8080):0

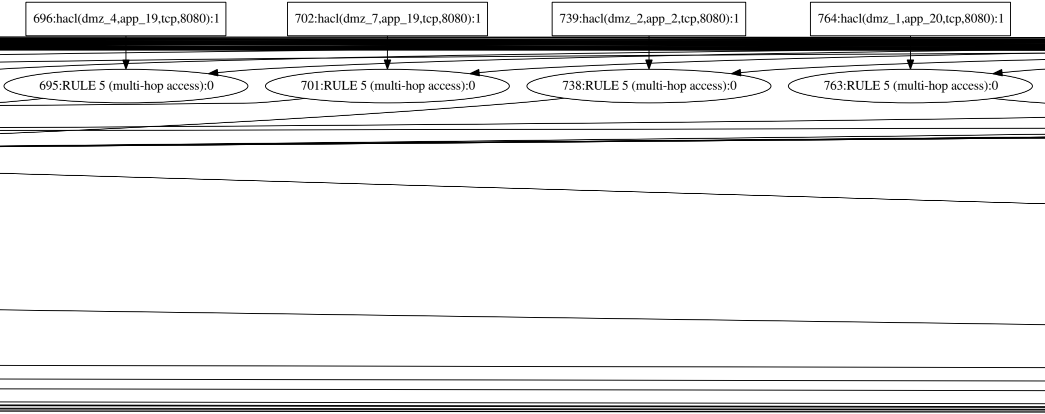

649:hacl(dmz_4,app_18,tcp,8080):1
655:hacl(dmz_7,app_18,tcp,8080):1
692:hacl(dmz_2,app_19,tcp,8080):1
717:hacl(dmz_1,app_2,tcp,8080):1
648:RULE 5 (multi-hop access):0
654:RULE 5 (multi-hop access):0
691:RULE 5 (multi-hop access):0
716:RULE 5 (multi-hop access):0
616:hacl(app_
615:RULE 5

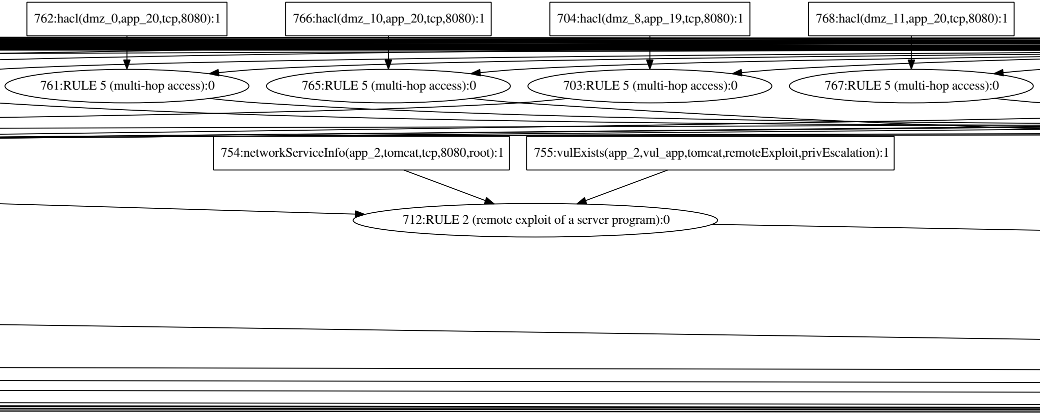

715:hacl(dmz_0,app_2,tcp,8080):1
719:hacl(dmz_10,app_2,tcp,8080):1
657:hacl(dmz_8,app_18,tcp,8080):1
721:hacl(dmz_11,app_2,tcp,8080):1
714:RULE 5 (multi-hop access):0
718:RULE 5 (multi-hop access):0
656:RULE 5 (multi-hop access):0
720:RULE 5 (multi-hop access):0
707:networkServiceInfo(app_19,tomcat,tcp,8080,root):1
708:vulExists(app_19,vul_app,tomcat,remoteExploit,privEscalation):1
665:RULE 2 (remote exploit of a server program):0
_18,db_39,tcp,5432):1
(multi-hop access):0

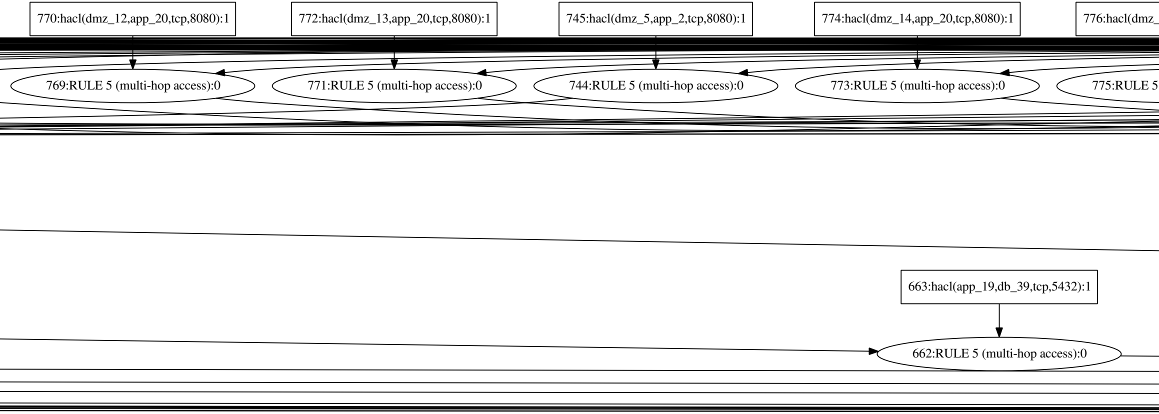

723:hacl(dmz_12,app_2,tcp,8080):1
725:hacl(dmz_13,app_2,tcp,8080):1
698:hacl(dmz_5,app_19,tcp,8080):1
727:hacl(dmz_14,app_2,tcp,8080):1
722:RULE 5 (multi-hop access):0
724:RULE 5 (multi-hop access):0
697:RULE 5 (multi-hop access):0
726:RULE 5 (multi-hop access):0

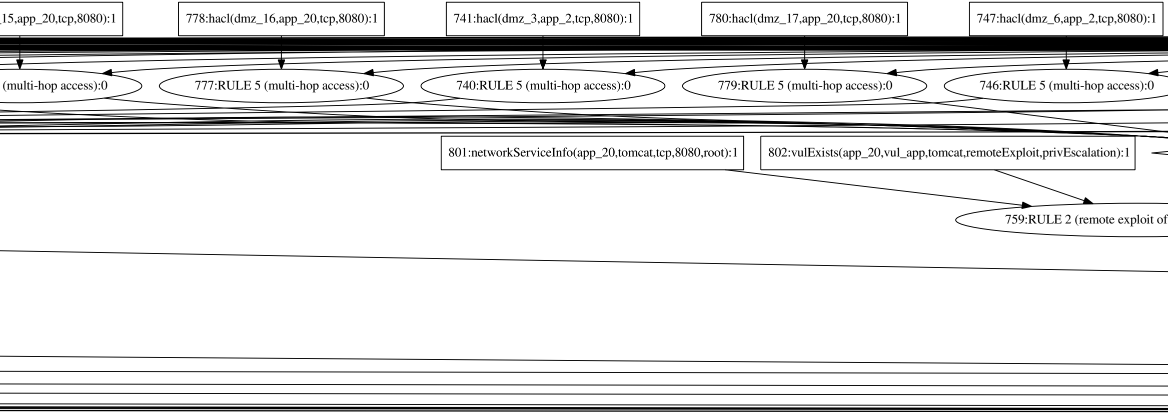

729:hacl(dmz_15,app_2,tcp,8080):1
731:hacl(dmz_16,app_2,tcp,8080):1
694:hacl(dmz_3,app_19,tcp,8080):1
733:hacl(dmz_17,app_2,tcp,8080):1
728:RULE 5 (multi-hop access):0
730:RULE 5 (multi-hop access):0
693:RULE 5 (multi-hop access):0
732:RULE 5 (multi-hop access):0
664:execCode(app_19,root):0

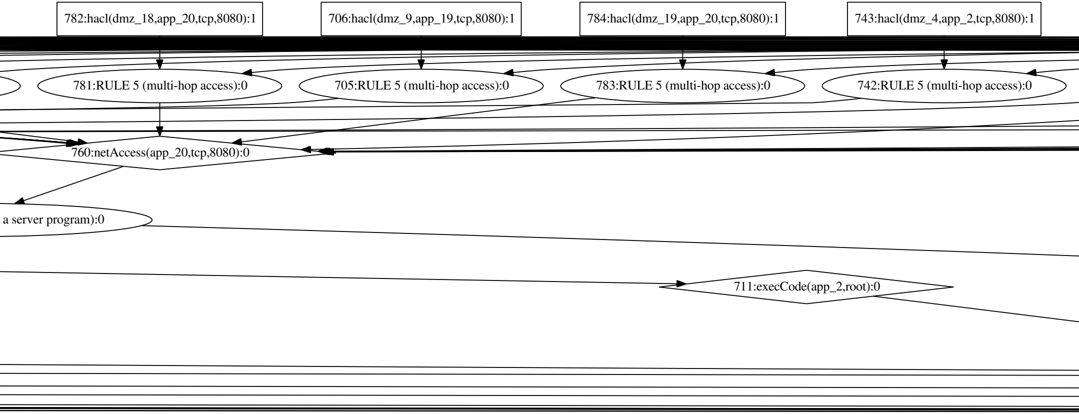

700:hacl(dmz_6,app_19,tcp,8080):1
735:hacl(dmz_18,app_2,tcp,8080):1
659:hacl(dmz_9,app_18,tcp,8080):1
737:hacl(dmz_19,app_2,tcp,8080):1
699:RULE 5 (multi-hop access):0
734:RULE 5 (multi-hop access):0
658:RULE 5 (multi-hop access):0
736:RULE 5 (multi-hop access):0
713:netAccess(app_2,tcp,8080):0

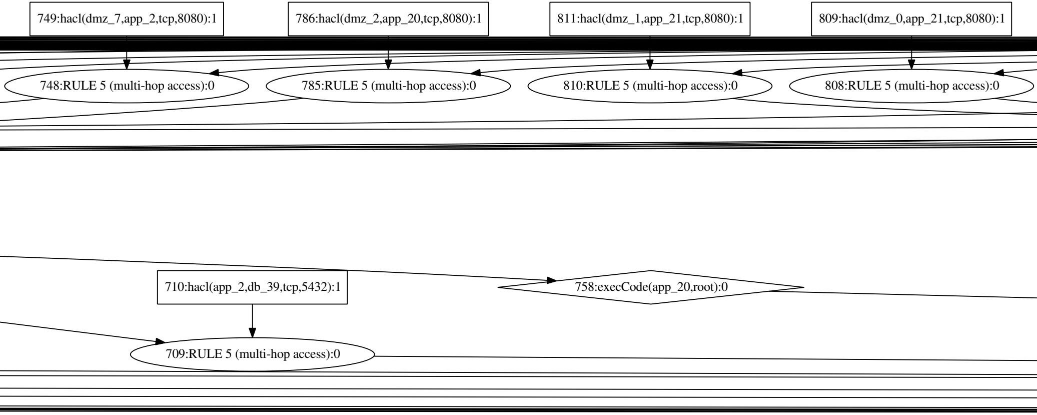

696:hacl(dmz_4,app_19,tcp,8080):1
702:hacl(dmz_7,app_19,tcp,8080):1
739:hacl(dmz_2,app_2,tcp,8080):1
764:hacl(dmz_1,app_20,tcp,8080):1
695:RULE 5 (multi-hop access):0
701:RULE 5 (multi-hop access):0
738:RULE 5 (multi-hop access):0
763:RULE 5 (multi-hop access):0

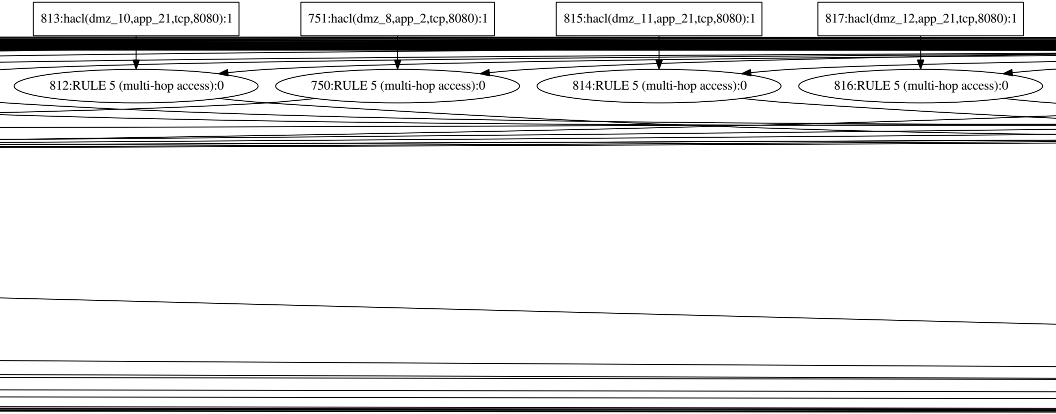

762:hacl(dmz_0,app_20,tcp,8080):1
766:hacl(dmz_10,app_20,tcp,8080):1
704:hacl(dmz_8,app_19,tcp,8080):1
768:hacl(dmz_11,app_20,tcp,8080):1
761:RULE 5 (multi-hop access):0
765:RULE 5 (multi-hop access):0
703:RULE 5 (multi-hop access):0
767:RULE 5 (multi-hop access):0
754:networkServiceInfo(app_2,tomcat,tcp,8080,root):1
755:vulExists(app_2,vul_app,tomcat,remoteExploit,privEscalation):1
712:RULE 2 (remote exploit of a server program):0

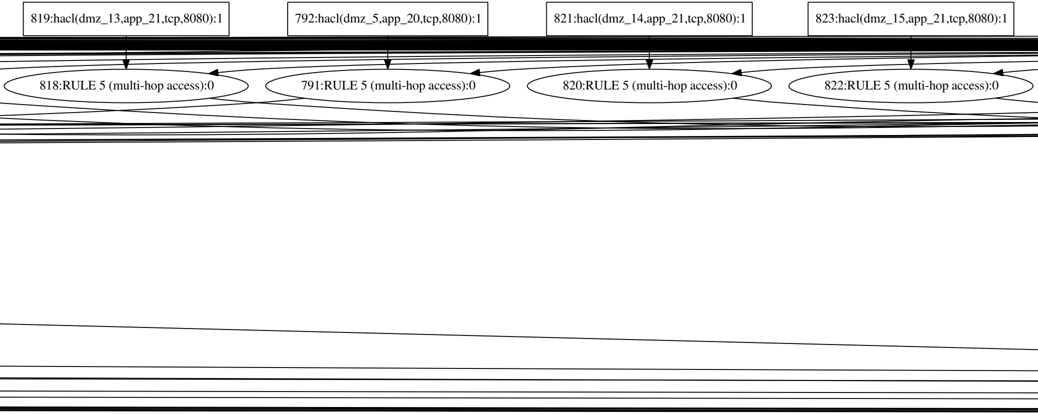

770:hacl(dmz_12,app_20,tcp,8080):1
772:hacl(dmz_13,app_20,tcp,8080):1
745:hacl(dmz_5,app_2,tcp,8080):1
774:hacl(dmz_14,app_20,tcp,8080):1
776:hacl(dmz_
769:RULE 5 (multi-hop access):0
771:RULE 5 (multi-hop access):0
744:RULE 5 (multi-hop access):0
773:RULE 5 (multi-hop access):0
775:RULE 5
663:hacl(app_19,db_39,tcp,5432):1
662:RULE 5 (multi-hop access):0

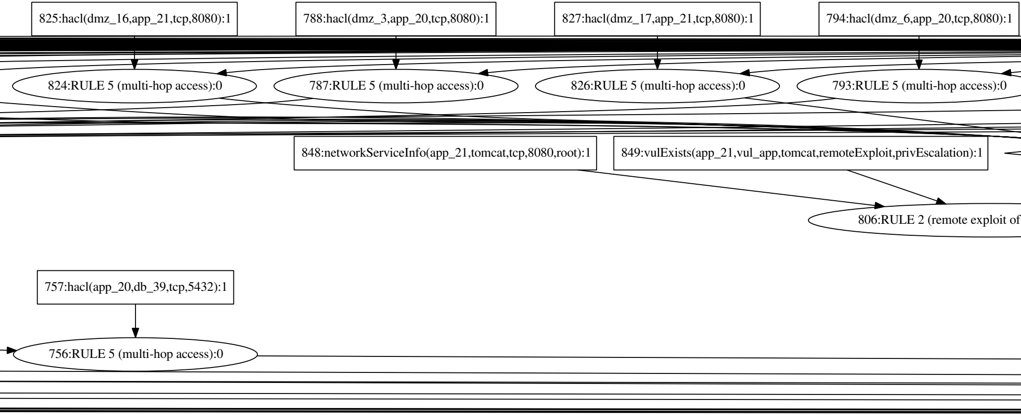

15,app_20,tcp,8080):1
778:hacl(dmz_16,app_20,tcp,8080):1
741:hacl(dmz_3,app_2,tcp,8080):1
780:hacl(dmz_17,app_20,tcp,8080):1
747:hacl(dmz_6,app_2,tcp,8080):1
(multi-hop access):0
777:RULE 5 (multi-hop access):0
740:RULE 5 (multi-hop access):0
779:RULE 5 (multi-hop access):0
746:RULE 5 (multi-hop access):0
801:networkServiceInfo(app_20,tomcat,tcp,8080,root):1
802:vulExists(app_20,vul_app,tomcat,remoteExploit,privEscalation):1
759:RULE 2 (remote exploit of

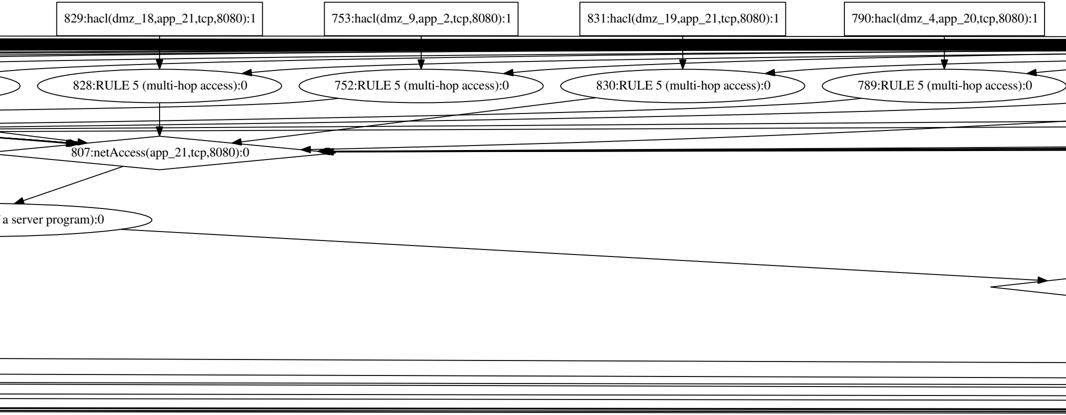

782:hacl(dmz_18,app_20,tcp,8080):1
706:hacl(dmz_9,app_19,tcp,8080):1
784:hacl(dmz_19,app_20,tcp,8080):1
743:hacl(dmz_4,app_2,tcp,8080):1
781:RULE 5 (multi-hop access):0
705:RULE 5 (multi-hop access):0
783:RULE 5 (multi-hop access):0
742:RULE 5 (multi-hop access):0
760:netAccess(app_20,tcp,8080):0
a server program):0
711:execCode(app_2,root):0

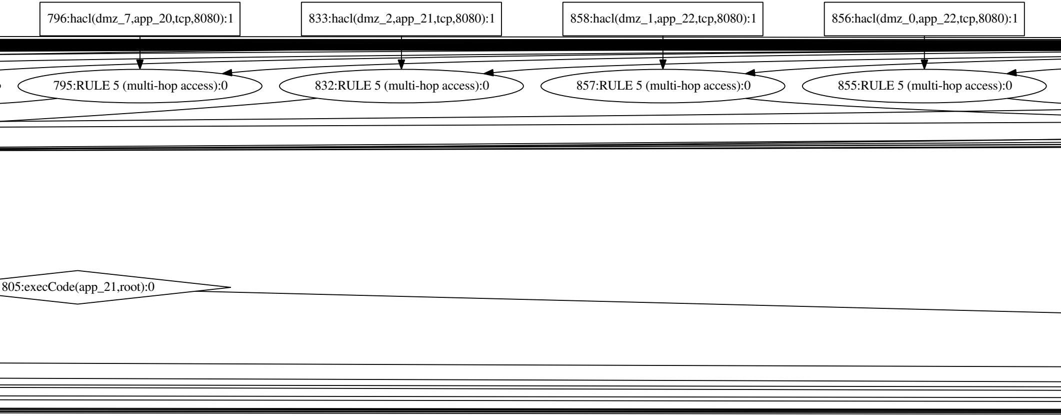

749:hacl(dmz_7,app_2,tcp,8080):1
786:hacl(dmz_2,app_20,tcp,8080):1
811:hacl(dmz_1,app_21,tcp,8080):1
809:hacl(dmz_0,app_21,tcp,8080):1
748:RULE 5 (multi-hop access):0
785:RULE 5 (multi-hop access):0
810:RULE 5 (multi-hop access):0
808:RULE 5 (multi-hop access):0
710:hacl(app_2,db_39,tcp,5432):1
758:execCode(app_20,root):0
709:RULE 5 (multi-hop access):0

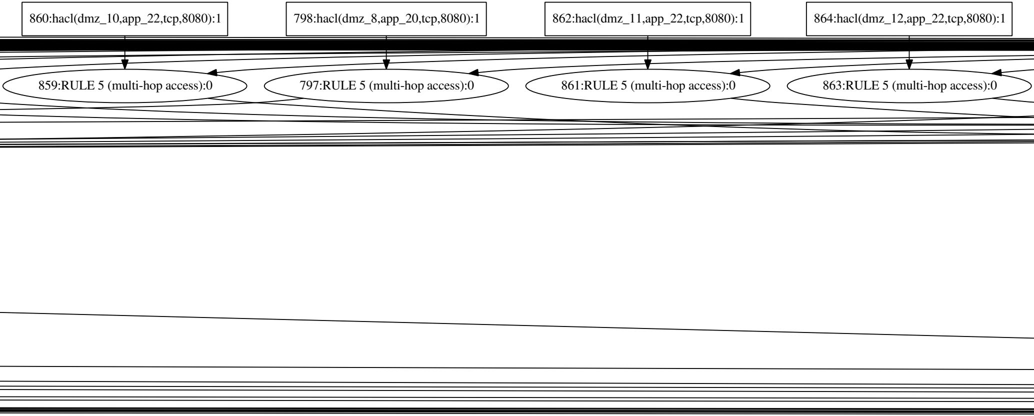

813:hacl(dmz_10,app_21,tcp,8080):1
751:hacl(dmz_8,app_2,tcp,8080):1
815:hacl(dmz_11,app_21,tcp,8080):1
817:hacl(dmz_12,app_21,tcp,8080):1
812:RULE 5 (multi-hop access):0
750:RULE 5 (multi-hop access):0
814:RULE 5 (multi-hop access):0
816:RULE 5 (multi-hop access):0

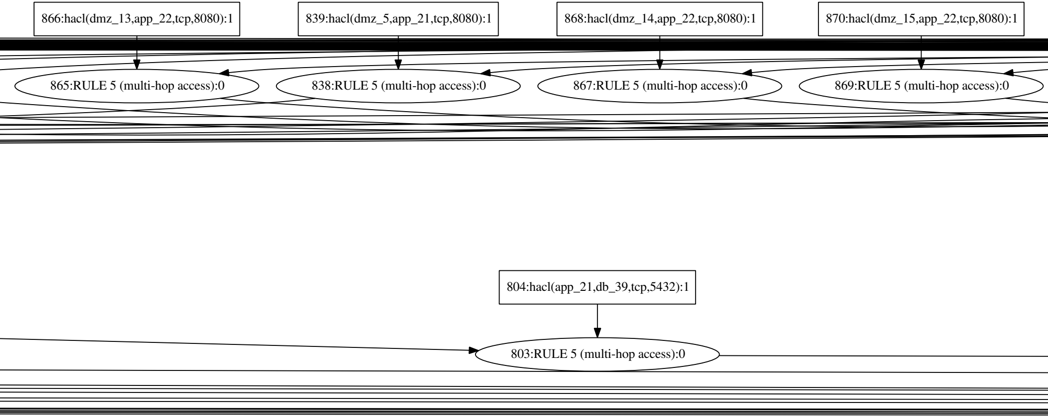

819:hacl(dmz_13,app_21,tcp,8080):1
792:hacl(dmz_5,app_20,tcp,8080):1
821:hacl(dmz_14,app_21,tcp,8080):1
823:hacl(dmz_15,app_21,tcp,8080):1
818:RULE 5 (multi-hop access):0
791:RULE 5 (multi-hop access):0
820:RULE 5 (multi-hop access):0
822:RULE 5 (multi-hop access):0

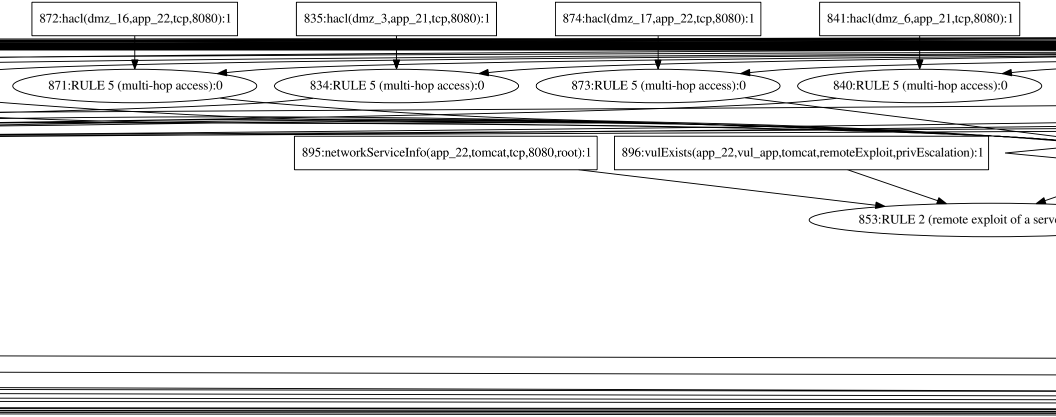

825:hacl(dmz_16,app_21,tcp,8080):1
788:hacl(dmz_3,app_20,tcp,8080):1
827:hacl(dmz_17,app_21,tcp,8080):1
794:hacl(dmz_6,app_20,tcp,8080):1
824:RULE 5 (multi-hop access):0
787:RULE 5 (multi-hop access):0
826:RULE 5 (multi-hop access):0
793:RULE 5 (multi-hop access):0
848:networkServiceInfo(app_21,tomcat,tcp,8080,root):1
849:vulExists(app_21,vul_app,tomcat,remoteExploit,privEscalation):1
806:RULE 2 (remote exploit of
757:hacl(app_20,db_39,tcp,5432):1
756:RULE 5 (multi-hop access):0

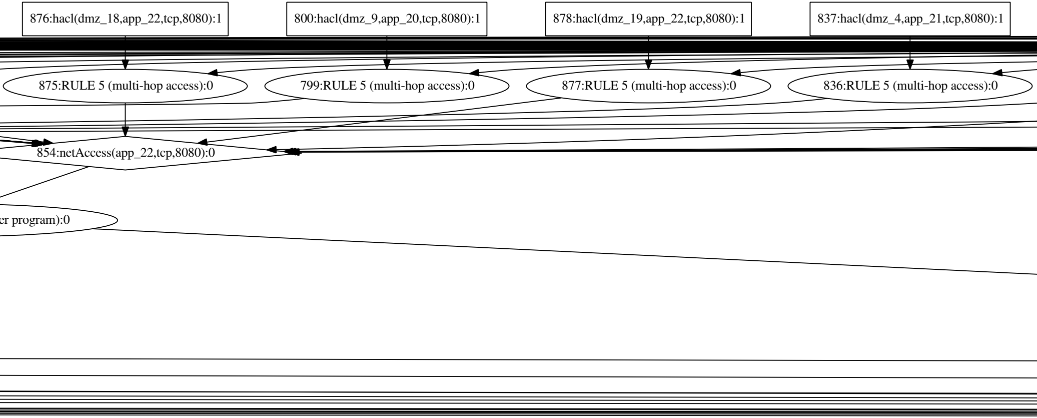

829:hacl(dmz_18,app_21,tcp,8080):1
753:hacl(dmz_9,app_2,tcp,8080):1
831:hacl(dmz_19,app_21,tcp,8080):1
790:hacl(dmz_4,app_20,tcp,8080):1
828:RULE 5 (multi-hop access):0
752:RULE 5 (multi-hop access):0
830:RULE 5 (multi-hop access):0
789:RULE 5 (multi-hop access):0
807:netAccess(app_21,tcp,8080):0
a server program):0

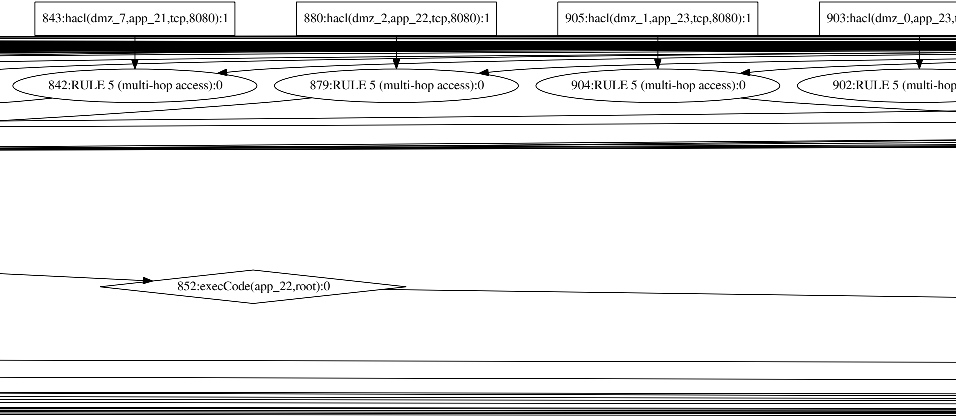

796:hacl(dmz_7,app_20,tcp,8080):1
833:hacl(dmz_2,app_21,tcp,8080):1
858:hacl(dmz_1,app_22,tcp,8080):1
856:hacl(dmz_0,app_22,tcp,8080):1
795:RULE 5 (multi-hop access):0
832:RULE 5 (multi-hop access):0
857:RULE 5 (multi-hop access):0
855:RULE 5 (multi-hop access):0
805:execCode(app_21,root):0

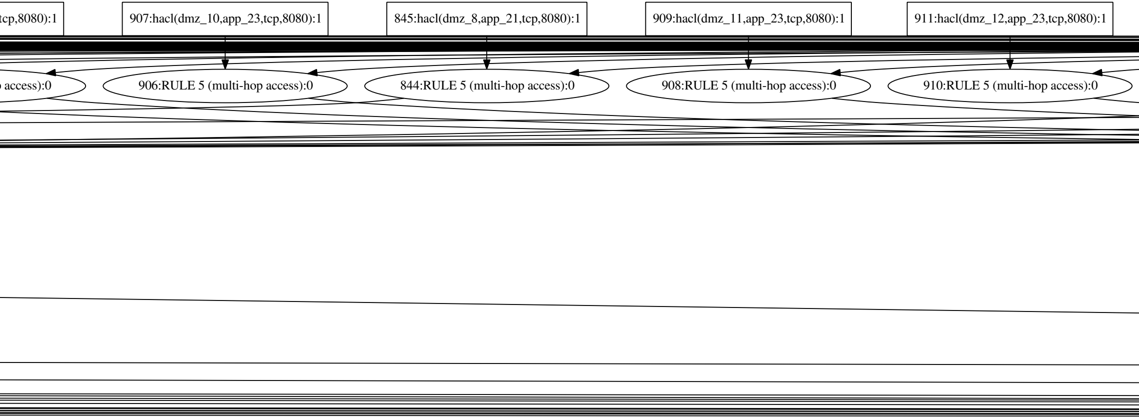

860:hacl(dmz_10,app_22,tcp,8080):1
798:hacl(dmz_8,app_20,tcp,8080):1
862:hacl(dmz_11,app_22,tcp,8080):1
864:hacl(dmz_12,app_22,tcp,8080):1
859:RULE 5 (multi-hop access):0
797:RULE 5 (multi-hop access):0
861:RULE 5 (multi-hop access):0
863:RULE 5 (multi-hop access):0

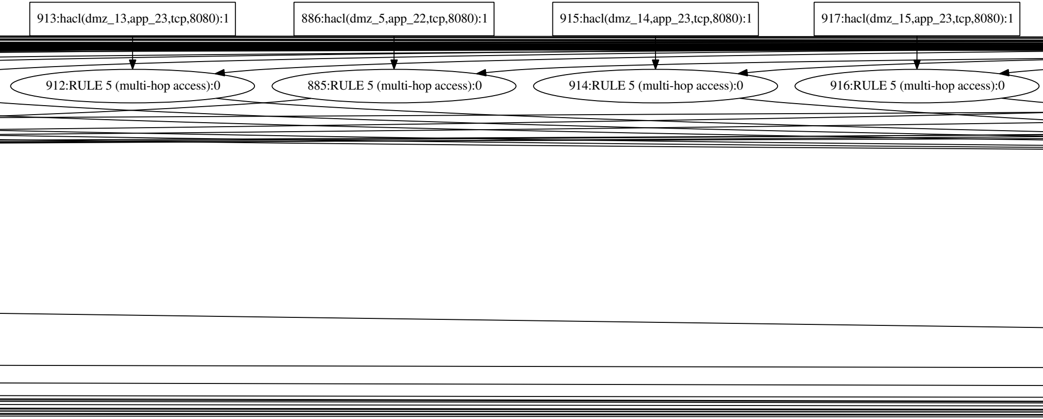

866:hacl(dmz_13,app_22,tcp,8080):1
839:hacl(dmz_5,app_21,tcp,8080):1
868:hacl(dmz_14,app_22,tcp,8080):1
870:hacl(dmz_15,app_22,tcp,8080):1
865:RULE 5 (multi-hop access):0
838:RULE 5 (multi-hop access):0
867:RULE 5 (multi-hop access):0
869:RULE 5 (multi-hop access):0
804:hacl(app_21,db_39,tcp,5432):1
803:RULE 5 (multi-hop access):0

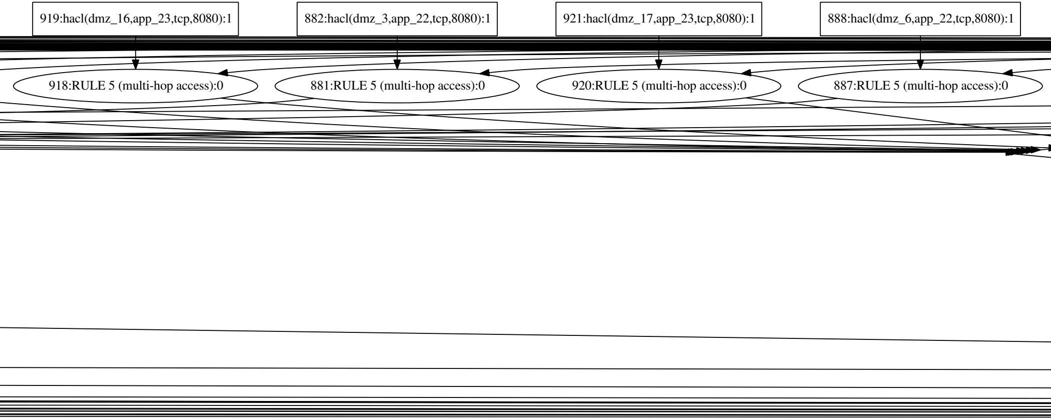

872:hacl(dmz_16,app_22,tcp,8080):1
835:hacl(dmz_3,app_21,tcp,8080):1
874:hacl(dmz_17,app_22,tcp,8080):1
841:hacl(dmz_6,app_21,tcp,8080):1
871:RULE 5 (multi-hop access):0
834:RULE 5 (multi-hop access):0
873:RULE 5 (multi-hop access):0
840:RULE 5 (multi-hop access):0
895:networkServiceInfo(app_22,tomcat,tcp,8080,root):1
896:vulExists(app_22,vul_app,tomcat,remoteExploit,privEscalation):1
853:RULE 2 (remote exploit of a serv

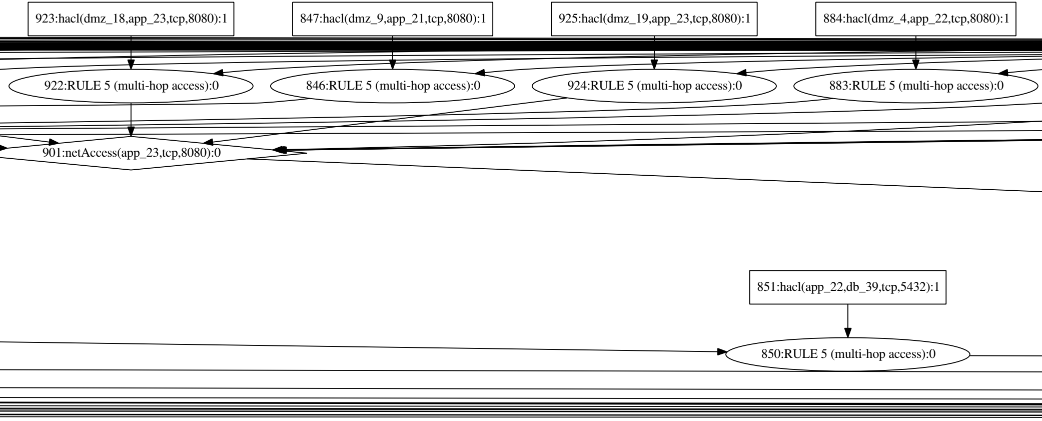

876:hacl(dmz_18,app_22,tcp,8080):1
800:hacl(dmz_9,app_20,tcp,8080):1
878:hacl(dmz_19,app_22,tcp,8080):1
837:hacl(dmz_4,app_21,tcp,8080):1
875:RULE 5 (multi-hop access):0
799:RULE 5 (multi-hop access):0
877:RULE 5 (multi-hop access):0
836:RULE 5 (multi-hop access):0
854:netAccess(app_22,tcp,8080):0
er program):0

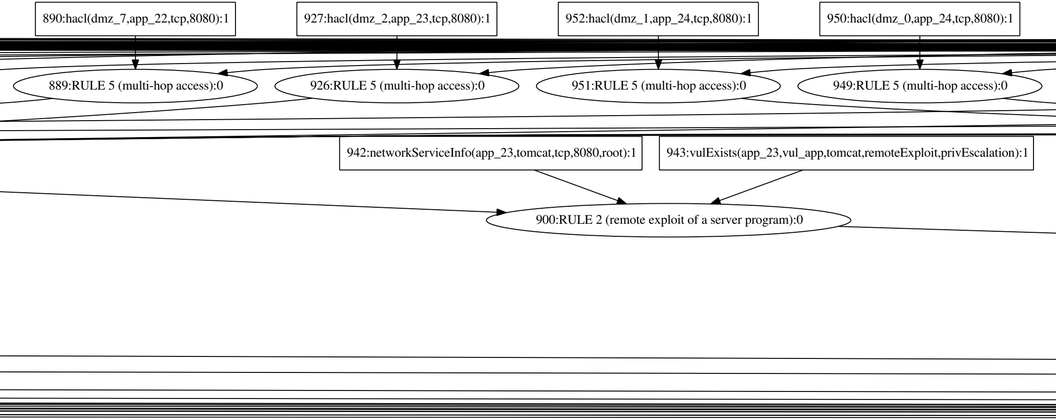

843:hacl(dmz_7,app_21,tcp,8080):1
880:hacl(dmz_2,app_22,tcp,8080):1
905:hacl(dmz_1,app_23,tcp,8080):1
903:hacl(dmz_0,app_23,t
842:RULE 5 (multi-hop access):0
879:RULE 5 (multi-hop access):0
904:RULE 5 (multi-hop access):0
902:RULE 5 (multi-hop
852:execCode(app_22,root):0

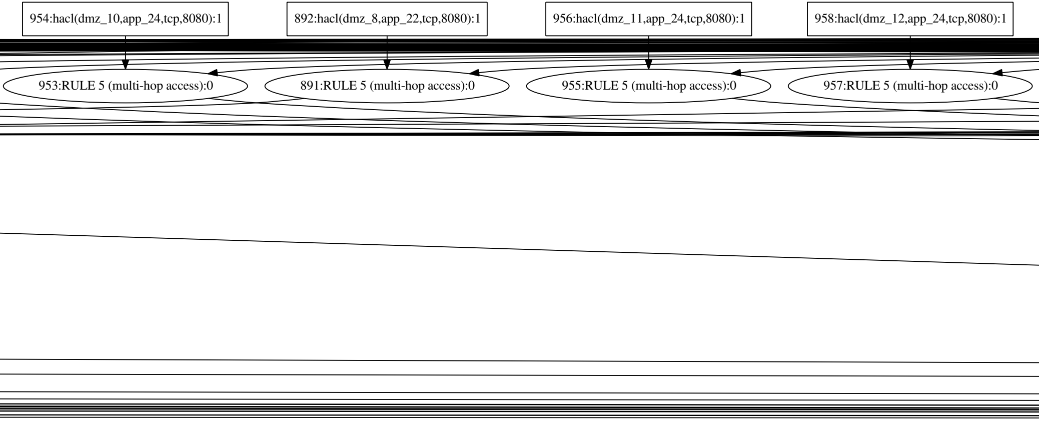

tcp,8080):1
907:hacl(dmz_10,app_23,tcp,8080):1
845:hacl(dmz_8,app_21,tcp,8080):1
909:hacl(dmz_11,app_23,tcp,8080):1
911:hacl(dmz_12,app_23,tcp,8080):1
access):0
906:RULE 5 (multi-hop access):0
844:RULE 5 (multi-hop access):0
908:RULE 5 (multi-hop access):0
910:RULE 5 (multi-hop access):0

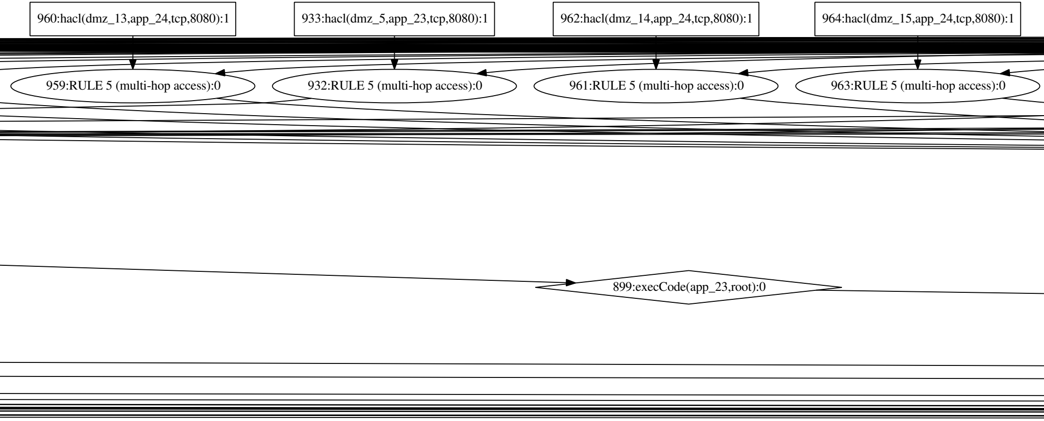

913:hacl(dmz_13,app_23,tcp,8080):1
886:hacl(dmz_5,app_22,tcp,8080):1
915:hacl(dmz_14,app_23,tcp,8080):1
917:hacl(dmz_15,app_23,tcp,8080):1
912:RULE 5 (multi-hop access):0
885:RULE 5 (multi-hop access):0
914:RULE 5 (multi-hop access):0
916:RULE 5 (multi-hop access):0

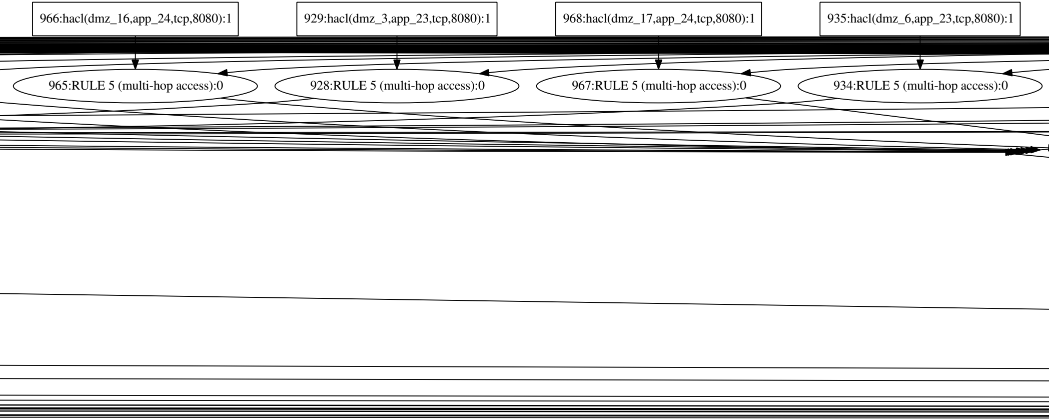

919:hacl(dmz_16,app_23,tcp,8080):1
882:hacl(dmz_3,app_22,tcp,8080):1
921:hacl(dmz_17,app_23,tcp,8080):1
888:hacl(dmz_6,app_22,tcp,8080):1
918:RULE 5 (multi-hop access):0
881:RULE 5 (multi-hop access):0
920:RULE 5 (multi-hop access):0
887:RULE 5 (multi-hop access):0

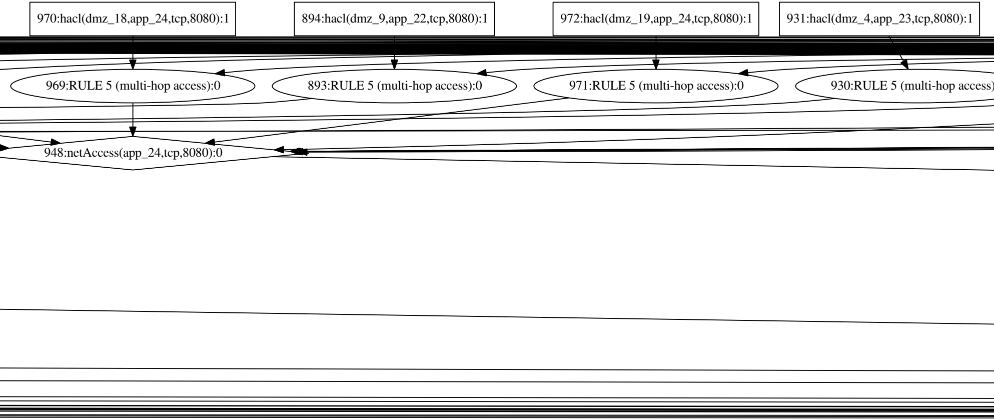

923:hacl(dmz_18,app_23,tcp,8080):1
847:hacl(dmz_9,app_21,tcp,8080):1
925:hacl(dmz_19,app_23,tcp,8080):1
884:hacl(dmz_4,app_22,tcp,8080):1
922:RULE 5 (multi-hop access):0
846:RULE 5 (multi-hop access):0
924:RULE 5 (multi-hop access):0
883:RULE 5 (multi-hop access):0
901:netAccess(app_23,tcp,8080):0
851:hacl(app_22,db_39,tcp,5432):1
850:RULE 5 (multi-hop access):0

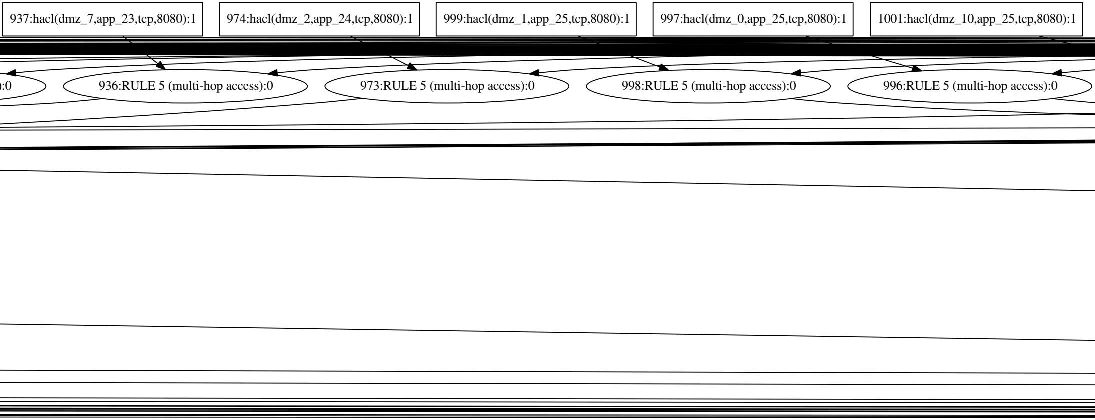

890:hacl(dmz_7,app_22,tcp,8080):1
927:hacl(dmz_2,app_23,tcp,8080):1
952:hacl(dmz_1,app_24,tcp,8080):1
950:hacl(dmz_0,app_24,tcp,8080):1
889:RULE 5 (multi-hop access):0
926:RULE 5 (multi-hop access):0
951:RULE 5 (multi-hop access):0
949:RULE 5 (multi-hop access):0
942:networkServiceInfo(app_23,tomcat,tcp,8080,root):1
943:vulExists(app_23,vul_app,tomcat,remoteExploit,privEscalation):1
900:RULE 2 (remote exploit of a server program):0

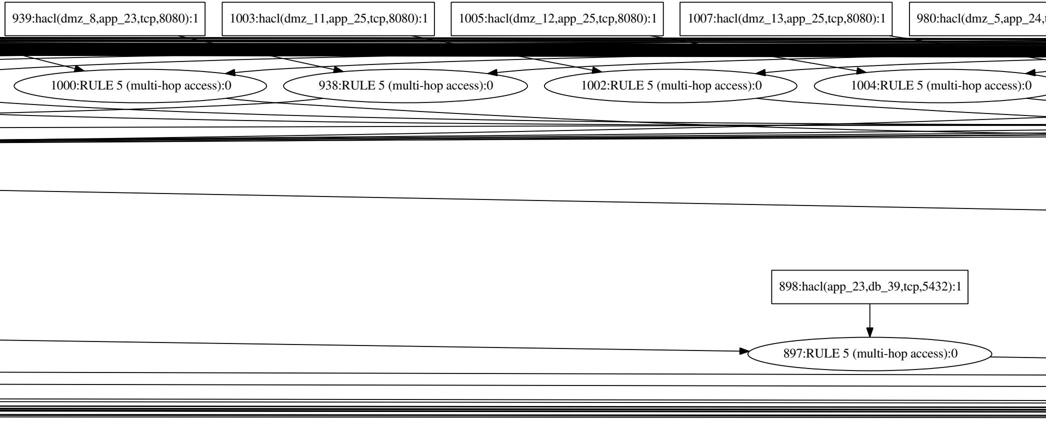

954:hacl(dmz_10,app_24,tcp,8080):1
892:hacl(dmz_8,app_22,tcp,8080):1
956:hacl(dmz_11,app_24,tcp,8080):1
958:hacl(dmz_12,app_24,tcp,8080):1
953:RULE 5 (multi-hop access):0
891:RULE 5 (multi-hop access):0
955:RULE 5 (multi-hop access):0
957:RULE 5 (multi-hop access):0

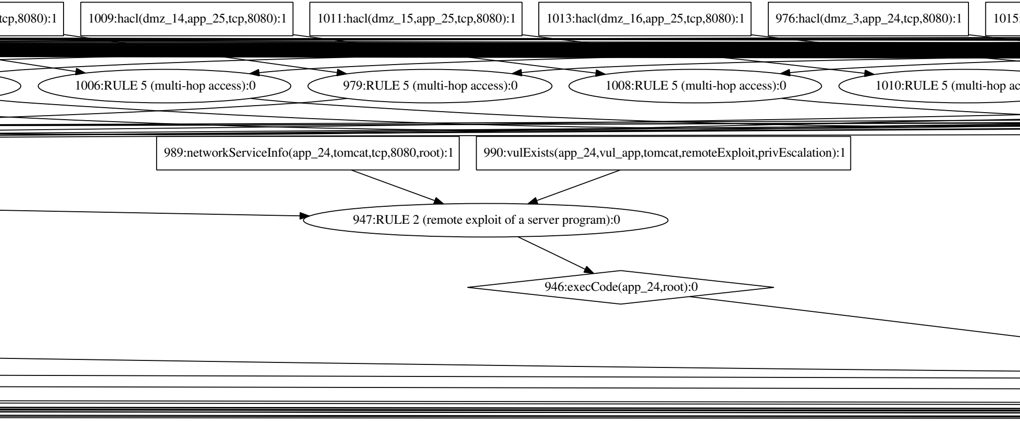

960:hacl(dmz_13,app_24,tcp,8080):1
933:hacl(dmz_5,app_23,tcp,8080):1
962:hacl(dmz_14,app_24,tcp,8080):1
964:hacl(dmz_15,app_24,tcp,8080):1
959:RULE 5 (multi-hop access):0
932:RULE 5 (multi-hop access):0
961:RULE 5 (multi-hop access):0
963:RULE 5 (multi-hop access):0
899:execCode(app_23,root):0

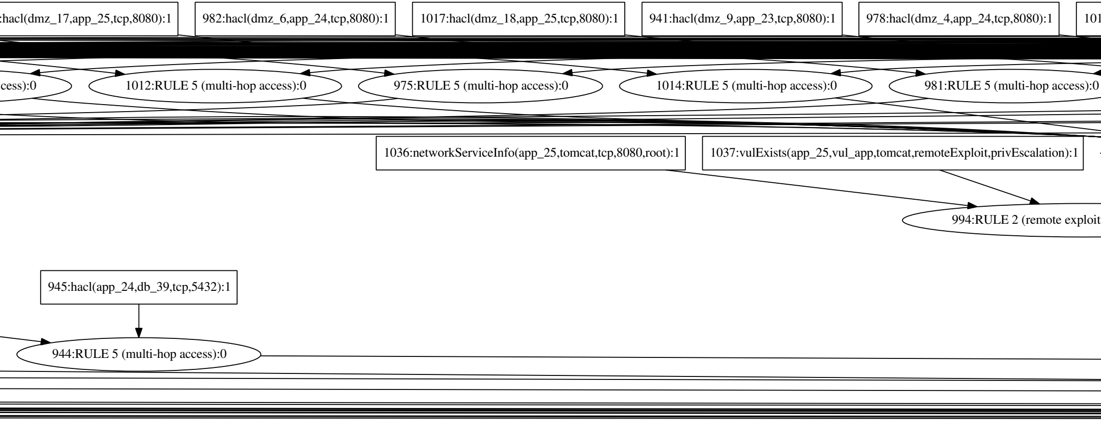

966:hacl(dmz_16,app_24,tcp,8080):1
929:hacl(dmz_3,app_23,tcp,8080):1
968:hacl(dmz_17,app_24,tcp,8080):1
935:hacl(dmz_6,app_23,tcp,8080):1
965:RULE 5 (multi-hop access):0
928:RULE 5 (multi-hop access):0
967:RULE 5 (multi-hop access):0
934:RULE 5 (multi-hop access):0

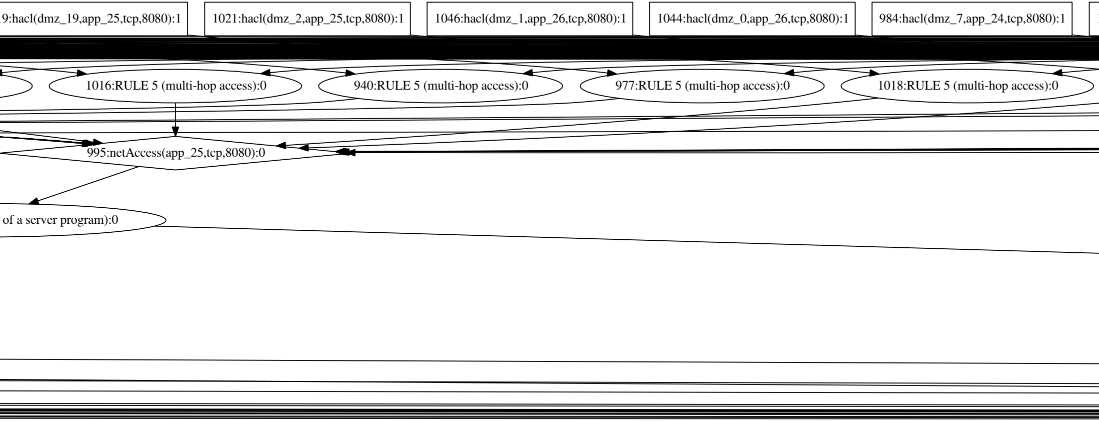

970:hacl(dmz_18,app_24,tcp,8080):1
894:hacl(dmz_9,app_22,tcp,8080):1
972:hacl(dmz_19,app_24,tcp,8080):1
931:hacl(dmz_4,app_23,tcp,8080):1
969:RULE 5 (multi-hop access):0
893:RULE 5 (multi-hop access):0
971:RULE 5 (multi-hop access):0
930:RULE 5 (multi-hop access)
948:netAccess(app_24,tcp,8080):0


937:hacl(dmz_7,app_23,tcp,8080):1
974:hacl(dmz_2,app_24,tcp,8080):1
999:hacl(dmz_1,app_25,tcp,8080):1
997:hacl(dmz_0,app_25,tcp,8080):1
1001:hacl(dmz_10,app_25,tcp,8080):1
:0
936:RULE 5 (multi-hop access):0
973:RULE 5 (multi-hop access):0
998:RULE 5 (multi-hop access):0
996:RULE 5 (multi-hop access):0

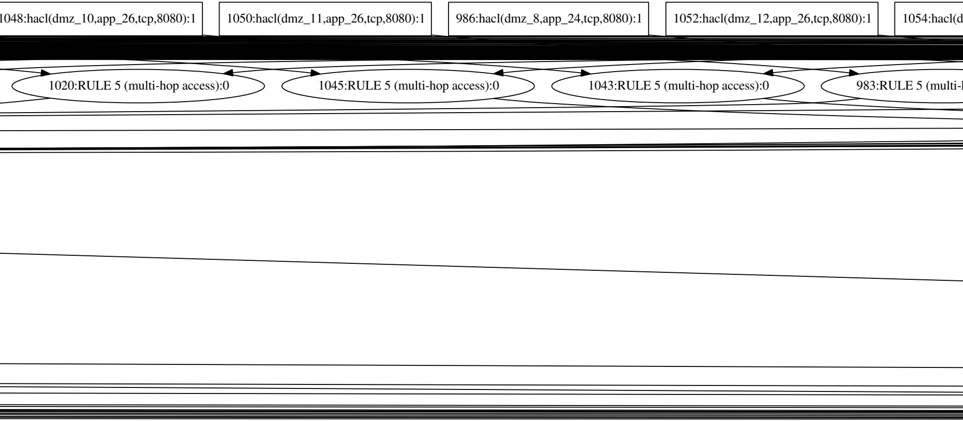

939:hacl(dmz_8,app_23,tcp,8080):1
1003:hacl(dmz_11,app_25,tcp,8080):1
1005:hacl(dmz_12,app_25,tcp,8080):1
1007:hacl(dmz_13,app_25,tcp,8080):1
980:hacl(dmz_5,app_24,t
1000:RULE 5 (multi-hop access):0
938:RULE 5 (multi-hop access):0
1002:RULE 5 (multi-hop access):0
1004:RULE 5 (multi-hop access):0
898:hacl(app_23,db_39,tcp,5432):1
897:RULE 5 (multi-hop access):0

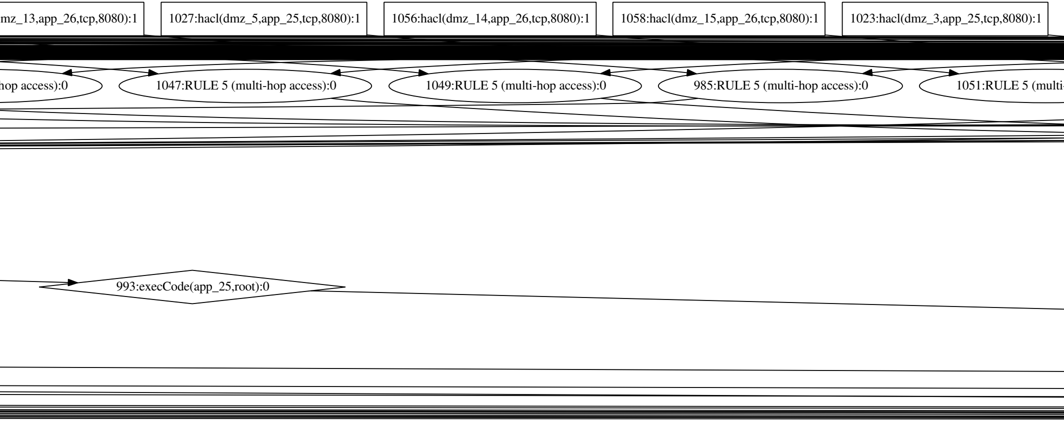

tcp,8080):1
1009:hacl(dmz_14,app_25,tcp,8080):1
1011:hacl(dmz_15,app_25,tcp,8080):1
1013:hacl(dmz_16,app_25,tcp,8080):1
976:hacl(dmz_3,app_24,tcp,8080):1
1015:
1006:RULE 5 (multi-hop access):0
979:RULE 5 (multi-hop access):0
1008:RULE 5 (multi-hop access):0
1010:RULE 5 (multi-hop ac
989:networkServiceInfo(app_24,tomcat,tcp,8080,root):1
990:vulExists(app_24,vul_app,tomcat,remoteExploit,privEscalation):1
947:RULE 2 (remote exploit of a server program):0
946:execCode(app_24,root):0

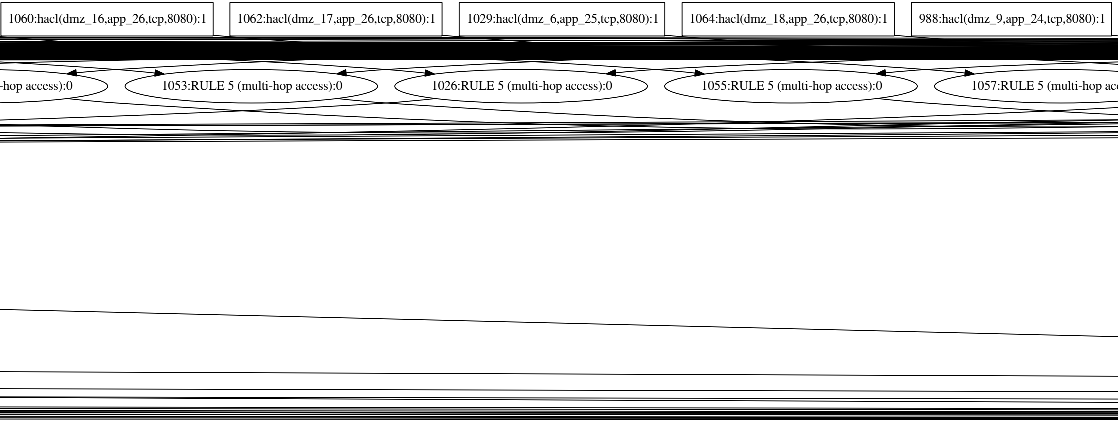

:hacl(dmz_17,app_25,tcp,8080):1
982:hacl(dmz_6,app_24,tcp,8080):1
1017:hacl(dmz_18,app_25,tcp,8080):1
941:hacl(dmz_9,app_23,tcp,8080):1
978:hacl(dmz_4,app_24,tcp,8080):1
101
cess):0
1012:RULE 5 (multi-hop access):0
975:RULE 5 (multi-hop access):0
1014:RULE 5 (multi-hop access):0
981:RULE 5 (multi-hop access):0
1036:networkServiceInfo(app_25,tomcat,tcp,8080,root):1
1037:vulExists(app_25,vul_app,tomcat,remoteExploit,privEscalation):1
994:RULE 2 (remote exploit
945:hacl(app_24,db_39,tcp,5432):1
944:RULE 5 (multi-hop access):0

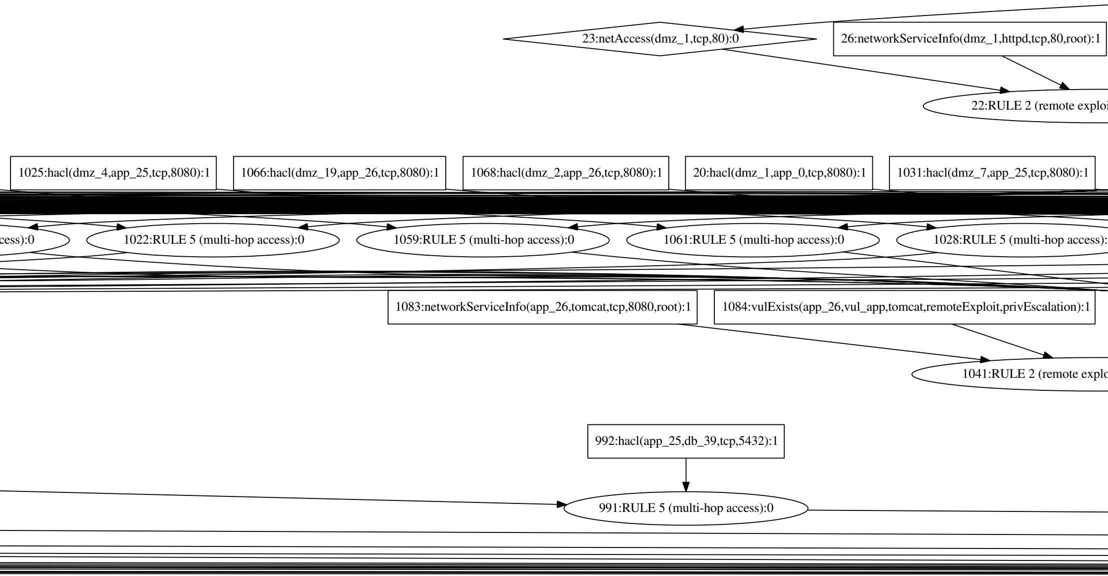


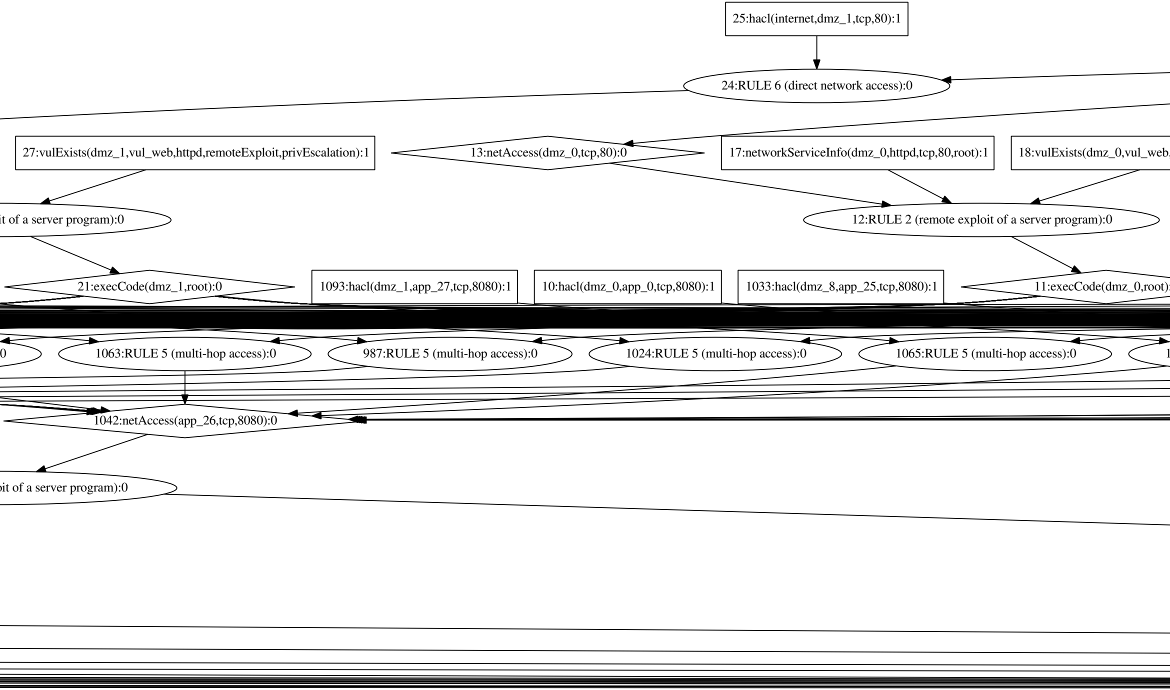
9:hacl(dmz_19,app_25,tcp,8080):1
1021:hacl(dmz_2,app_25,tcp,8080):1
1046:hacl(dmz_1,app_26,tcp,8080):1
1044:hacl(dmz_0,app_26,tcp,8080):1
984:hacl(dmz_7,app_24,tcp,8080):1
1016:RULE 5 (multi-hop access):0
940:RULE 5 (multi-hop access):0
977:RULE 5 (multi-hop access):0
1018:RULE 5 (multi-hop access):0
995:netAccess(app_25,tcp,8080):0
of a server program):0

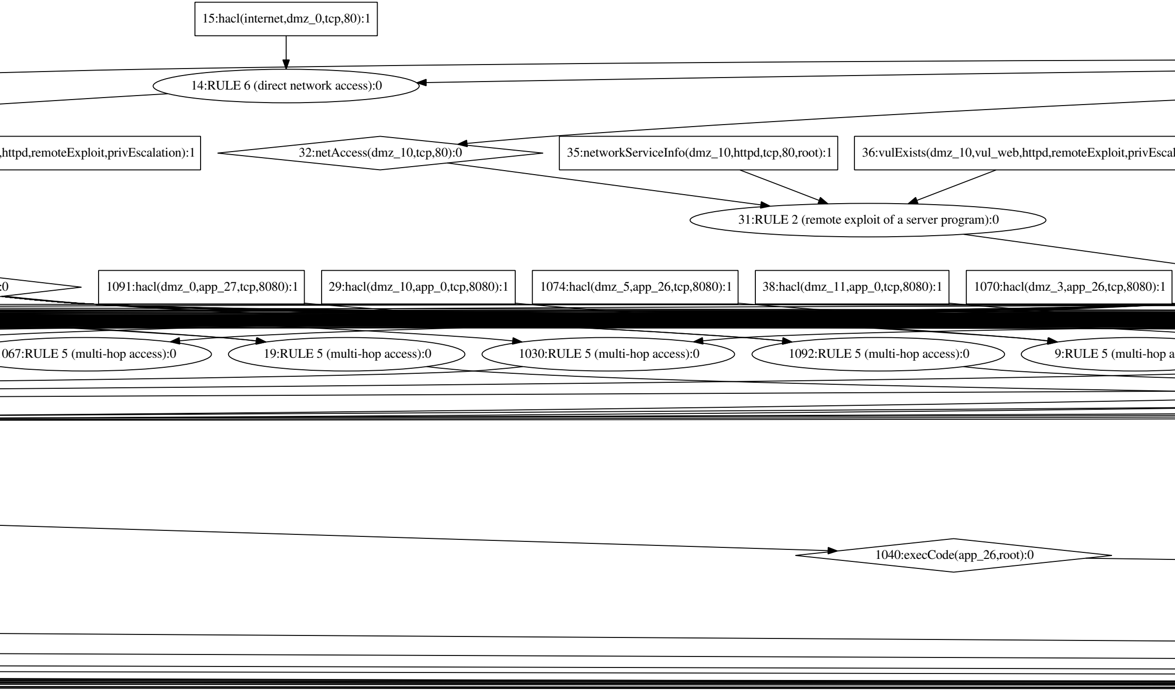

1048:hacl(dmz_10,app_26,tcp,8080):1
1050:hacl(dmz_11,app_26,tcp,8080):1
986:hacl(dmz_8,app_24,tcp,8080):1
1052:hacl(dmz_12,app_26,tcp,8080):1
1054:hacl(d
1020:RULE 5 (multi-hop access):0
1045:RULE 5 (multi-hop access):0
1043:RULE 5 (multi-hop access):0
983:RULE 5 (multi-

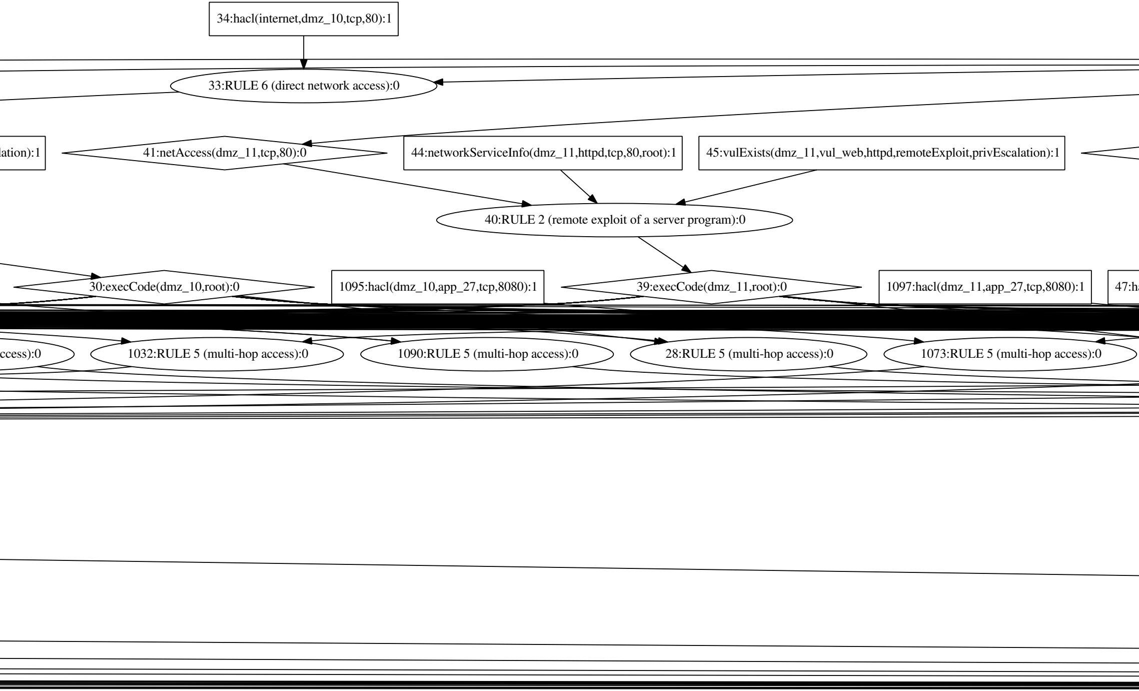

mz_13,app_26,tcp,8080):1
1027:hacl(dmz_5,app_25,tcp,8080):1
1056:hacl(dmz_14,app_26,tcp,8080):1
1058:hacl(dmz_15,app_26,tcp,8080):1
1023:hacl(dmz_3,app_25,tcp,8080):1
hop access):0
1047:RULE 5 (multi-hop access):0
1049:RULE 5 (multi-hop access):0
985:RULE 5 (multi-hop access):0
1051:RULE 5 (multi-
993:execCode(app_25,root):0

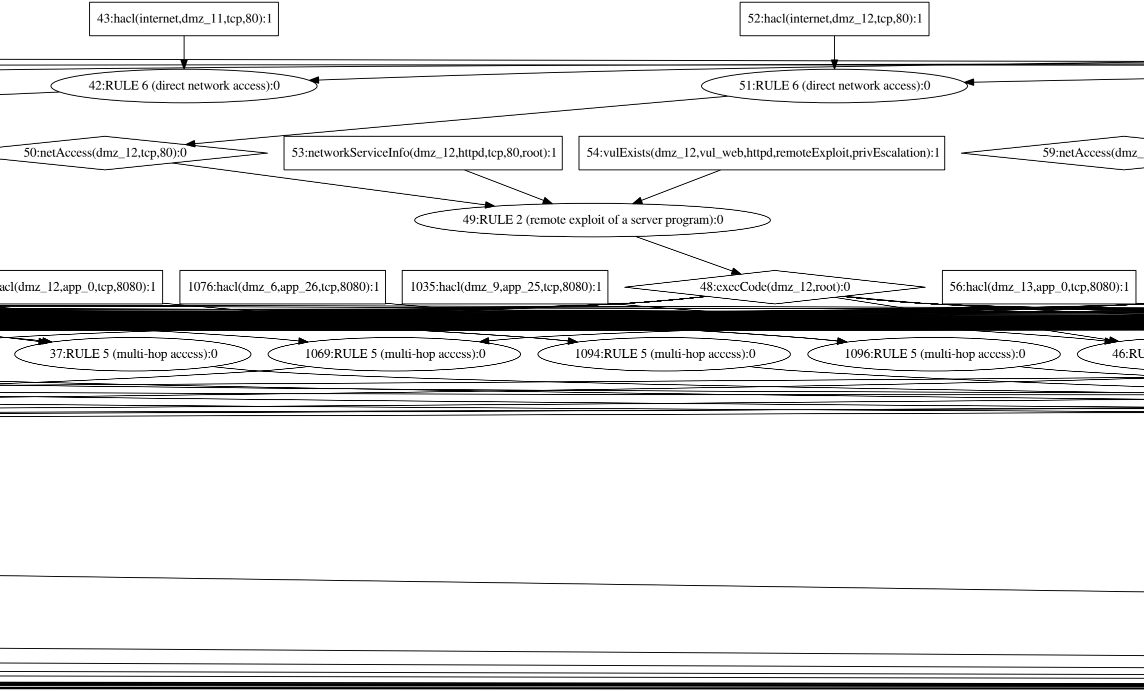

1060:hacl(dmz_16,app_26,tcp,8080):1
1062:hacl(dmz_17,app_26,tcp,8080):1
1029:hacl(dmz_6,app_25,tcp,8080):1
1064:hacl(dmz_18,app_26,tcp,8080):1
988:hacl(dmz_9,app_24,tcp,8080):1
-hop access):0
1053:RULE 5 (multi-hop access):0
1026:RULE 5 (multi-hop access):0
1055:RULE 5 (multi-hop access):0
1057:RULE 5 (multi-hop acc

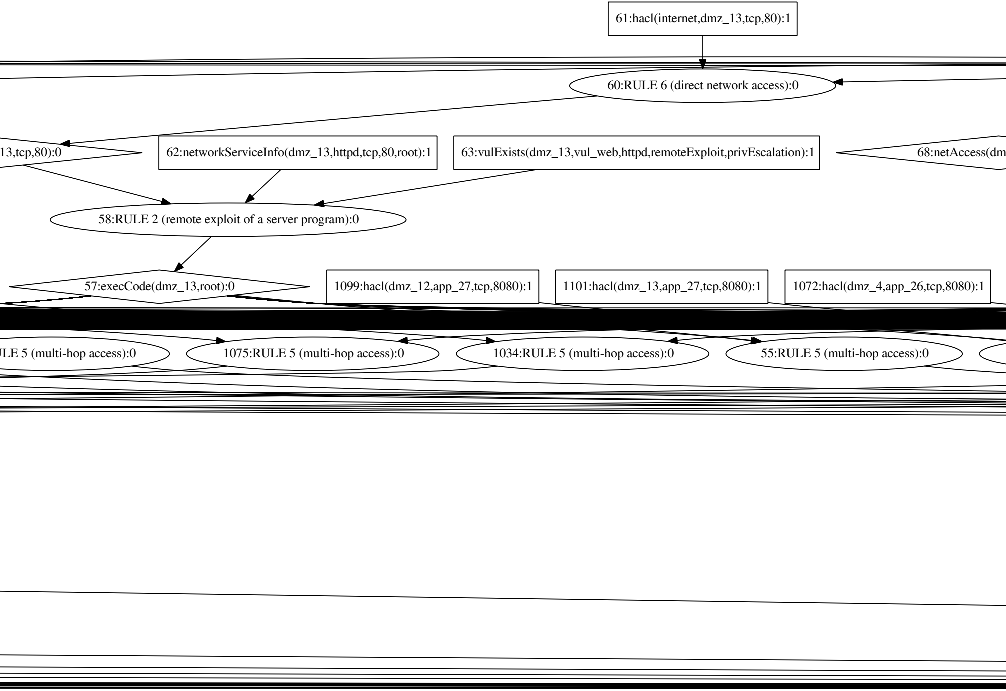

23:netAccess(dmz_1,tcp,80):0
26:networkServiceInfo(dmz_1,httpd,tcp,80,root):1
22:RULE 2 (remote exploi
1025:hacl(dmz_4,app_25,tcp,8080):1
1066:hacl(dmz_19,app_26,tcp,8080):1
1068:hacl(dmz_2,app_26,tcp,8080):1
20:hacl(dmz_1,app_0,tcp,8080):1
1031:hacl(dmz_7,app_25,tcp,8080):1
cess):0
1022:RULE 5 (multi-hop access):0
1059:RULE 5 (multi-hop access):0
1061:RULE 5 (multi-hop access):0
1028:RULE 5 (multi-hop access):
1083:networkServiceInfo(app_26,tomcat,tcp,8080,root):1
1084:vulExists(app_26,vul_app,tomcat,remoteExploit,privEscalation):1
1041:RULE 2 (remote explo
992:hacl(app_25,db_39,tcp,5432):1
991:RULE 5 (multi-hop access):0

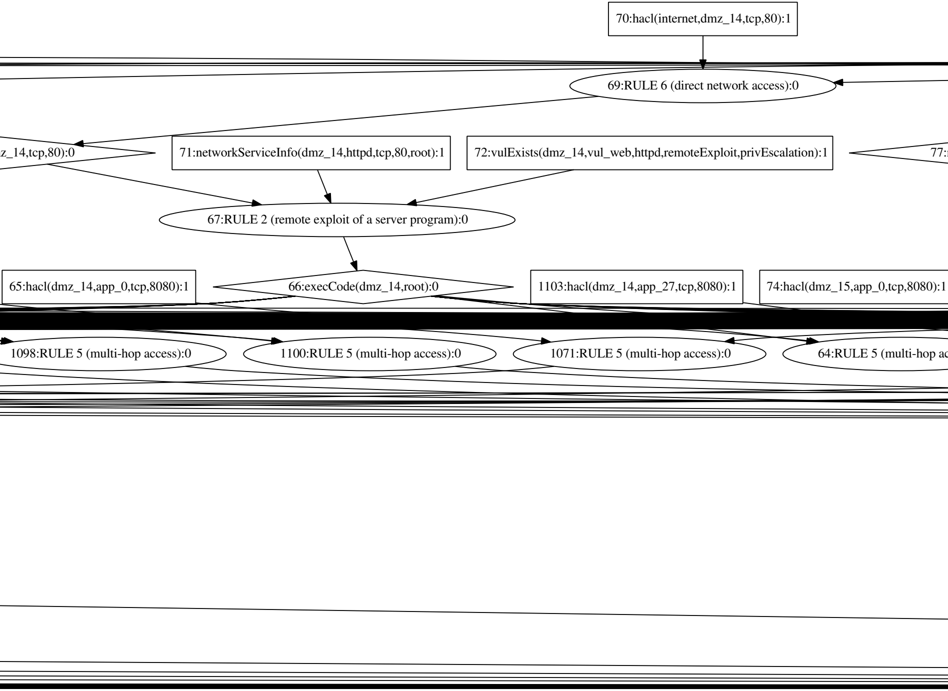

25:hacl(internet,dmz_1,tcp,80):1
24:RULE 6 (direct network access):0
27:vulExists(dmz_1,vul_web,httpd,remoteExploit,privEscalation):1
13:netAccess(dmz_0,tcp,80):0
17:networkServiceInfo(dmz_0,httpd,tcp,80,root):1
18:vulExists(dmz_0,vul_web,
it of a server program):0
12:RULE 2 (remote exploit of a server program):0
21:execCode(dmz_1,root):0
1093:hacl(dmz_1,app_27,tcp,8080):1
10:hacl(dmz_0,app_0,tcp,8080):1
1033:hacl(dmz_8,app_25,tcp,8080):1
11:execCode(dmz_0,root):
0
1063:RULE 5 (multi-hop access):0
987:RULE 5 (multi-hop access):0
1024:RULE 5 (multi-hop access):0
1065:RULE 5 (multi-hop access):0
1
1042:netAccess(app_26,tcp,8080):0
oit of a server program):0

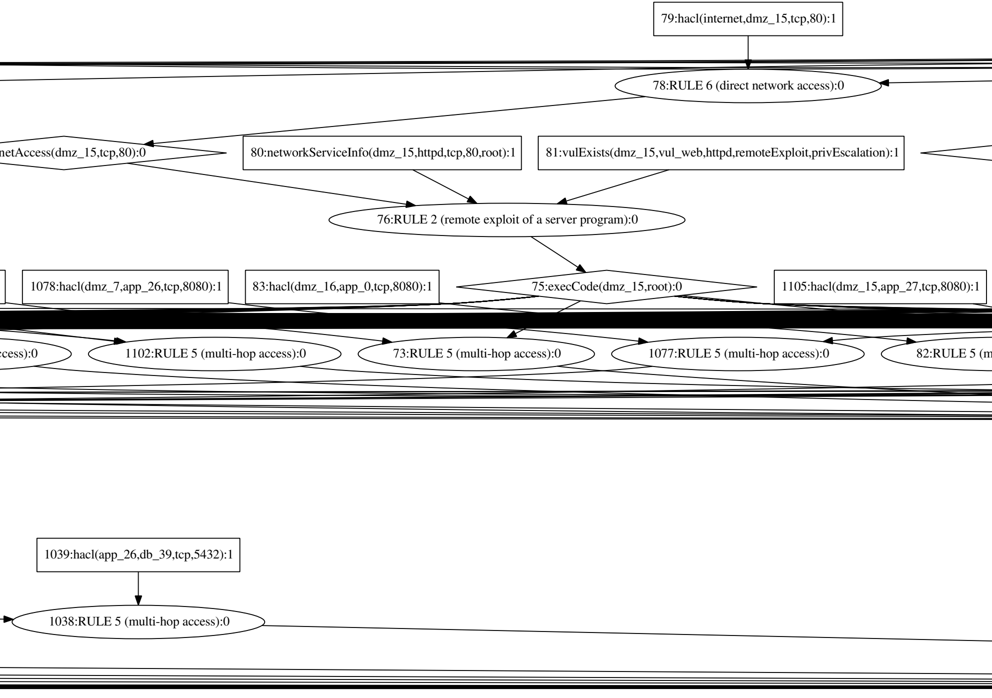

15:hacl(internet,dmz_0,tcp,80):1
14:RULE 6 (direct network access):0
,httpd,remoteExploit,privEscalation):1
32:netAccess(dmz_10,tcp,80):0
35:networkServiceInfo(dmz_10,httpd,tcp,80,root):1
36:vulExists(dmz_10,vul_web,httpd,remoteExploit,privEscal
31:RULE 2 (remote exploit of a server program):0
:0
1091:hacl(dmz_0,app_27,tcp,8080):1
29:hacl(dmz_10,app_0,tcp,8080):1
1074:hacl(dmz_5,app_26,tcp,8080):1
38:hacl(dmz_11,app_0,tcp,8080):1
1070:hacl(dmz_3,app_26,tcp,8080):1
067:RULE 5 (multi-hop access):0
19:RULE 5 (multi-hop access):0
1030:RULE 5 (multi-hop access):0
1092:RULE 5 (multi-hop access):0
9:RULE 5 (multi-hop a
1040:execCode(app_26,root):0

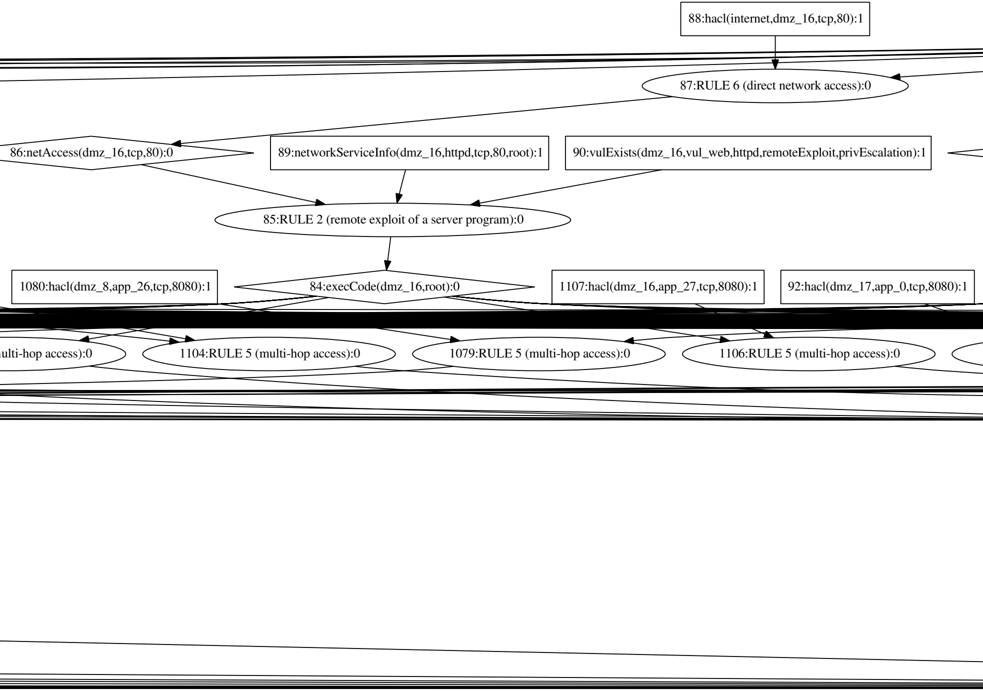

34:hacl(internet,dmz_10,tcp,80):1
33:RULE 6 (direct network access):0
lation):1
41:netAccess(dmz_11,tcp,80):0
44:networkServiceInfo(dmz_11,httpd,tcp,80,root):1
45:vulExists(dmz_11,vul_web,httpd,remoteExploit,privEscalation):1
40:RULE 2 (remote exploit of a server program):0
30:execCode(dmz_10,root):0
1095:hacl(dmz_10,app_27,tcp,8080):1
39:execCode(dmz_11,root):0
1097:hacl(dmz_11,app_27,tcp,8080):1
47:ha
ccess):0
1032:RULE 5 (multi-hop access):0
1090:RULE 5 (multi-hop access):0
28:RULE 5 (multi-hop access):0
1073:RULE 5 (multi-hop access):0

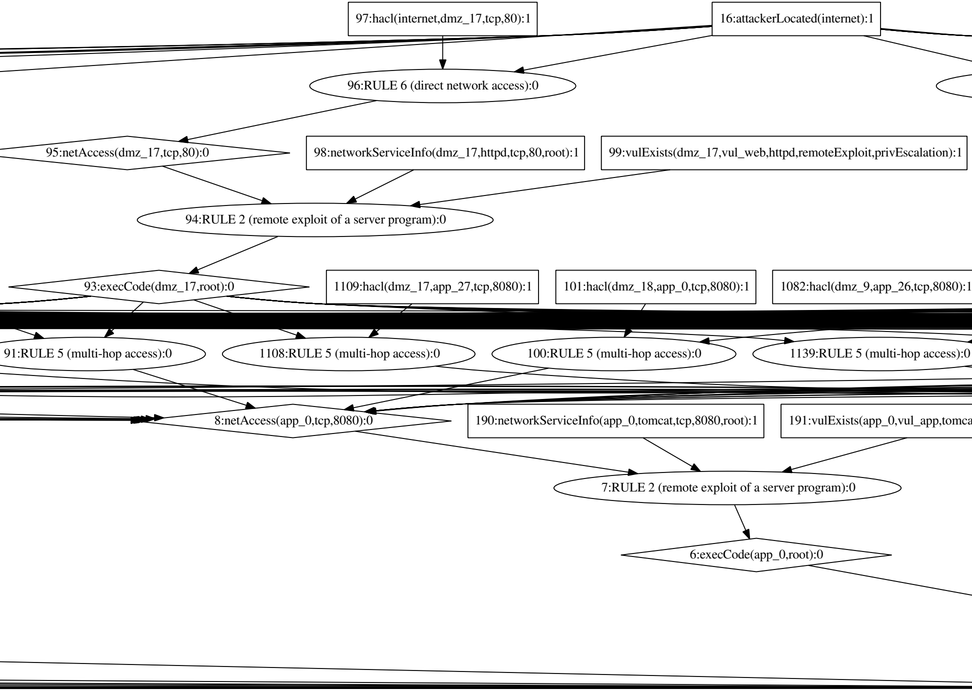

43:hacl(internet,dmz_11,tcp,80):1
52:hacl(internet,dmz_12,tcp,80):1
42:RULE 6 (direct network access):0
51:RULE 6 (direct network access):0
50:netAccess(dmz_12,tcp,80):0
53:networkServiceInfo(dmz_12,httpd,tcp,80,root):1
54:vulExists(dmz_12,vul_web,httpd,remoteExploit,privEscalation):1
59:netAccess(dmz_
49:RULE 2 (remote exploit of a server program):0
acl(dmz_12,app_0,tcp,8080):1
1076:hacl(dmz_6,app_26,tcp,8080):1
1035:hacl(dmz_9,app_25,tcp,8080):1
48:execCode(dmz_12,root):0
56:hacl(dmz_13,app_0,tcp,8080):1
37:RULE 5 (multi-hop access):0
1069:RULE 5 (multi-hop access):0
1094:RULE 5 (multi-hop access):0
1096:RULE 5 (multi-hop access):0
46:RU

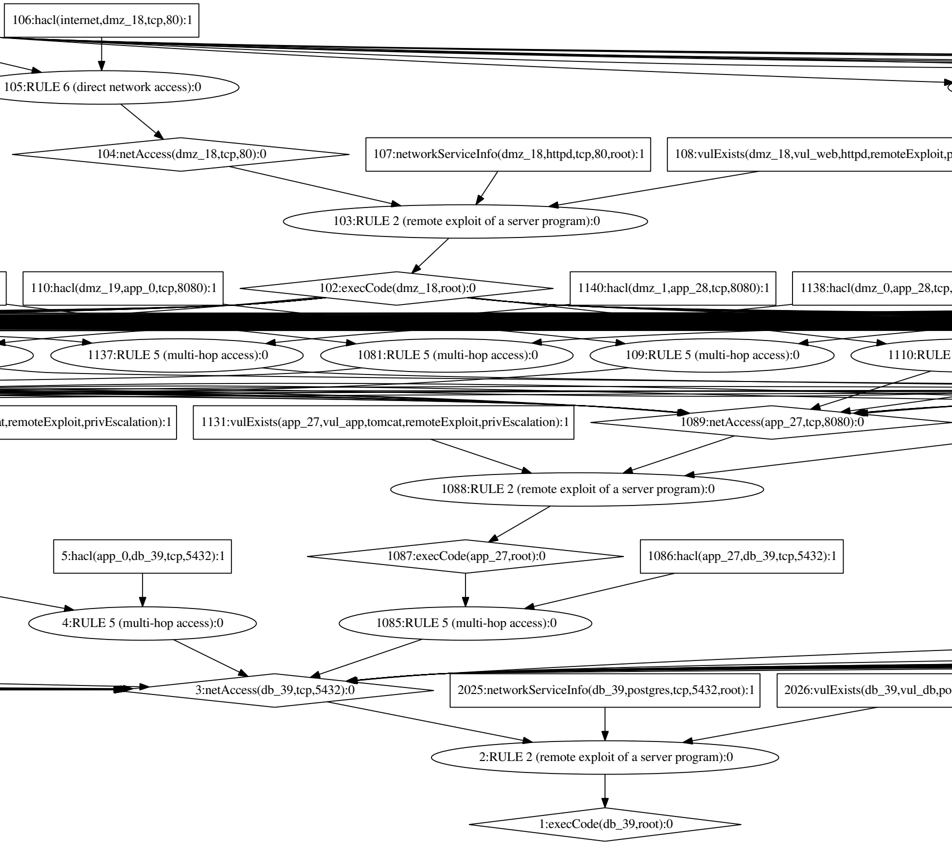

61:hacl(internet,dmz_13,tcp,80):1
60:RULE 6 (direct network access):0
13,tcp,80):0
62:networkServiceInfo(dmz_13,httpd,tcp,80,root):1
63:vulExists(dmz_13,vul_web,httpd,remoteExploit,privEscalation):1
68:netAccess(dm
58:RULE 2 (remote exploit of a server program):0
57:execCode(dmz_13,root):0
1099:hacl(dmz_12,app_27,tcp,8080):1
1101:hacl(dmz_13,app_27,tcp,8080):1
1072:hacl(dmz_4,app_26,tcp,8080):1
JLE 5 (multi-hop access):0
1075:RULE 5 (multi-hop access):0
1034:RULE 5 (multi-hop access):0
55:RULE 5 (multi-hop access):0

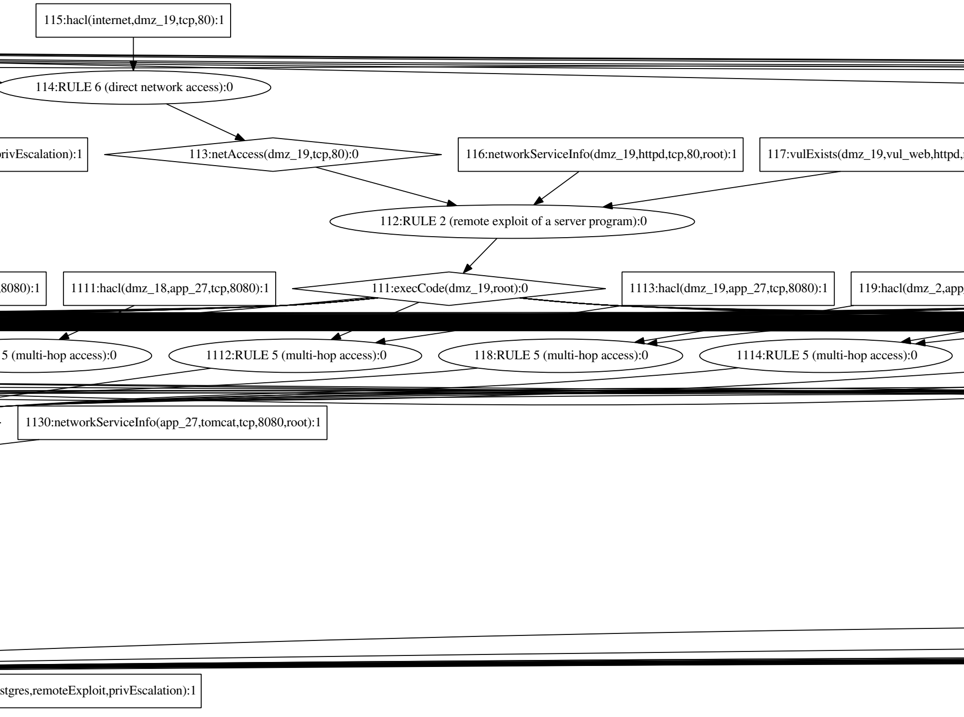

70:hacl(internet,dmz_14,tcp,80):1
69:RULE 6 (direct network access):0
z_14,tcp,80):0
71:networkServiceInfo(dmz_14,httpd,tcp,80,root):1
72:vulExists(dmz_14,vul_web,httpd,remoteExploit,privEscalation):1
77:
67:RULE 2 (remote exploit of a server program):0
65:hacl(dmz_14,app_0,tcp,8080):1
66:execCode(dmz_14,root):0
1103:hacl(dmz_14,app_27,tcp,8080):1
74:hacl(dmz_15,app_0,tcp,8080):1
1098:RULE 5 (multi-hop access):0
1100:RULE 5 (multi-hop access):0
1071:RULE 5 (multi-hop access):0
64:RULE 5 (multi-hop ac

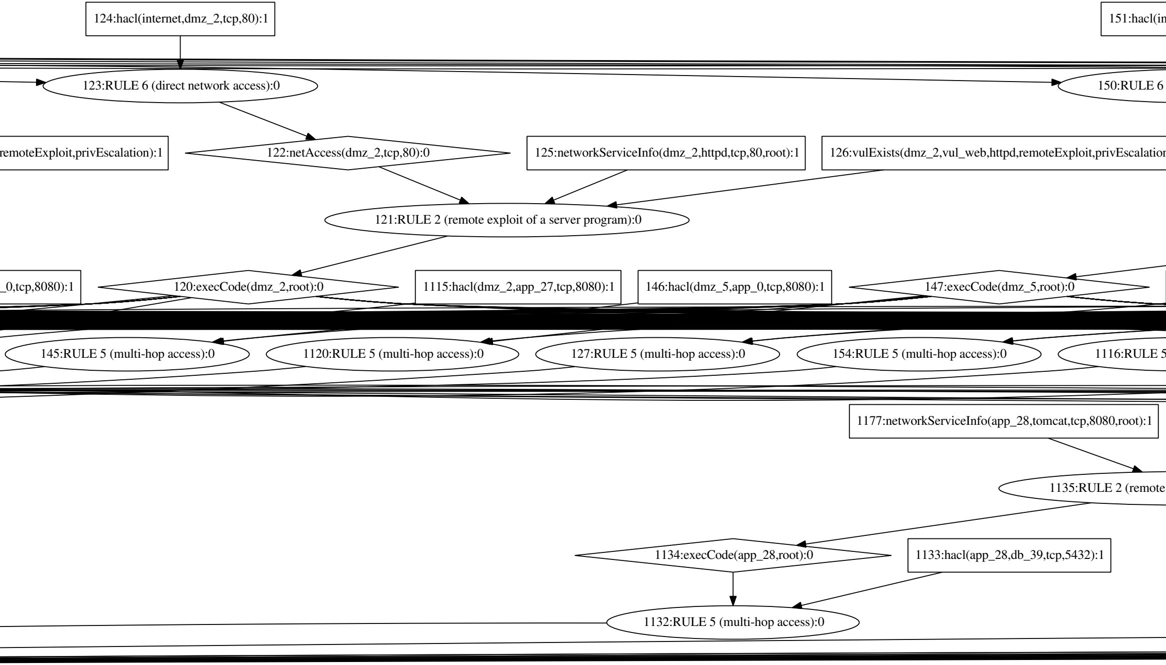

79:hacl(internet,dmz_15,tcp,80):1
78:RULE 6 (direct network access):0
netAccess(dmz_15,tcp,80):0
80:networkServiceInfo(dmz_15,httpd,tcp,80,root):1
81:vulExists(dmz_15,vul_web,httpd,remoteExploit,privEscalation):1
76:RULE 2 (remote exploit of a server program):0
1078:hacl(dmz_7,app_26,tcp,8080):1
83:hacl(dmz_16,app_0,tcp,8080):1
75:execCode(dmz_15,root):0
1105:hacl(dmz_15,app_27,tcp,8080):1
ccess):0
1102:RULE 5 (multi-hop access):0
73:RULE 5 (multi-hop access):0
1077:RULE 5 (multi-hop access):0
82:RULE 5 (m
1039:hacl(app_26,db_39,tcp,5432):1
1038:RULE 5 (multi-hop access):0

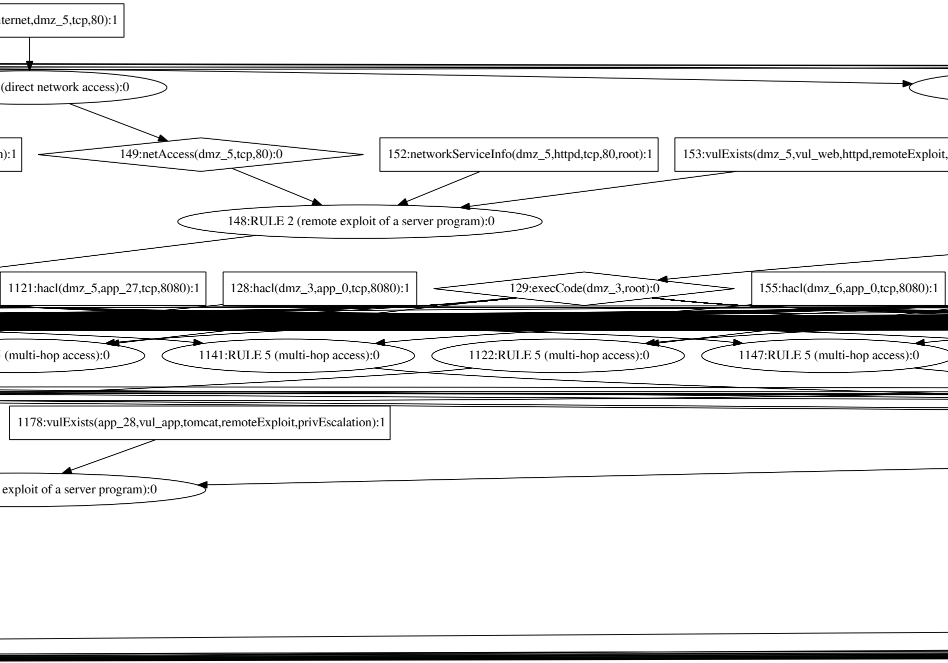

88:hacl(internet,dmz_16,tcp,80):1
87:RULE 6 (direct network access):0
86:netAccess(dmz_16,tcp,80):0
89:networkServiceInfo(dmz_16,httpd,tcp,80,root):1
90:vulExists(dmz_16,vul_web,httpd,remoteExploit,privEscalation):1
85:RULE 2 (remote exploit of a server program):0
1080:hacl(dmz_8,app_26,tcp,8080):1
84:execCode(dmz_16,root):0
1107:hacl(dmz_16,app_27,tcp,8080):1
92:hacl(dmz_17,app_0,tcp,8080):1
ulti-hop access):0
1104:RULE 5 (multi-hop access):0
1079:RULE 5 (multi-hop access):0
1106:RULE 5 (multi-hop access):0

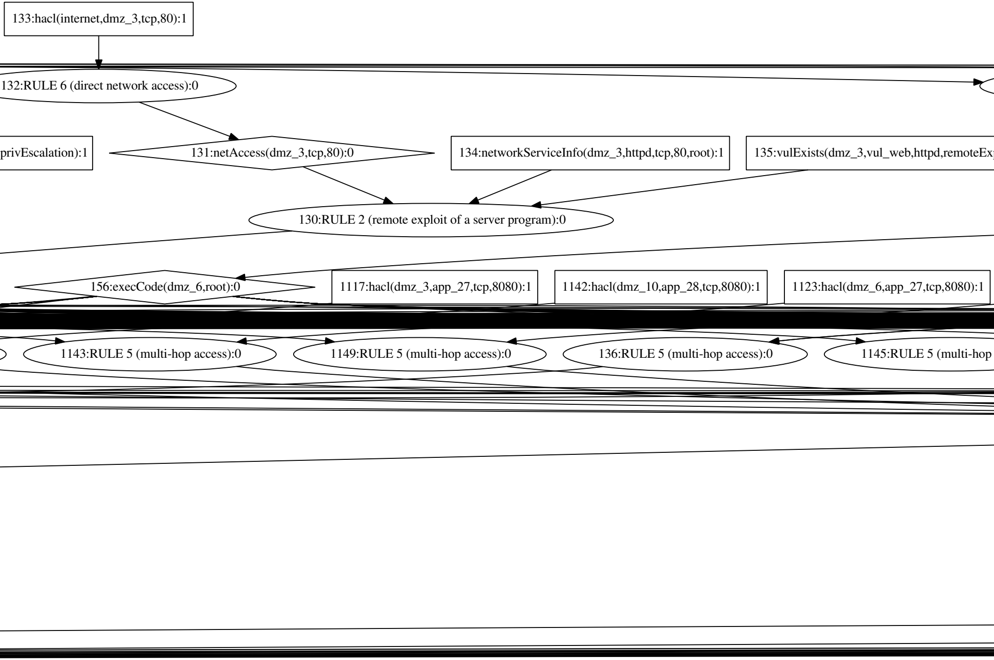

97:hacl(internet,dmz_17,tcp,80):1
16:attackerLocated(internet):1
96:RULE 6 (direct network access):0
95:netAccess(dmz_17,tcp,80):0
98:networkServiceInfo(dmz_17,httpd,tcp,80,root):1
99:vulExists(dmz_17,vul_web,httpd,remoteExploit,privEscalation):1
94:RULE 2 (remote exploit of a server program):0
93:execCode(dmz_17,root):0
1109:hacl(dmz_17,app_27,tcp,8080):1
101:hacl(dmz_18,app_0,tcp,8080):1
1082:hacl(dmz_9,app_26,tcp,8080):1
91:RULE 5 (multi-hop access):0
1108:RULE 5 (multi-hop access):0
100:RULE 5 (multi-hop access):0
1139:RULE 5 (multi-hop access):0
8:netAccess(app_0,tcp,8080):0
190:networkServiceInfo(app_0,tomcat,tcp,8080,root):1
191:vulExists(app_0,vul_app,tomca
7:RULE 2 (remote exploit of a server program):0
6:execCode(app_0,root):0

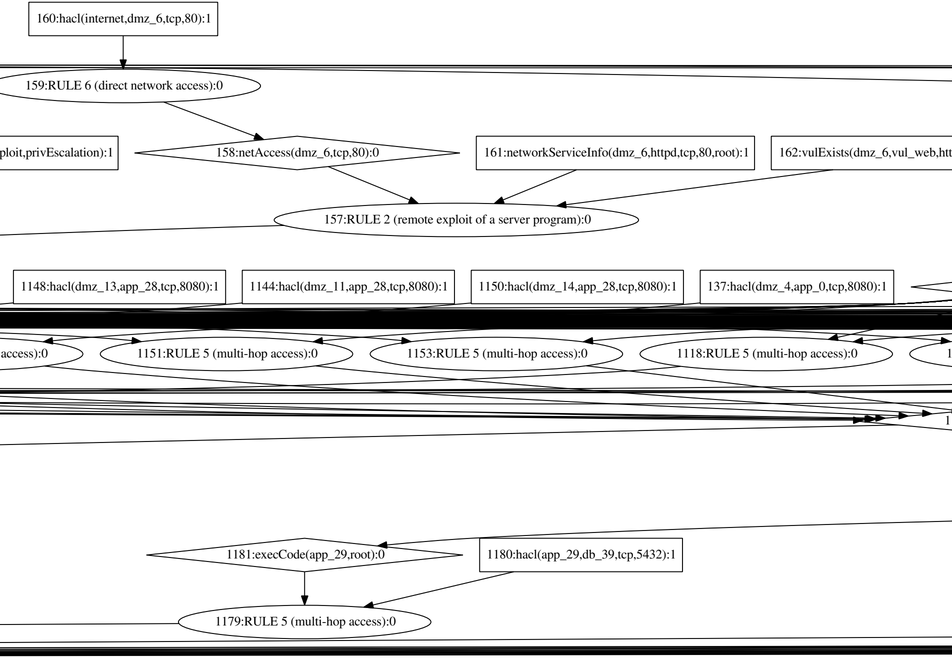

106:hacl(internet,dmz_18,tcp,80):1
105:RULE 6 (direct network access):0
104:netAccess(dmz_18,tcp,80):0
107:networkServiceInfo(dmz_18,httpd,tcp,80,root):1
108:vulExists(dmz_18,vul_web,httpd,remoteExploit,p
103:RULE 2 (remote exploit of a server program):0
110:hacl(dmz_19,app_0,tcp,8080):1
102:execCode(dmz_18,root):0
1140:hacl(dmz_1,app_28,tcp,8080):1
1138:hacl(dmz_0,app_28,tcp,
1137:RULE 5 (multi-hop access):0
1081:RULE 5 (multi-hop access):0
109:RULE 5 (multi-hop access):0
1110:RULE
t,remoteExploit,privEscalation):1
1131:vulExists(app_27,vul_app,tomcat,remoteExploit,privEscalation):1
1089:netAccess(app_27,tcp,8080):0
1088:RULE 2 (remote exploit of a server program):0
5:hacl(app_0,db_39,tcp,5432):1
1087:execCode(app_27,root):0
1086:hacl(app_27,db_39,tcp,5432):1
4:RULE 5 (multi-hop access):0
1085:RULE 5 (multi-hop access):0
3:netAccess(db_39,tcp,5432):0
2025:networkServiceInfo(db_39,postgres,tcp,5432,root):1
2026:vulExists(db_39,vul_db,po
2:RULE 2 (remote exploit of a server program):0
1:execCode(db_39,root):0

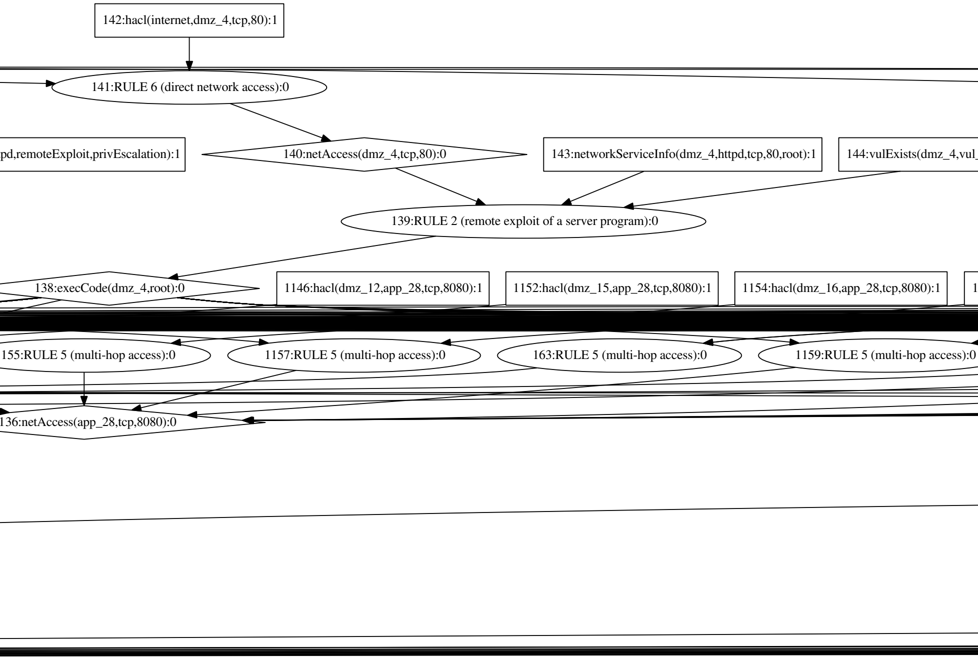

115:hacl(internet,dmz_19,tcp,80):1
114:RULE 6 (direct network access):0
rivEscalation):1
113:netAccess(dmz_19,tcp,80):0
116:networkServiceInfo(dmz_19,httpd,tcp,80,root):1
117:vulExists(dmz_19,vul_web,httpd,
112:RULE 2 (remote exploit of a server program):0
8080):1
1111:hacl(dmz_18,app_27,tcp,8080):1
111:execCode(dmz_19,root):0
1113:hacl(dmz_19,app_27,tcp,8080):1
119:hacl(dmz_2,app
5 (multi-hop access):0
1112:RULE 5 (multi-hop access):0
118:RULE 5 (multi-hop access):0
1114:RULE 5 (multi-hop access):0
1130:networkServiceInfo(app_27,tomcat,tcp,8080,root):1
stgres,remoteExploit,privEscalation):1

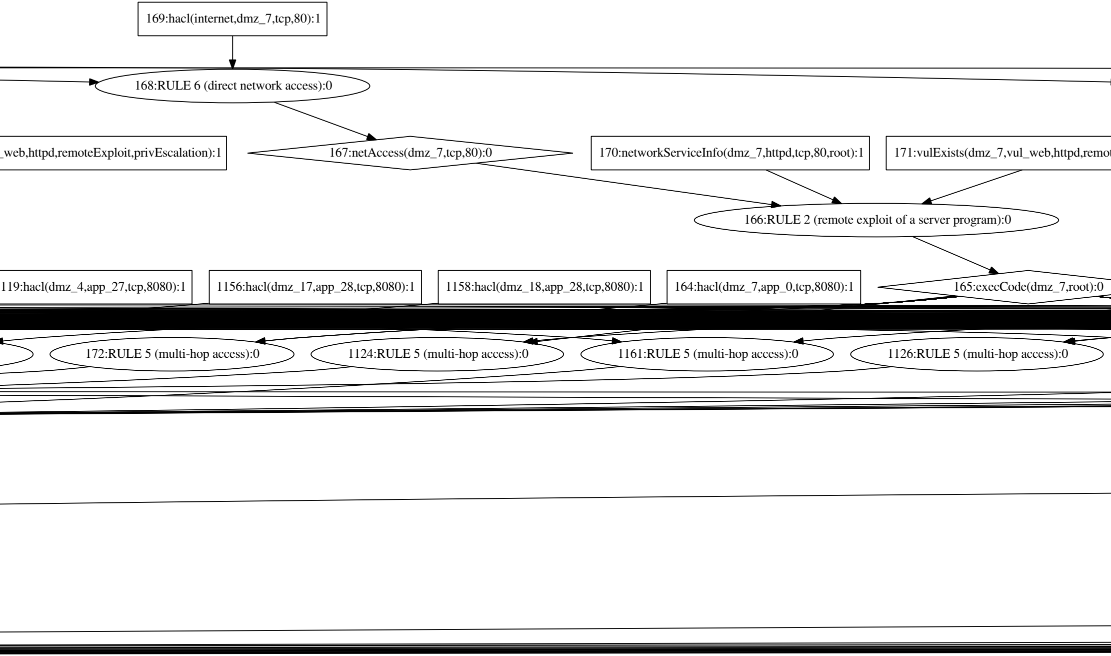

124:hacl(internet,dmz_2,tcp,80):1
151:hacl(in
123:RULE 6 (direct network access):0
150:RULE 6
remoteExploit,privEscalation):1
122:netAccess(dmz_2,tcp,80):0
125:networkServiceInfo(dmz_2,httpd,tcp,80,root):1
126:vulExists(dmz_2,vul_web,httpd,remoteExploit,privEscalation
121:RULE 2 (remote exploit of a server program):0
_0,tcp,8080):1
120:execCode(dmz_2,root):0
1115:hacl(dmz_2,app_27,tcp,8080):1
146:hacl(dmz_5,app_0,tcp,8080):1
147:execCode(dmz_5,root):0
145:RULE 5 (multi-hop access):0
1120:RULE 5 (multi-hop access):0
127:RULE 5 (multi-hop access):0
154:RULE 5 (multi-hop access):0
1116:RULE 5
1177:networkServiceInfo(app_28,tomcat,tcp,8080,root):1
1135:RULE 2 (remote
1134:execCode(app_28,root):0
1133:hacl(app_28,db_39,tcp,5432):1
1132:RULE 5 (multi-hop access):0

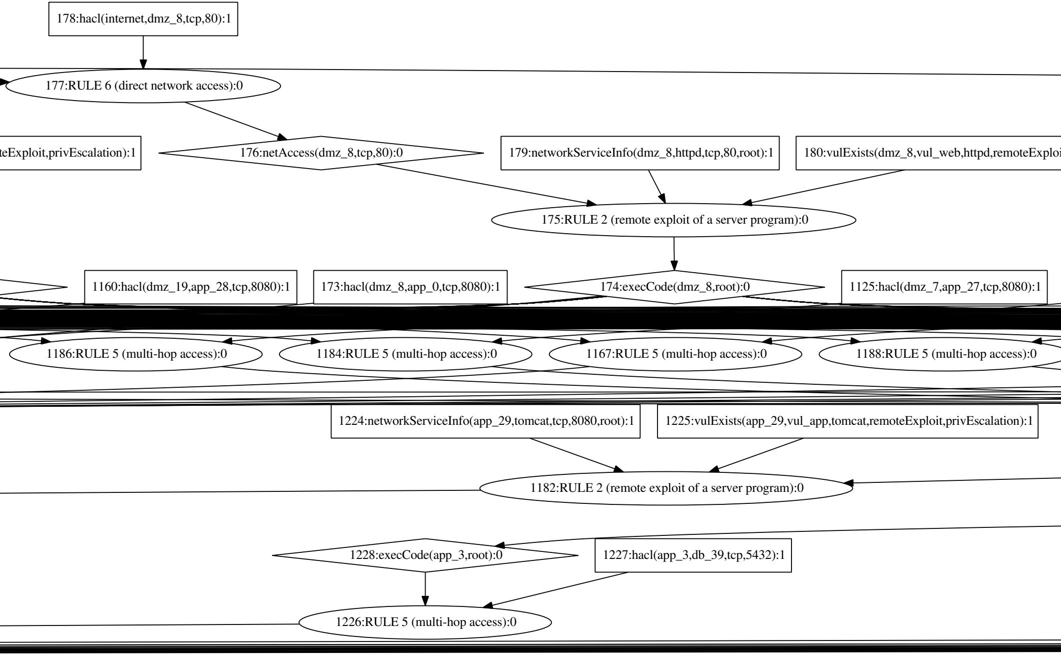

ternet,dmz_5,tcp,80):1
(direct network access):0
):1
149:netAccess(dmz_5,tcp,80):0
152:networkServiceInfo(dmz_5,httpd,tcp,80,root):1
153:vulExists(dmz_5,vul_web,httpd,remoteExploit,
148:RULE 2 (remote exploit of a server program):0
1121:hacl(dmz_5,app_27,tcp,8080):1
128:hacl(dmz_3,app_0,tcp,8080):1
129:execCode(dmz_3,root):0
155:hacl(dmz_6,app_0,tcp,8080):1
(multi-hop access):0
1141:RULE 5 (multi-hop access):0
1122:RULE 5 (multi-hop access):0
1147:RULE 5 (multi-hop access):0
1178:vulExists(app_28,vul_app,tomcat,remoteExploit,privEscalation):1
exploit of a server program):0

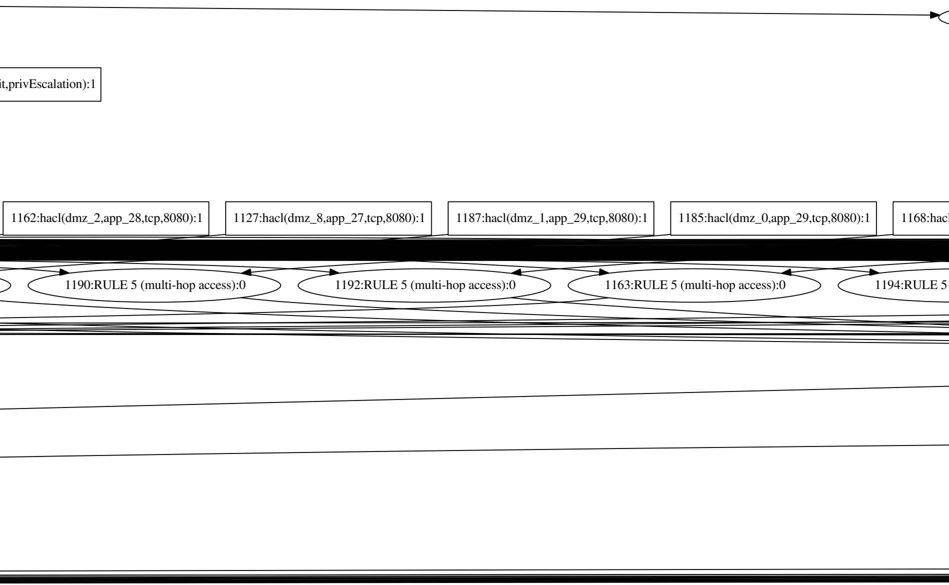

133:hacl(internet,dmz_3,tcp,80):1
132:RULE 6 (direct network access):0
privEscalation):1
131:netAccess(dmz_3,tcp,80):0
134:networkServiceInfo(dmz_3,httpd,tcp,80,root):1
135:vulExists(dmz_3,vul_web,httpd,remoteEx
130:RULE 2 (remote exploit of a server program):0
156:execCode(dmz_6,root):0
1117:hacl(dmz_3,app_27,tcp,8080):1
1142:hacl(dmz_10,app_28,tcp,8080):1
1123:hacl(dmz_6,app_27,tcp,8080):1
1143:RULE 5 (multi-hop access):0
1149:RULE 5 (multi-hop access):0
136:RULE 5 (multi-hop access):0
1145:RULE 5 (multi-hop

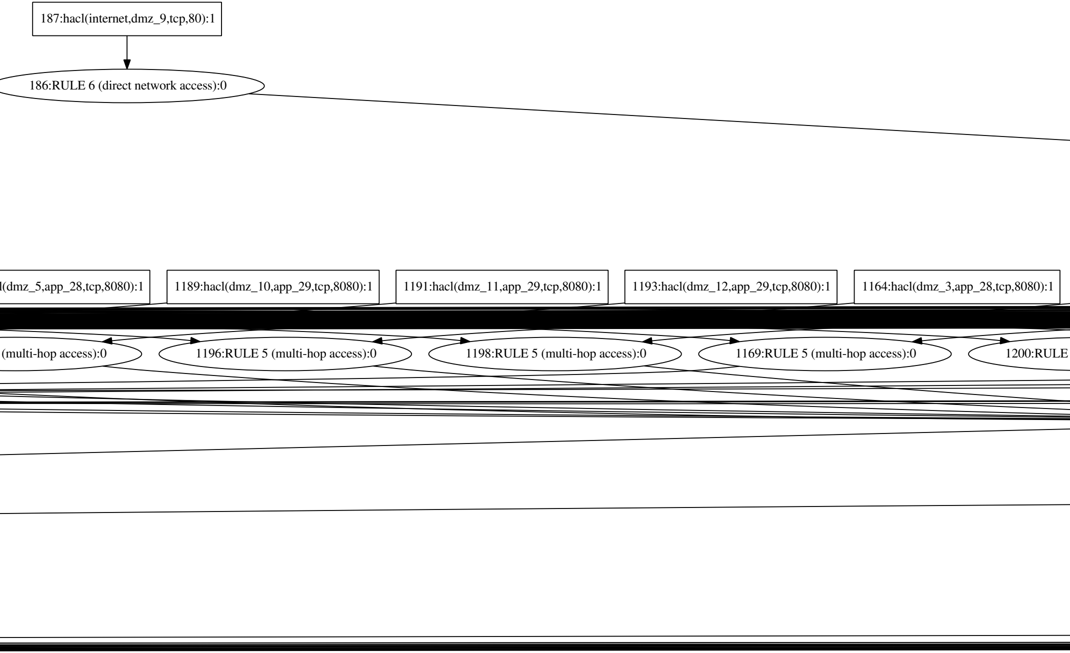

160:hacl(internet,dmz_6,tcp,80):1
159:RULE 6 (direct network access):0
ploit,privEscalation):1
158:netAccess(dmz_6,tcp,80):0
161:networkServiceInfo(dmz_6,httpd,tcp,80,root):1
162:vulExists(dmz_6,vul_web,htt
157:RULE 2 (remote exploit of a server program):0
1148:hacl(dmz_13,app_28,tcp,8080):1
1144:hacl(dmz_11,app_28,tcp,8080):1
1150:hacl(dmz_14,app_28,tcp,8080):1
137:hacl(dmz_4,app_0,tcp,8080):1
access):0
1151:RULE 5 (multi-hop access):0
1153:RULE 5 (multi-hop access):0
1118:RULE 5 (multi-hop access):0
1181:execCode(app_29,root):0
1180:hacl(app_29,db_39,tcp,5432):1
1179:RULE 5 (multi-hop access):0

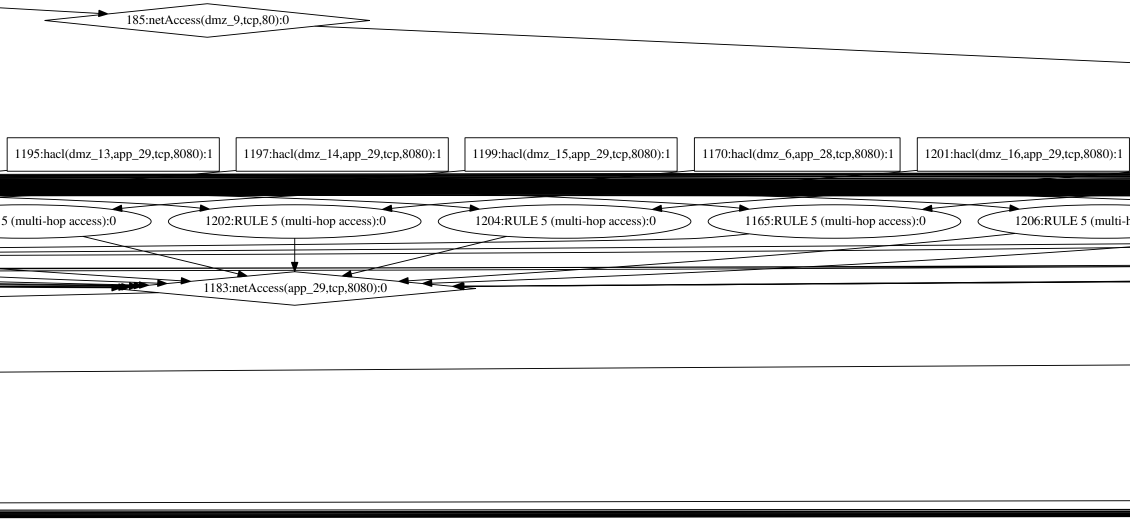

142:hacl(internet,dmz_4,tcp,80):1
141:RULE 6 (direct network access):0
pd,remoteExploit,privEscalation):1
140:netAccess(dmz_4,tcp,80):0
143:networkServiceInfo(dmz_4,httpd,tcp,80,root):1
144:vulExists(dmz_4,vul_
139:RULE 2 (remote exploit of a server program):0
138:execCode(dmz_4,root):0
1146:hacl(dmz_12,app_28,tcp,8080):1
1152:hacl(dmz_15,app_28,tcp,8080):1
1154:hacl(dmz_16,app_28,tcp,8080):1
1
155:RULE 5 (multi-hop access):0
1157:RULE 5 (multi-hop access):0
163:RULE 5 (multi-hop access):0
1159:RULE 5 (multi-hop access):0
136:netAccess(app_28,tcp,8080):0

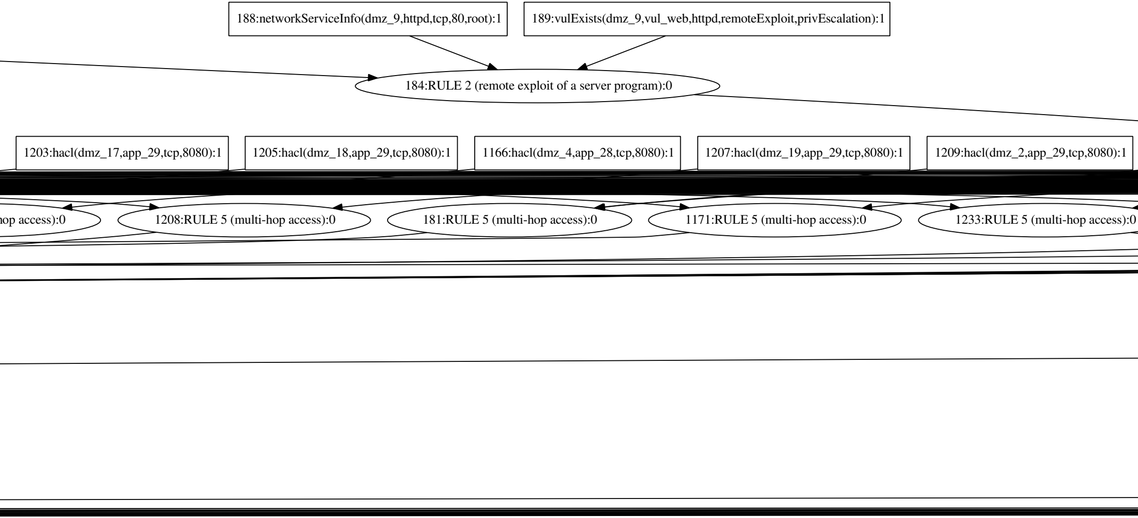

169:hacl(internet,dmz_7,tcp,80):1
168:RULE 6 (direct network access):0
_web,httpd,remoteExploit,privEscalation):1
167:netAccess(dmz_7,tcp,80):0
170:networkServiceInfo(dmz_7,httpd,tcp,80,root):1
171:vulExists(dmz_7,vul_web,httpd,remot
166:RULE 2 (remote exploit of a server program):0
119:hacl(dmz_4,app_27,tcp,8080):1
1156:hacl(dmz_17,app_28,tcp,8080):1
1158:hacl(dmz_18,app_28,tcp,8080):1
164:hacl(dmz_7,app_0,tcp,8080):1
165:execCode(dmz_7,root):0
172:RULE 5 (multi-hop access):0
1124:RULE 5 (multi-hop access):0
1161:RULE 5 (multi-hop access):0
1126:RULE 5 (multi-hop access):0

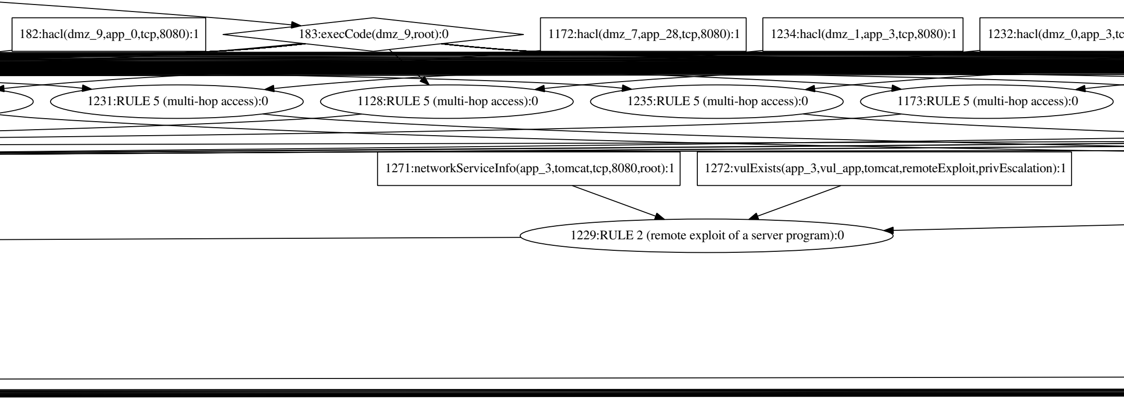

178:hacl(internet,dmz_8,tcp,80):1
177:RULE 6 (direct network access):0
teExploit,privEscalation):1
176:netAccess(dmz_8,tcp,80):0
179:networkServiceInfo(dmz_8,httpd,tcp,80,root):1
180:vulExists(dmz_8,vul_web,httpd,remoteExploi
175:RULE 2 (remote exploit of a server program):0
1160:hacl(dmz_19,app_28,tcp,8080):1
173:hacl(dmz_8,app_0,tcp,8080):1
174:execCode(dmz_8,root):0
1125:hacl(dmz_7,app_27,tcp,8080):1
1186:RULE 5 (multi-hop access):0
1184:RULE 5 (multi-hop access):0
1167:RULE 5 (multi-hop access):0
1188:RULE 5 (multi-hop access):0
1224:networkServiceInfo(app_29,tomcat,tcp,8080,root):1
1225:vulExists(app_29,vul_app,tomcat,remoteExploit,privEscalation):1
1182:RULE 2 (remote exploit of a server program):0
1228:execCode(app_3,root):0
1227:hacl(app_3,db_39,tcp,5432):1
1226:RULE 5 (multi-hop access):0

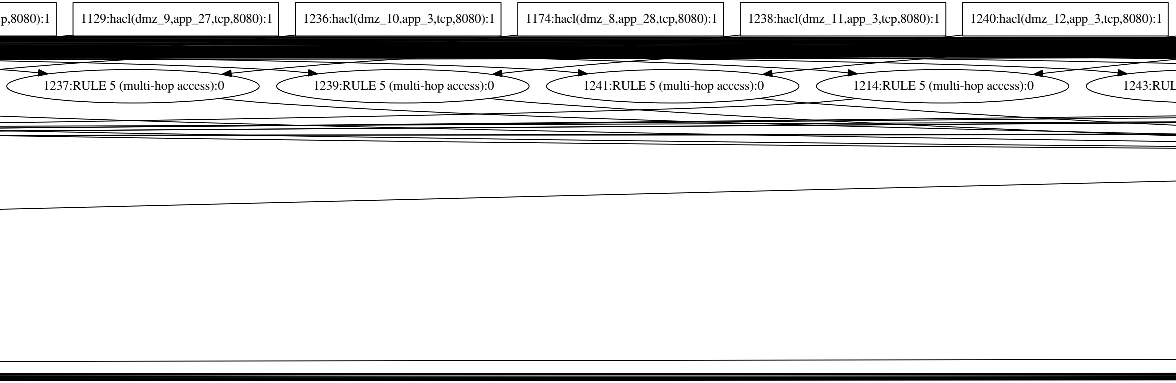

it,privEscalation):1
1162:hacl(dmz_2,app_28,tcp,8080):1
1127:hacl(dmz_8,app_27,tcp,8080):1
1187:hacl(dmz_1,app_29,tcp,8080):1
1185:hacl(dmz_0,app_29,tcp,8080):1
1168:hacl
1190:RULE 5 (multi-hop access):0
1192:RULE 5 (multi-hop access):0
1163:RULE 5 (multi-hop access):0
1194:RULE 5

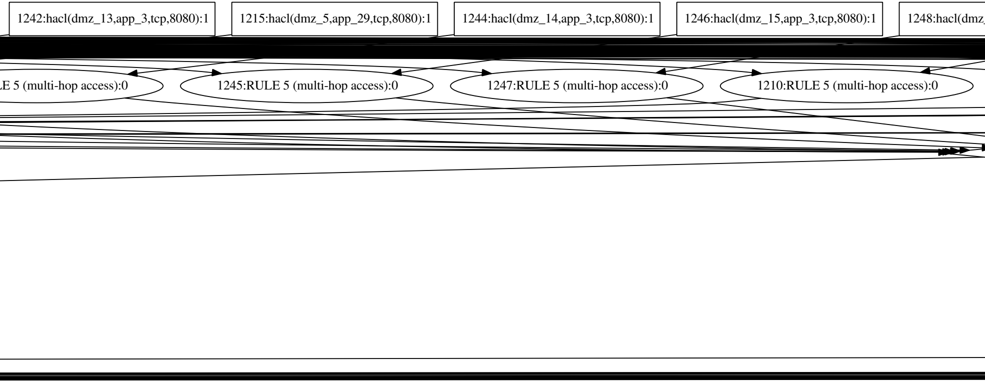

187:hacl(internet,dmz_9,tcp,80):1
186:RULE 6 (direct network access):0
l(dmz_5,app_28,tcp,8080):1
1189:hacl(dmz_10,app_29,tcp,8080):1
1191:hacl(dmz_11,app_29,tcp,8080):1
1193:hacl(dmz_12,app_29,tcp,8080):1
1164:hacl(dmz_3,app_28,tcp,8080):1
(multi-hop access):0
1196:RULE 5 (multi-hop access):0
1198:RULE 5 (multi-hop access):0
1169:RULE 5 (multi-hop access):0
1200:RULE

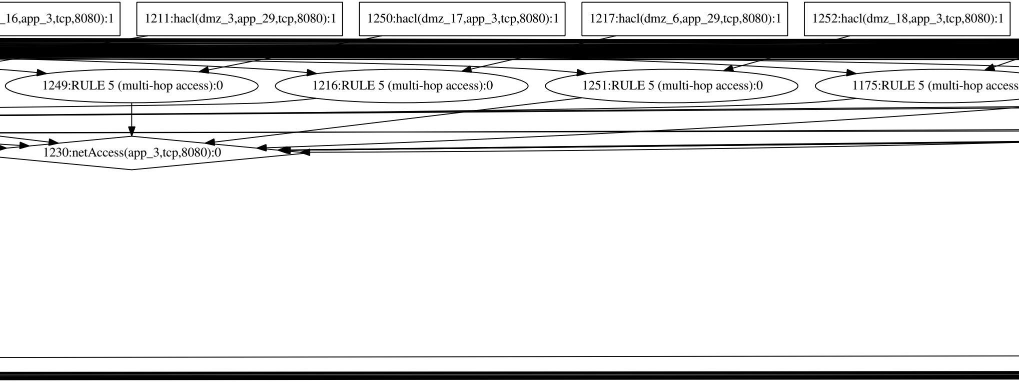

185:netAccess(dmz_9,tcp,80):0
1195:hacl(dmz_13,app_29,tcp,8080):1
1197:hacl(dmz_14,app_29,tcp,8080):1
1199:hacl(dmz_15,app_29,tcp,8080):1
1170:hacl(dmz_6,app_28,tcp,8080):1
1201:hacl(dmz_16,app_29,tcp,8080):1
5 (multi-hop access):0
1202:RULE 5 (multi-hop access):0
1204:RULE 5 (multi-hop access):0
1165:RULE 5 (multi-hop access):0
1206:RULE 5 (multi-h
1183:netAccess(app_29,tcp,8080):0

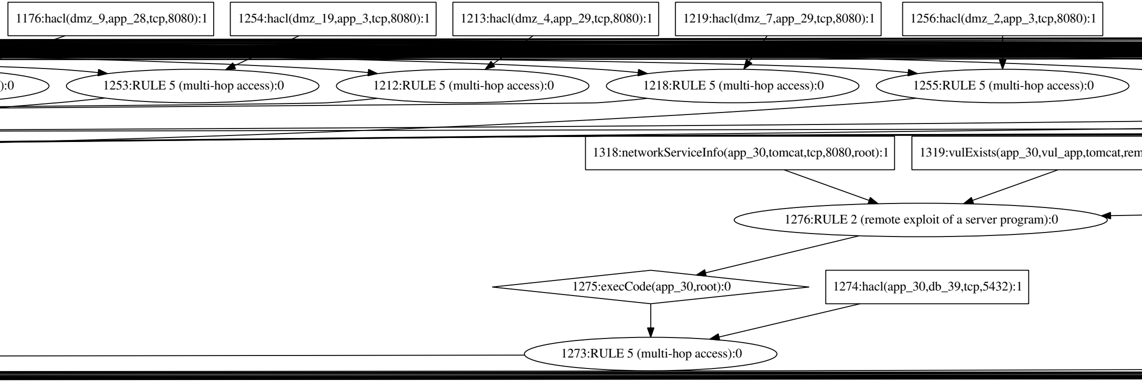

188:networkServiceInfo(dmz_9,httpd,tcp,80,root):1
189:vulExists(dmz_9,vul_web,httpd,remoteExploit,privEscalation):1
184:RULE 2 (remote exploit of a server program):0
1203:hacl(dmz_17,app_29,tcp,8080):1
1205:hacl(dmz_18,app_29,tcp,8080):1
1166:hacl(dmz_4,app_28,tcp,8080):1
1207:hacl(dmz_19,app_29,tcp,8080):1
1209:hacl(dmz_2,app_29,tcp,8080):1
op access):0
1208:RULE 5 (multi-hop access):0
181:RULE 5 (multi-hop access):0
1171:RULE 5 (multi-hop access):0
1233:RULE 5 (multi-hop access):0

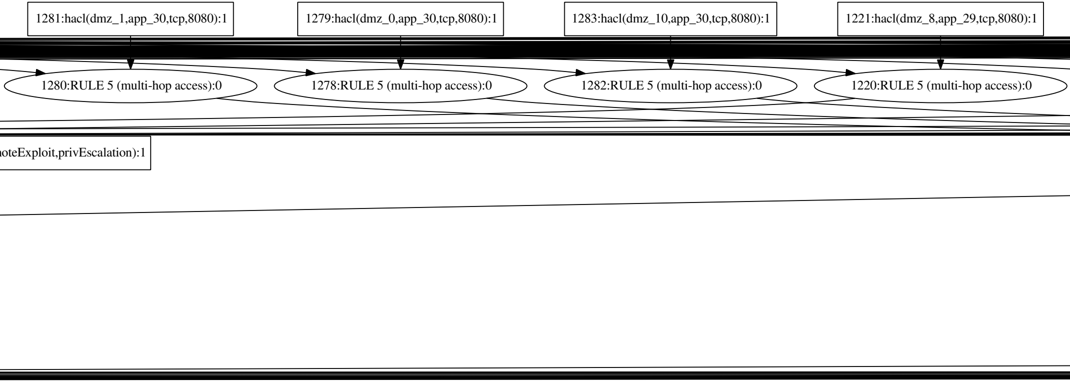

182:hacl(dmz_9,app_0,tcp,8080):1
183:execCode(dmz_9,root):0
1172:hacl(dmz_7,app_28,tcp,8080):1
1234:hacl(dmz_1,app_3,tcp,8080):1
1232:hacl(dmz_0,app_3,tc
1231:RULE 5 (multi-hop access):0
1128:RULE 5 (multi-hop access):0
1235:RULE 5 (multi-hop access):0
1173:RULE 5 (multi-hop access):0
1271:networkServiceInfo(app_3,tomcat,tcp,8080,root):1
1272:vulExists(app_3,vul_app,tomcat,remoteExploit,privEscalation):1
1229:RULE 2 (remote exploit of a server program):0

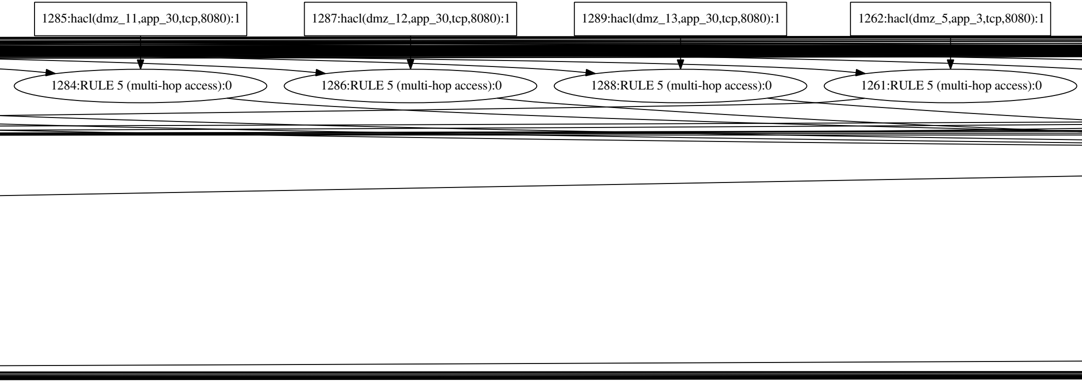

p,8080):1
1129:hacl(dmz_9,app_27,tcp,8080):1
1236:hacl(dmz_10,app_3,tcp,8080):1
1174:hacl(dmz_8,app_28,tcp,8080):1
1238:hacl(dmz_11,app_3,tcp,8080):1
1240:hacl(dmz_12,app_3,tcp,8080):1
1237:RULE 5 (multi-hop access):0
1239:RULE 5 (multi-hop access):0
1241:RULE 5 (multi-hop access):0
1214:RULE 5 (multi-hop access):0
1243:RUL

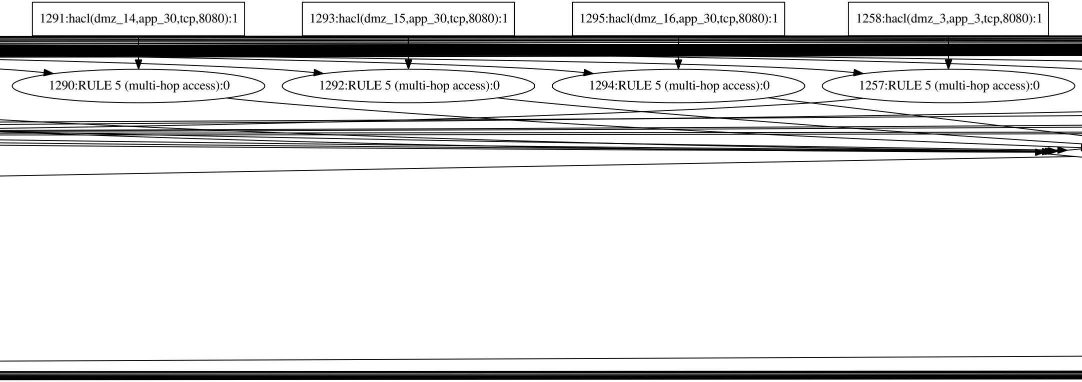

1242:hacl(dmz_13,app_3,tcp,8080):1
1215:hacl(dmz_5,app_29,tcp,8080):1
1244:hacl(dmz_14,app_3,tcp,8080):1
1246:hacl(dmz_15,app_3,tcp,8080):1
1248:hacl(dmz_
LE 5 (multi-hop access):0
1245:RULE 5 (multi-hop access):0
1247:RULE 5 (multi-hop access):0
1210:RULE 5 (multi-hop access):0

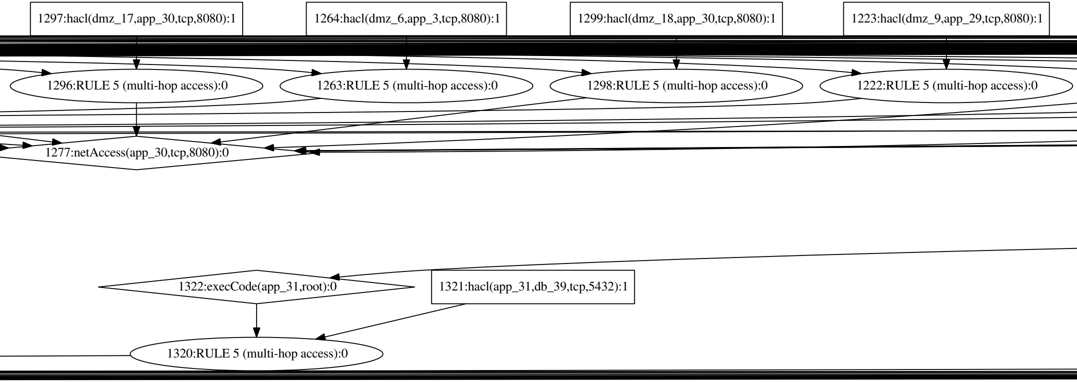

_16,app_3,tcp,8080):1
1211:hacl(dmz_3,app_29,tcp,8080):1
1250:hacl(dmz_17,app_3,tcp,8080):1
1217:hacl(dmz_6,app_29,tcp,8080):1
1252:hacl(dmz_18,app_3,tcp,8080):1
1249:RULE 5 (multi-hop access):0
1216:RULE 5 (multi-hop access):0
1251:RULE 5 (multi-hop access):0
1175:RULE 5 (multi-hop access
1230:netAccess(app_3,tcp,8080):0

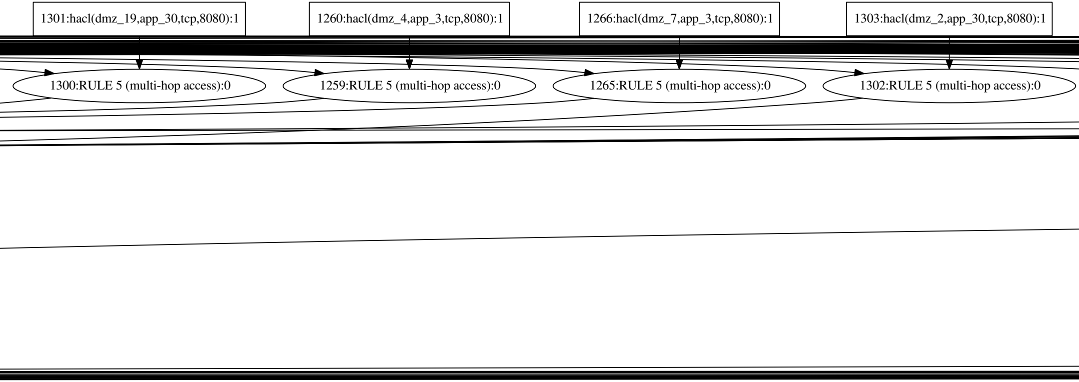

1176:hacl(dmz_9,app_28,tcp,8080):1
1254:hacl(dmz_19,app_3,tcp,8080):1
1213:hacl(dmz_4,app_29,tcp,8080):1
1219:hacl(dmz_7,app_29,tcp,8080):1
1256:hacl(dmz_2,app_3,tcp,8080):1
):0
1253:RULE 5 (multi-hop access):0
1212:RULE 5 (multi-hop access):0
1218:RULE 5 (multi-hop access):0
1255:RULE 5 (multi-hop access):0
1318:networkServiceInfo(app_30,tomcat,tcp,8080,root):1
1319:vulExists(app_30,vul_app,tomcat,rem
1276:RULE 2 (remote exploit of a server program):0
1275:execCode(app_30,root):0
1274:hacl(app_30,db_39,tcp,5432):1
1273:RULE 5 (multi-hop access):0

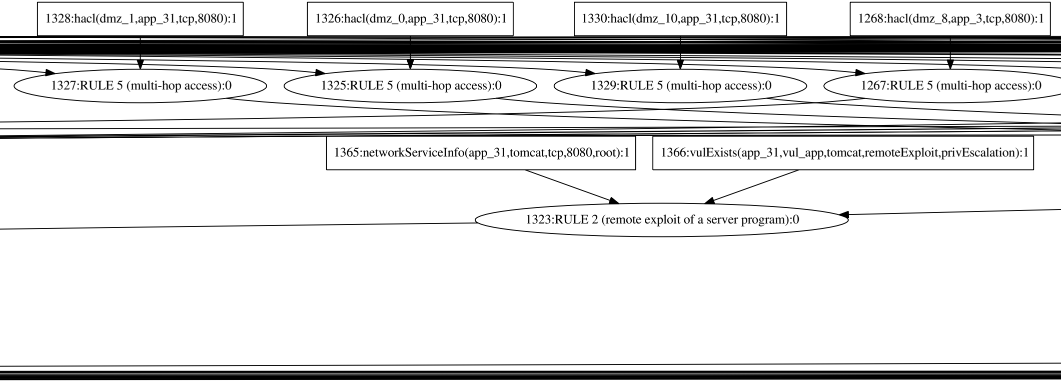

1281:hacl(dmz_1,app_30,tcp,8080):1
1279:hacl(dmz_0,app_30,tcp,8080):1
1283:hacl(dmz_10,app_30,tcp,8080):1
1221:hacl(dmz_8,app_29,tcp,8080):1
1280:RULE 5 (multi-hop access):0
1278:RULE 5 (multi-hop access):0
1282:RULE 5 (multi-hop access):0
1220:RULE 5 (multi-hop access):0
oteExploit,privEscalation):1

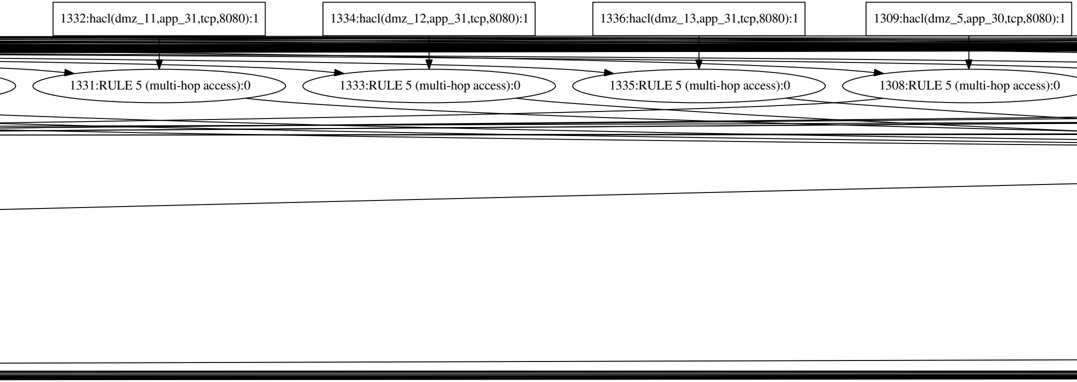

1285:hacl(dmz_11,app_30,tcp,8080):1
1287:hacl(dmz_12,app_30,tcp,8080):1
1289:hacl(dmz_13,app_30,tcp,8080):1
1262:hacl(dmz_5,app_3,tcp,8080):1
1284:RULE 5 (multi-hop access):0
1286:RULE 5 (multi-hop access):0
1288:RULE 5 (multi-hop access):0
1261:RULE 5 (multi-hop access):0

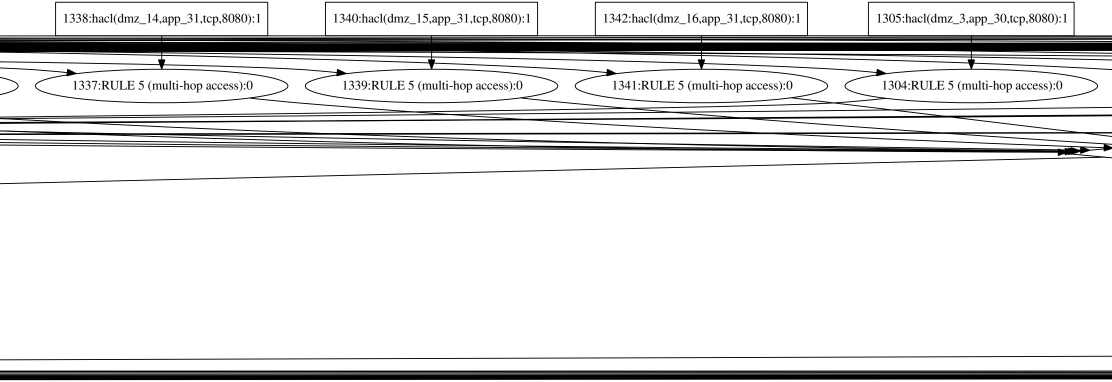

1291:hacl(dmz_14,app_30,tcp,8080):1
1293:hacl(dmz_15,app_30,tcp,8080):1
1295:hacl(dmz_16,app_30,tcp,8080):1
1258:hacl(dmz_3,app_3,tcp,8080):1
1290:RULE 5 (multi-hop access):0
1292:RULE 5 (multi-hop access):0
1294:RULE 5 (multi-hop access):0
1257:RULE 5 (multi-hop access):0

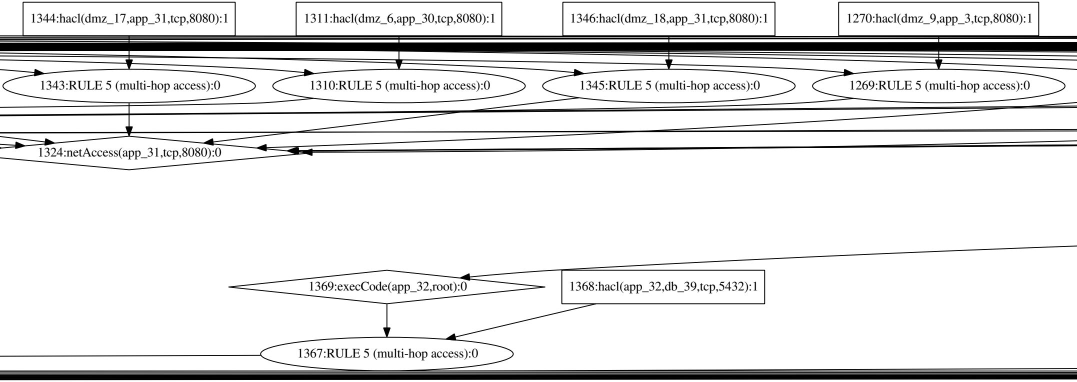

1297:hacl(dmz_17,app_30,tcp,8080):1
1264:hacl(dmz_6,app_3,tcp,8080):1
1299:hacl(dmz_18,app_30,tcp,8080):1
1223:hacl(dmz_9,app_29,tcp,8080):1
1296:RULE 5 (multi-hop access):0
1263:RULE 5 (multi-hop access):0
1298:RULE 5 (multi-hop access):0
1222:RULE 5 (multi-hop access):0
1277:netAccess(app_30,tcp,8080):0
1322:execCode(app_31,root):0
1321:hacl(app_31,db_39,tcp,5432):1
1320:RULE 5 (multi-hop access):0

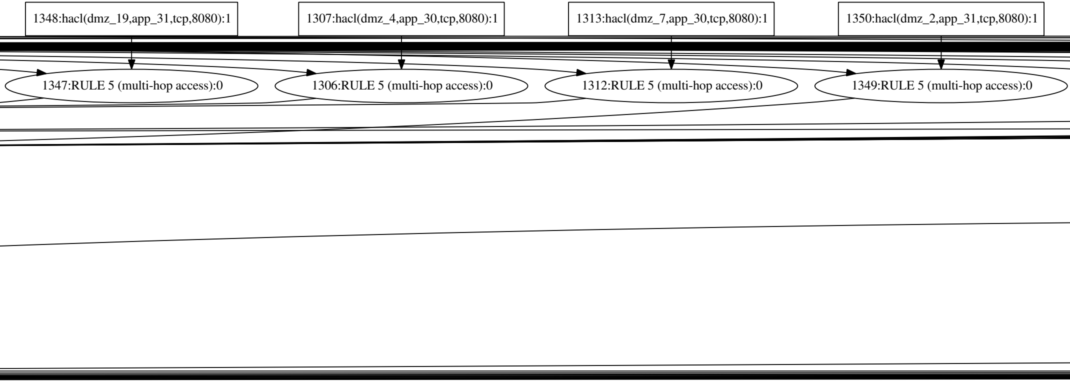

1301:hacl(dmz_19,app_30,tcp,8080):1
1260:hacl(dmz_4,app_3,tcp,8080):1
1266:hacl(dmz_7,app_3,tcp,8080):1
1303:hacl(dmz_2,app_30,tcp,8080):1
1300:RULE 5 (multi-hop access):0
1259:RULE 5 (multi-hop access):0
1265:RULE 5 (multi-hop access):0
1302:RULE 5 (multi-hop access):0

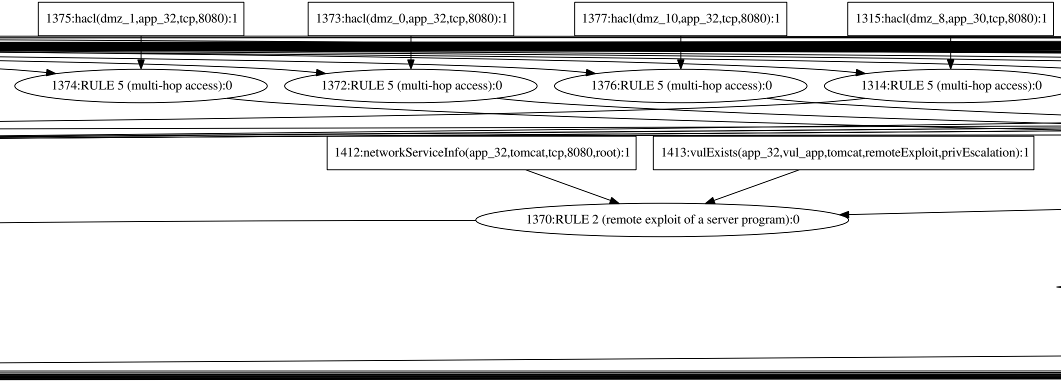

1328:hacl(dmz_1,app_31,tcp,8080):1
1326:hacl(dmz_0,app_31,tcp,8080):1
1330:hacl(dmz_10,app_31,tcp,8080):1
1268:hacl(dmz_8,app_3,tcp,8080):1
1327:RULE 5 (multi-hop access):0
1325:RULE 5 (multi-hop access):0
1329:RULE 5 (multi-hop access):0
1267:RULE 5 (multi-hop access):0
1365:networkServiceInfo(app_31,tomcat,tcp,8080,root):1
1366:vulExists(app_31,vul_app,tomcat,remoteExploit,privEscalation):1
1323:RULE 2 (remote exploit of a server program):0

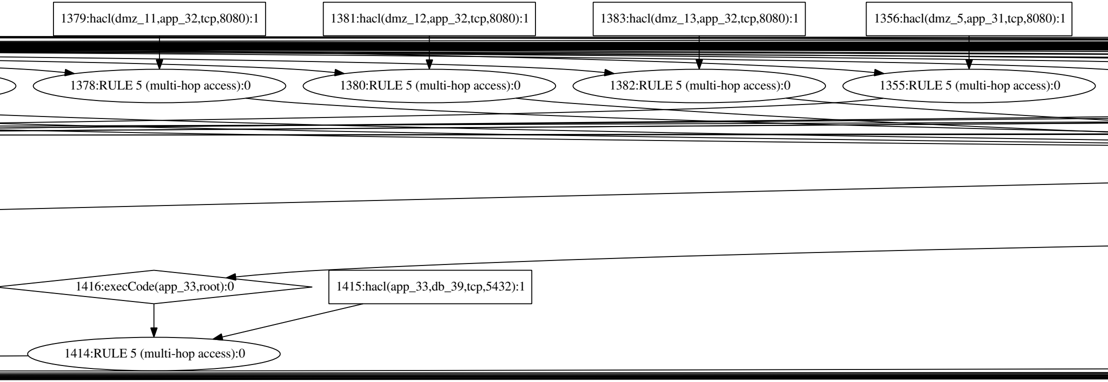

1332:hacl(dmz_11,app_31,tcp,8080):1
1334:hacl(dmz_12,app_31,tcp,8080):1
1336:hacl(dmz_13,app_31,tcp,8080):1
1309:hacl(dmz_5,app_30,tcp,8080):1
1331:RULE 5 (multi-hop access):0
1333:RULE 5 (multi-hop access):0
1335:RULE 5 (multi-hop access):0
1308:RULE 5 (multi-hop access):0

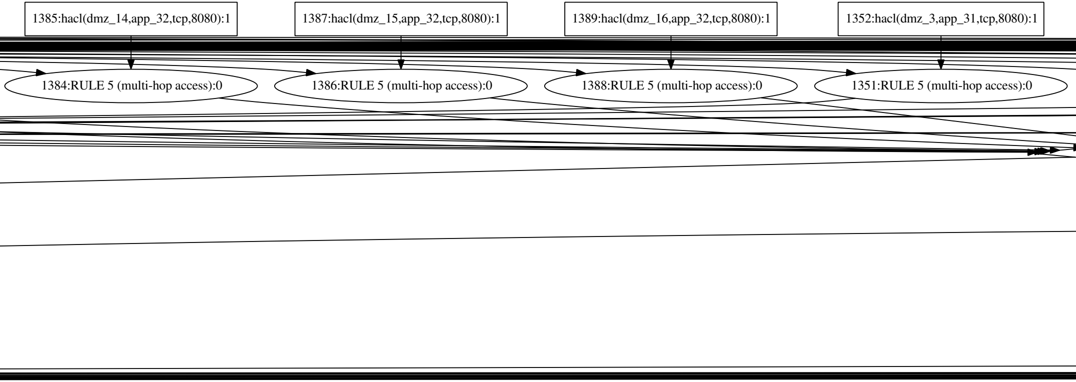

1338:hacl(dmz_14,app_31,tcp,8080):1
1340:hacl(dmz_15,app_31,tcp,8080):1
1342:hacl(dmz_16,app_31,tcp,8080):1
1305:hacl(dmz_3,app_30,tcp,8080):1
1337:RULE 5 (multi-hop access):0
1339:RULE 5 (multi-hop access):0
1341:RULE 5 (multi-hop access):0
1304:RULE 5 (multi-hop access):0

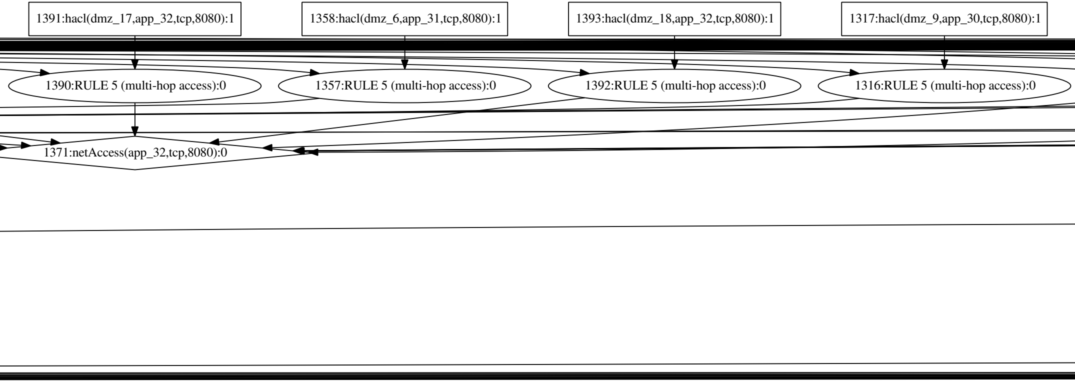

1344:hacl(dmz_17,app_31,tcp,8080):1
1311:hacl(dmz_6,app_30,tcp,8080):1
1346:hacl(dmz_18,app_31,tcp,8080):1
1270:hacl(dmz_9,app_3,tcp,8080):1
1343:RULE 5 (multi-hop access):0
1310:RULE 5 (multi-hop access):0
1345:RULE 5 (multi-hop access):0
1269:RULE 5 (multi-hop access):0
1324:netAccess(app_31,tcp,8080):0
1369:execCode(app_32,root):0
1368:hacl(app_32,db_39,tcp,5432):1
1367:RULE 5 (multi-hop access):0

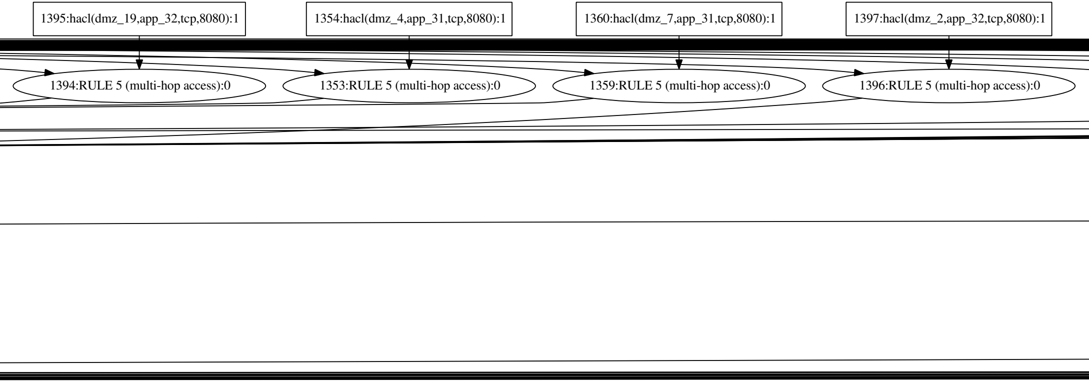

1348:hacl(dmz_19,app_31,tcp,8080):1
1307:hacl(dmz_4,app_30,tcp,8080):1
1313:hacl(dmz_7,app_30,tcp,8080):1
1350:hacl(dmz_2,app_31,tcp,8080):1
1347:RULE 5 (multi-hop access):0
1306:RULE 5 (multi-hop access):0
1312:RULE 5 (multi-hop access):0
1349:RULE 5 (multi-hop access):0

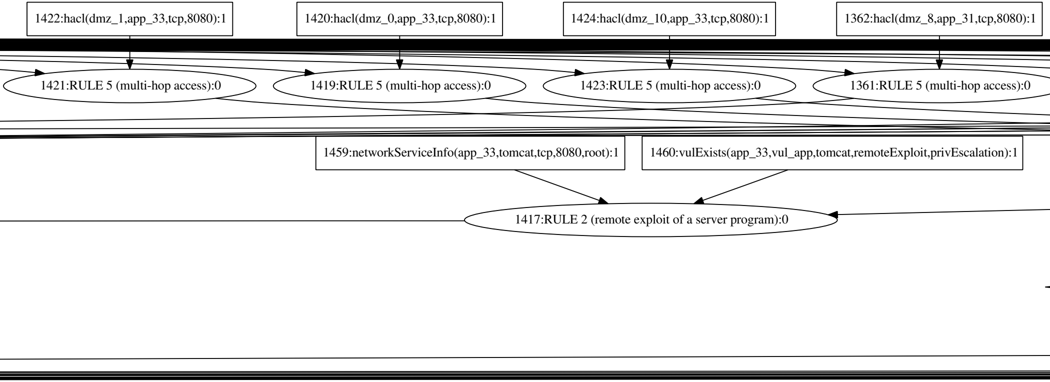

1375:hacl(dmz_1,app_32,tcp,8080):1
1373:hacl(dmz_0,app_32,tcp,8080):1
1377:hacl(dmz_10,app_32,tcp,8080):1
1315:hacl(dmz_8,app_30,tcp,8080):1
1374:RULE 5 (multi-hop access):0
1372:RULE 5 (multi-hop access):0
1376:RULE 5 (multi-hop access):0
1314:RULE 5 (multi-hop access):0
1412:networkServiceInfo(app_32,tomcat,tcp,8080,root):1
1413:vulExists(app_32,vul_app,tomcat,remoteExploit,privEscalation):1
1370:RULE 2 (remote exploit of a server program):0

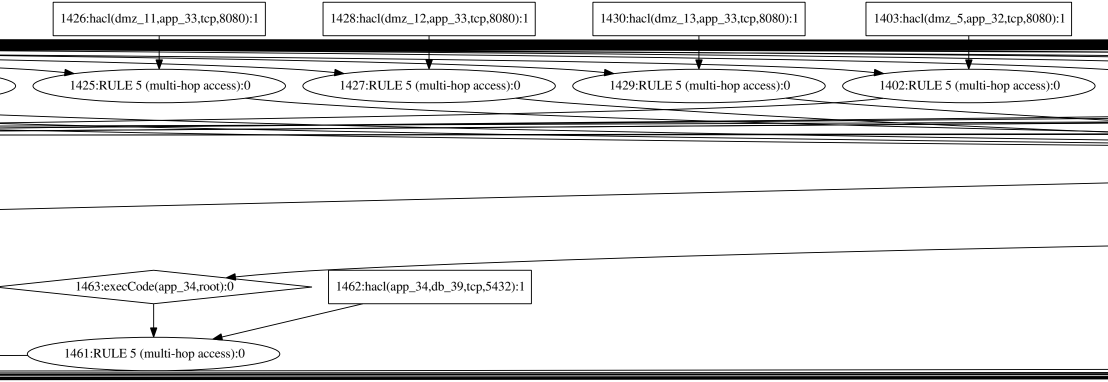

1379:hacl(dmz_11,app_32,tcp,8080):1
1381:hacl(dmz_12,app_32,tcp,8080):1
1383:hacl(dmz_13,app_32,tcp,8080):1
1356:hacl(dmz_5,app_31,tcp,8080):1
1378:RULE 5 (multi-hop access):0
1380:RULE 5 (multi-hop access):0
1382:RULE 5 (multi-hop access):0
1355:RULE 5 (multi-hop access):0
1416:execCode(app_33,root):0
1415:hacl(app_33,db_39,tcp,5432):1
1414:RULE 5 (multi-hop access):0

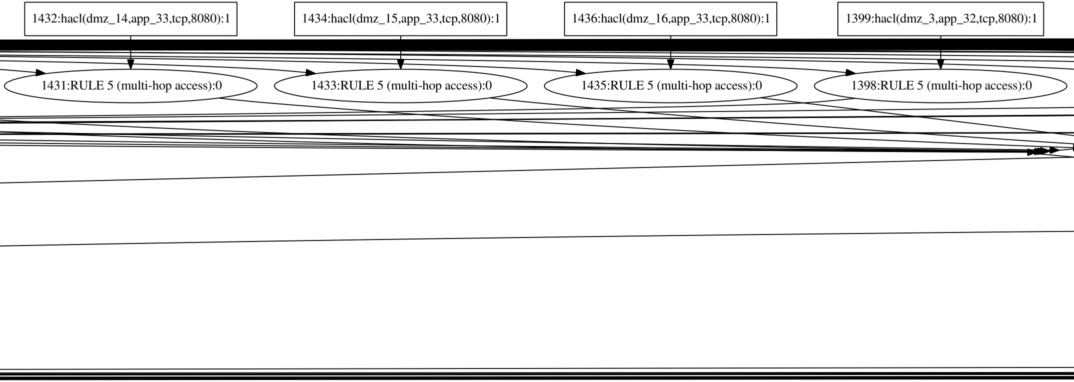

1385:hacl(dmz_14,app_32,tcp,8080):1
1387:hacl(dmz_15,app_32,tcp,8080):1
1389:hacl(dmz_16,app_32,tcp,8080):1
1352:hacl(dmz_3,app_31,tcp,8080):1
1384:RULE 5 (multi-hop access):0
1386:RULE 5 (multi-hop access):0
1388:RULE 5 (multi-hop access):0
1351:RULE 5 (multi-hop access):0

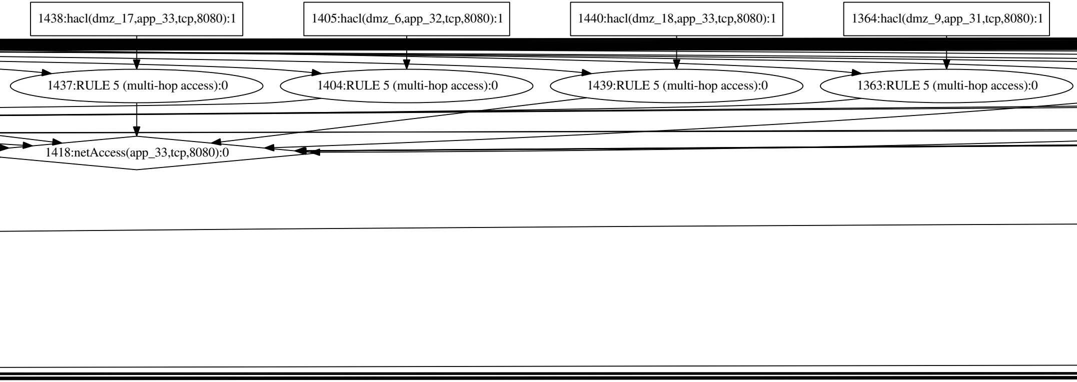

1391:hacl(dmz_17,app_32,tcp,8080):1
1358:hacl(dmz_6,app_31,tcp,8080):1
1393:hacl(dmz_18,app_32,tcp,8080):1
1317:hacl(dmz_9,app_30,tcp,8080):1
1390:RULE 5 (multi-hop access):0
1357:RULE 5 (multi-hop access):0
1392:RULE 5 (multi-hop access):0
1316:RULE 5 (multi-hop access):0
1371:netAccess(app_32,tcp,8080):0

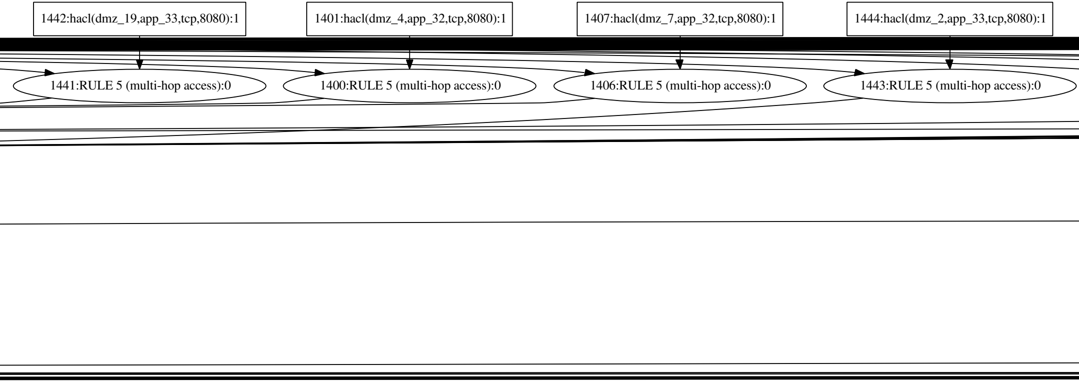

1395:hacl(dmz_19,app_32,tcp,8080):1
1354:hacl(dmz_4,app_31,tcp,8080):1
1360:hacl(dmz_7,app_31,tcp,8080):1
1397:hacl(dmz_2,app_32,tcp,8080):1
1394:RULE 5 (multi-hop access):0
1353:RULE 5 (multi-hop access):0
1359:RULE 5 (multi-hop access):0
1396:RULE 5 (multi-hop access):0

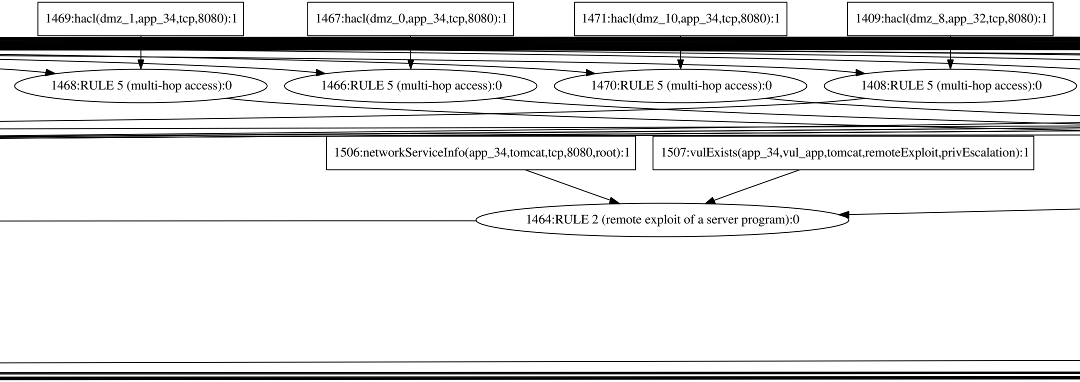

1422:hacl(dmz_1,app_33,tcp,8080):1
1420:hacl(dmz_0,app_33,tcp,8080):1
1424:hacl(dmz_10,app_33,tcp,8080):1
1362:hacl(dmz_8,app_31,tcp,8080):1
1421:RULE 5 (multi-hop access):0
1419:RULE 5 (multi-hop access):0
1423:RULE 5 (multi-hop access):0
1361:RULE 5 (multi-hop access):0
1459:networkServiceInfo(app_33,tomcat,tcp,8080,root):1
1460:vulExists(app_33,vul_app,tomcat,remoteExploit,privEscalation):1
1417:RULE 2 (remote exploit of a server program):0

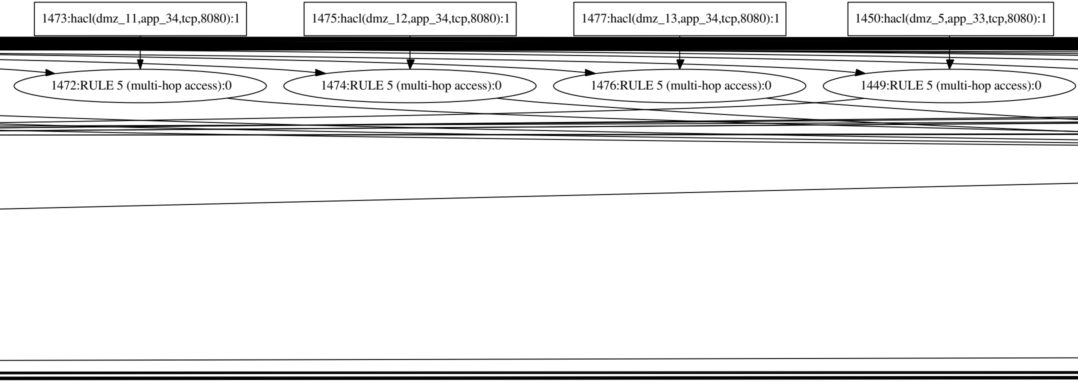

1426:hacl(dmz_11,app_33,tcp,8080):1
1428:hacl(dmz_12,app_33,tcp,8080):1
1430:hacl(dmz_13,app_33,tcp,8080):1
1403:hacl(dmz_5,app_32,tcp,8080):1
1425:RULE 5 (multi-hop access):0
1427:RULE 5 (multi-hop access):0
1429:RULE 5 (multi-hop access):0
1402:RULE 5 (multi-hop access):0
1463:execCode(app_34,root):0
1462:hacl(app_34,db_39,tcp,5432):1
1461:RULE 5 (multi-hop access):0

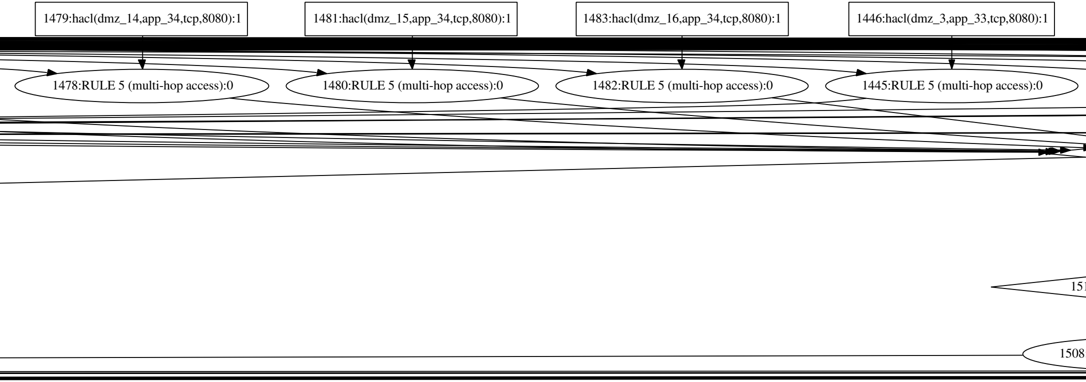

1432:hacl(dmz_14,app_33,tcp,8080):1
1434:hacl(dmz_15,app_33,tcp,8080):1
1436:hacl(dmz_16,app_33,tcp,8080):1
1399:hacl(dmz_3,app_32,tcp,8080):1
1431:RULE 5 (multi-hop access):0
1433:RULE 5 (multi-hop access):0
1435:RULE 5 (multi-hop access):0
1398:RULE 5 (multi-hop access):0

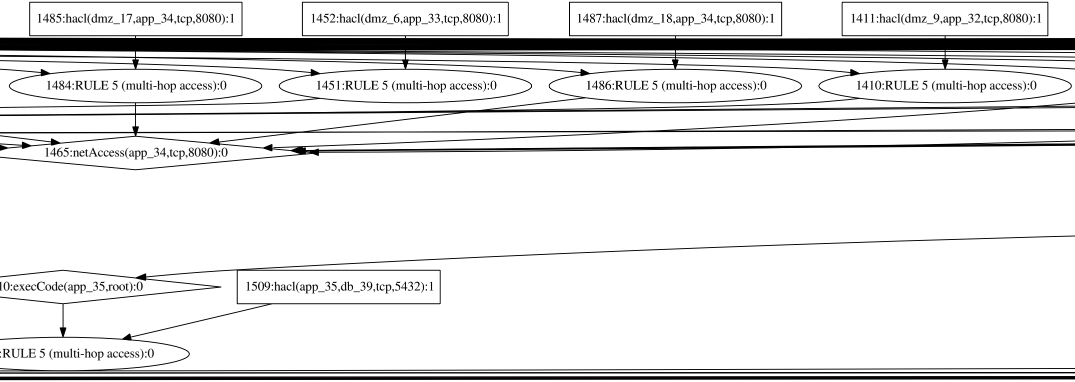

1438:hacl(dmz_17,app_33,tcp,8080):1
1405:hacl(dmz_6,app_32,tcp,8080):1
1440:hacl(dmz_18,app_33,tcp,8080):1
1364:hacl(dmz_9,app_31,tcp,8080):1
1437:RULE 5 (multi-hop access):0
1404:RULE 5 (multi-hop access):0
1439:RULE 5 (multi-hop access):0
1363:RULE 5 (multi-hop access):0
1418:netAccess(app_33,tcp,8080):0

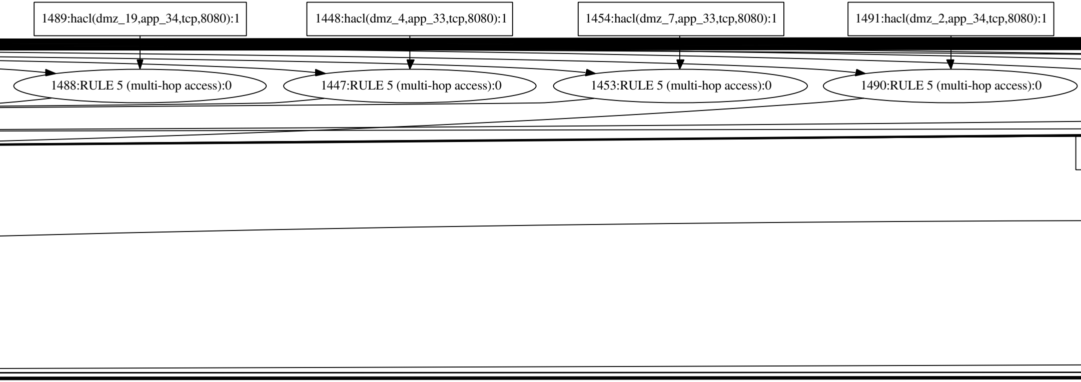

1442:hacl(dmz_19,app_33,tcp,8080):1
1401:hacl(dmz_4,app_32,tcp,8080):1
1407:hacl(dmz_7,app_32,tcp,8080):1
1444:hacl(dmz_2,app_33,tcp,8080):1
1441:RULE 5 (multi-hop access):0
1400:RULE 5 (multi-hop access):0
1406:RULE 5 (multi-hop access):0
1443:RULE 5 (multi-hop access):0

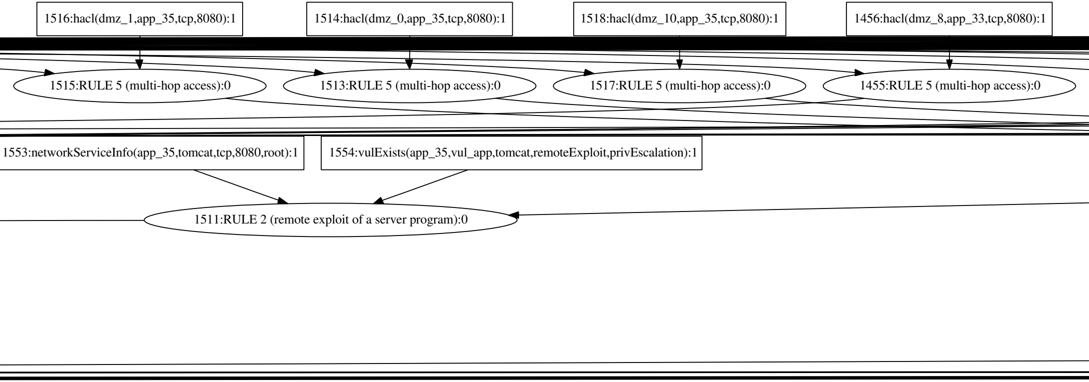

1469:hacl(dmz_1,app_34,tcp,8080):1
1467:hacl(dmz_0,app_34,tcp,8080):1
1471:hacl(dmz_10,app_34,tcp,8080):1
1409:hacl(dmz_8,app_32,tcp,8080):1
1468:RULE 5 (multi-hop access):0
1466:RULE 5 (multi-hop access):0
1470:RULE 5 (multi-hop access):0
1408:RULE 5 (multi-hop access):0
1506:networkServiceInfo(app_34,tomcat,tcp,8080,root):1
1507:vulExists(app_34,vul_app,tomcat,remoteExploit,privEscalation):1
1464:RULE 2 (remote exploit of a server program):0

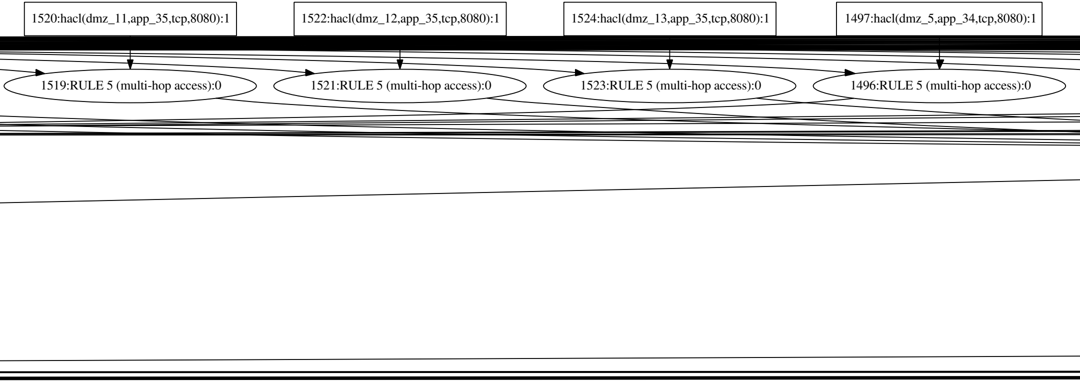

1473:hacl(dmz_11,app_34,tcp,8080):1
1475:hacl(dmz_12,app_34,tcp,8080):1
1477:hacl(dmz_13,app_34,tcp,8080):1
1450:hacl(dmz_5,app_33,tcp,8080):1
1472:RULE 5 (multi-hop access):0
1474:RULE 5 (multi-hop access):0
1476:RULE 5 (multi-hop access):0
1449:RULE 5 (multi-hop access):0

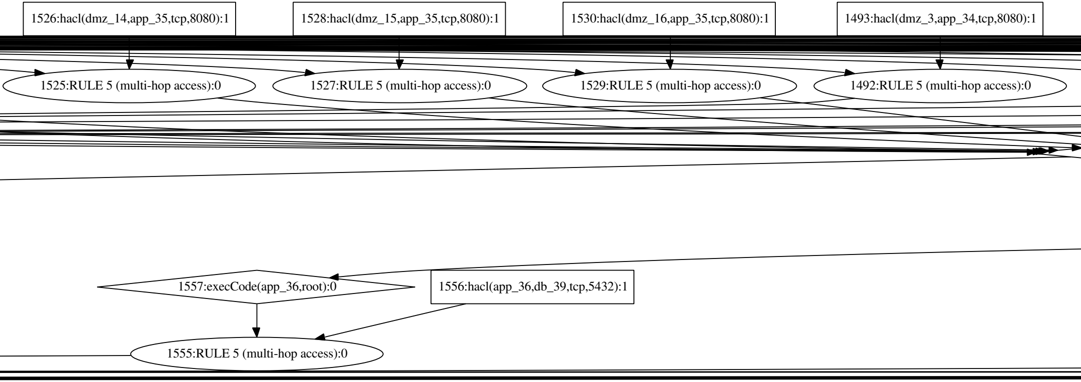

1479:hacl(dmz_14,app_34,tcp,8080):1
1481:hacl(dmz_15,app_34,tcp,8080):1
1483:hacl(dmz_16,app_34,tcp,8080):1
1446:hacl(dmz_3,app_33,tcp,8080):1
1478:RULE 5 (multi-hop access):0
1480:RULE 5 (multi-hop access):0
1482:RULE 5 (multi-hop access):0
1445:RULE 5 (multi-hop access):0
151
1508

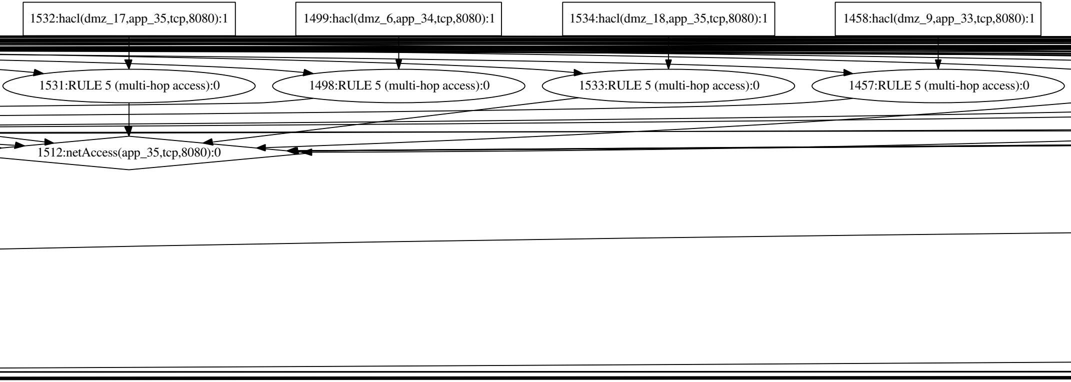

1485:hacl(dmz_17,app_34,tcp,8080):1
1452:hacl(dmz_6,app_33,tcp,8080):1
1487:hacl(dmz_18,app_34,tcp,8080):1
1411:hacl(dmz_9,app_32,tcp,8080):1
1484:RULE 5 (multi-hop access):0
1451:RULE 5 (multi-hop access):0
1486:RULE 5 (multi-hop access):0
1410:RULE 5 (multi-hop access):0
1465:netAccess(app_34,tcp,8080):0
10:execCode(app_35,root):0
1509:hacl(app_35,db_39,tcp,5432):1
:RULE 5 (multi-hop access):0

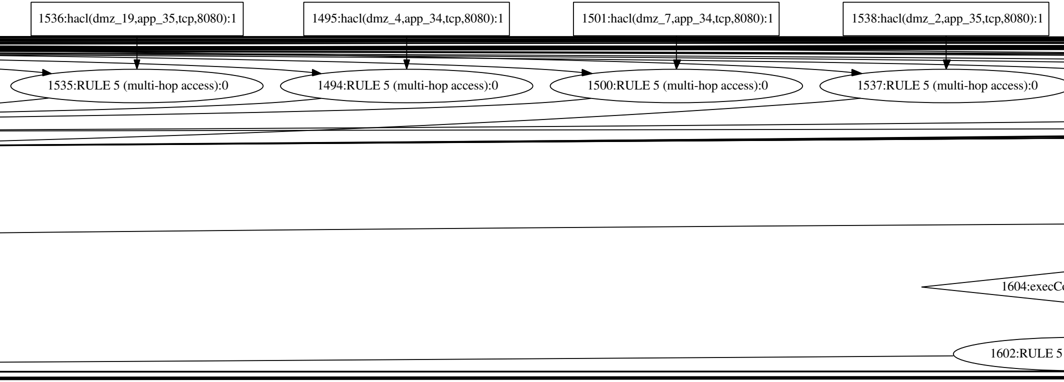

1489:hacl(dmz_19,app_34,tcp,8080):1
1448:hacl(dmz_4,app_33,tcp,8080):1
1454:hacl(dmz_7,app_33,tcp,8080):1
1491:hacl(dmz_2,app_34,tcp,8080):1
1488:RULE 5 (multi-hop access):0
1447:RULE 5 (multi-hop access):0
1453:RULE 5 (multi-hop access):0
1490:RULE 5 (multi-hop access):0

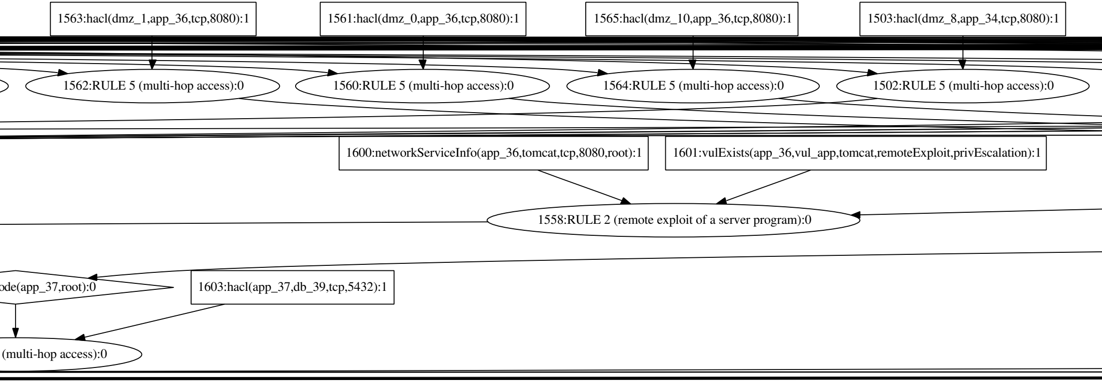

1516:hacl(dmz_1,app_35,tcp,8080):1
1514:hacl(dmz_0,app_35,tcp,8080):1
1518:hacl(dmz_10,app_35,tcp,8080):1
1456:hacl(dmz_8,app_33,tcp,8080):1
1515:RULE 5 (multi-hop access):0
1513:RULE 5 (multi-hop access):0
1517:RULE 5 (multi-hop access):0
1455:RULE 5 (multi-hop access):0
1553:networkServiceInfo(app_35,tomcat,tcp,8080,root):1
1554:vulExists(app_35,vul_app,tomcat,remoteExploit,privEscalation):1
1511:RULE 2 (remote exploit of a server program):0

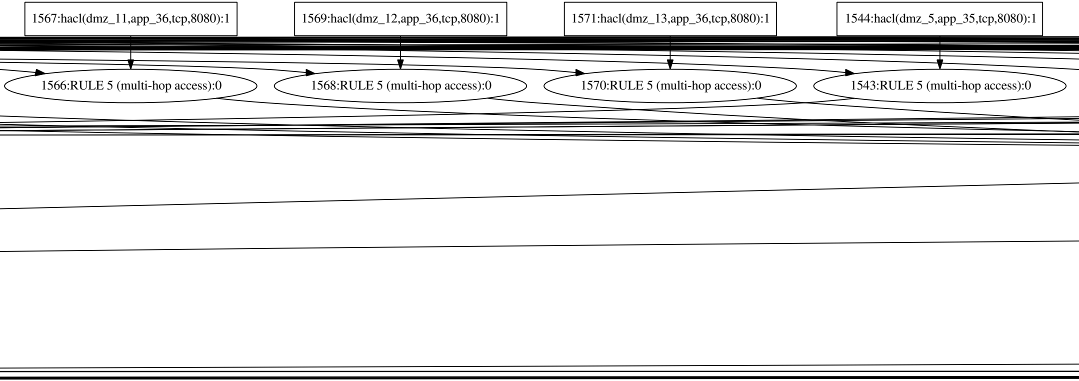

1520:hacl(dmz_11,app_35,tcp,8080):1
1522:hacl(dmz_12,app_35,tcp,8080):1
1524:hacl(dmz_13,app_35,tcp,8080):1
1497:hacl(dmz_5,app_34,tcp,8080):1
1519:RULE 5 (multi-hop access):0
1521:RULE 5 (multi-hop access):0
1523:RULE 5 (multi-hop access):0
1496:RULE 5 (multi-hop access):0

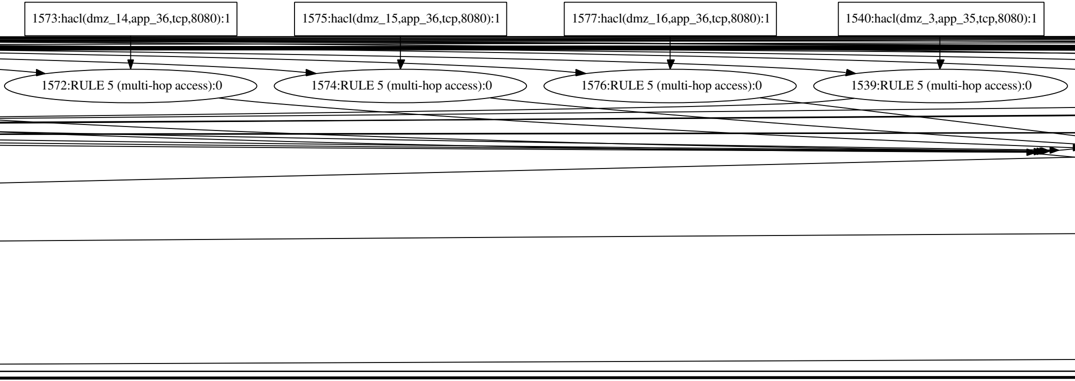

1526:hacl(dmz_14,app_35,tcp,8080):1
1528:hacl(dmz_15,app_35,tcp,8080):1
1530:hacl(dmz_16,app_35,tcp,8080):1
1493:hacl(dmz_3,app_34,tcp,8080):1
1525:RULE 5 (multi-hop access):0
1527:RULE 5 (multi-hop access):0
1529:RULE 5 (multi-hop access):0
1492:RULE 5 (multi-hop access):0
1557:execCode(app_36,root):0
1556:hacl(app_36,db_39,tcp,5432):1
1555:RULE 5 (multi-hop access):0

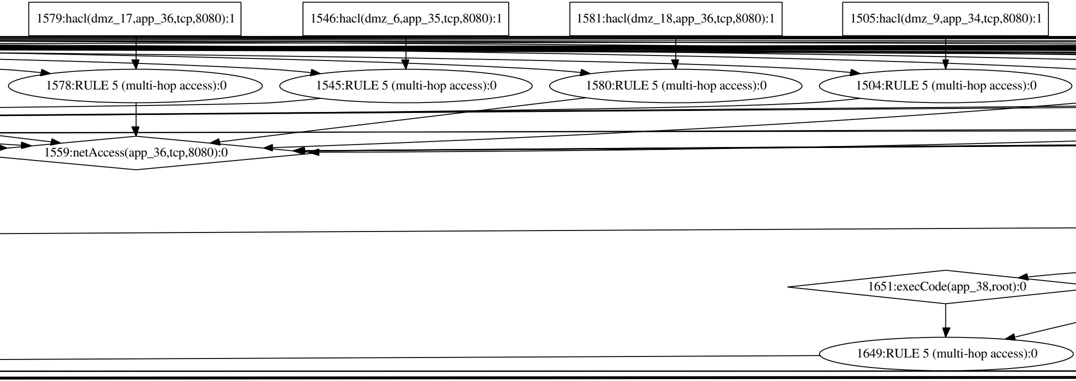

1532:hacl(dmz_17,app_35,tcp,8080):1
1499:hacl(dmz_6,app_34,tcp,8080):1
1534:hacl(dmz_18,app_35,tcp,8080):1
1458:hacl(dmz_9,app_33,tcp,8080):1
1531:RULE 5 (multi-hop access):0
1498:RULE 5 (multi-hop access):0
1533:RULE 5 (multi-hop access):0
1457:RULE 5 (multi-hop access):0
1512:netAccess(app_35,tcp,8080):0

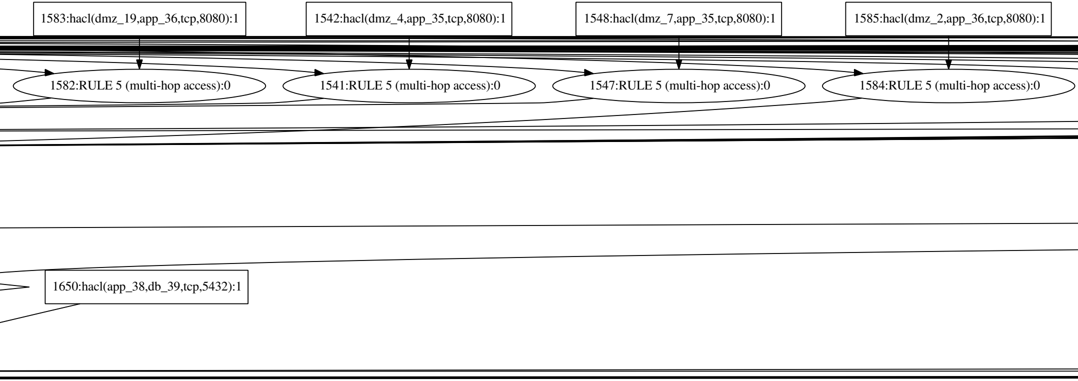

1536:hacl(dmz_19,app_35,tcp,8080):1
1495:hacl(dmz_4,app_34,tcp,8080):1
1501:hacl(dmz_7,app_34,tcp,8080):1
1538:hacl(dmz_2,app_35,tcp,8080):1
1535:RULE 5 (multi-hop access):0
1494:RULE 5 (multi-hop access):0
1500:RULE 5 (multi-hop access):0
1537:RULE 5 (multi-hop access):0
1604:execCo
1602:RULE 5

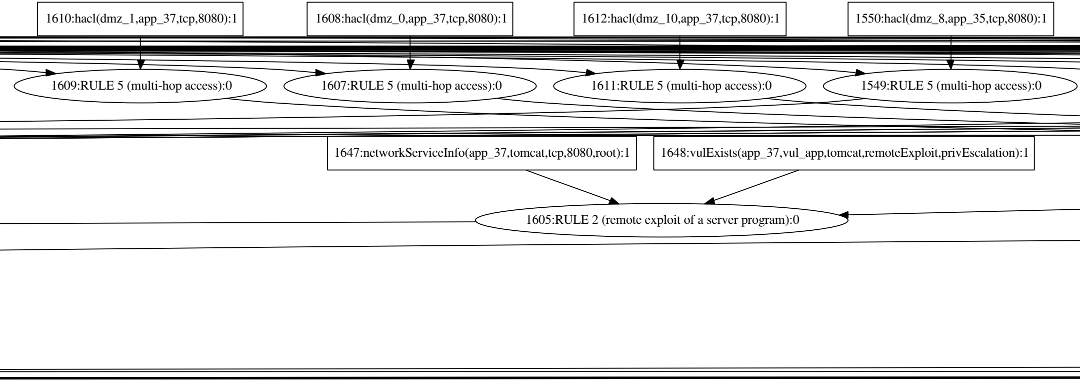

1563:hacl(dmz_1,app_36,tcp,8080):1
1561:hacl(dmz_0,app_36,tcp,8080):1
1565:hacl(dmz_10,app_36,tcp,8080):1
1503:hacl(dmz_8,app_34,tcp,8080):1
1562:RULE 5 (multi-hop access):0
1560:RULE 5 (multi-hop access):0
1564:RULE 5 (multi-hop access):0
1502:RULE 5 (multi-hop access):0
1600:networkServiceInfo(app_36,tomcat,tcp,8080,root):1
1601:vulExists(app_36,vul_app,tomcat,remoteExploit,privEscalation):1
1558:RULE 2 (remote exploit of a server program):0
ode(app_37,root):0
1603:hacl(app_37,db_39,tcp,5432):1
(multi-hop access):0

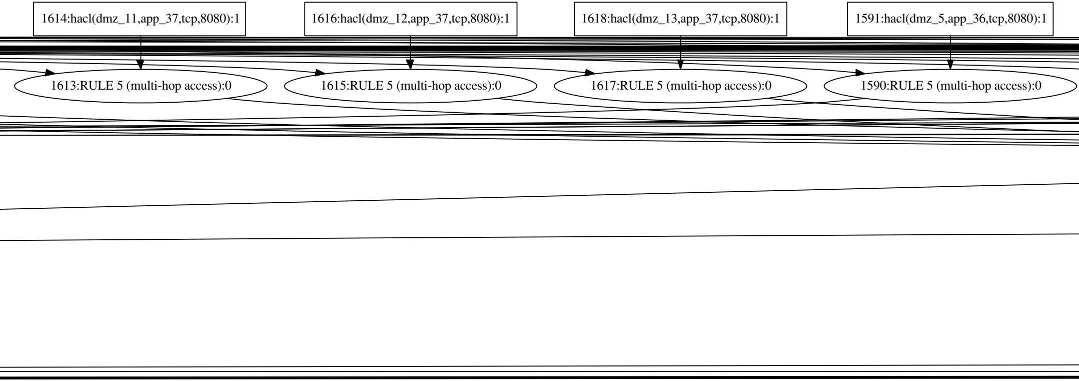

1567:hacl(dmz_11,app_36,tcp,8080):1
1569:hacl(dmz_12,app_36,tcp,8080):1
1571:hacl(dmz_13,app_36,tcp,8080):1
1544:hacl(dmz_5,app_35,tcp,8080):1
1566:RULE 5 (multi-hop access):0
1568:RULE 5 (multi-hop access):0
1570:RULE 5 (multi-hop access):0
1543:RULE 5 (multi-hop access):0

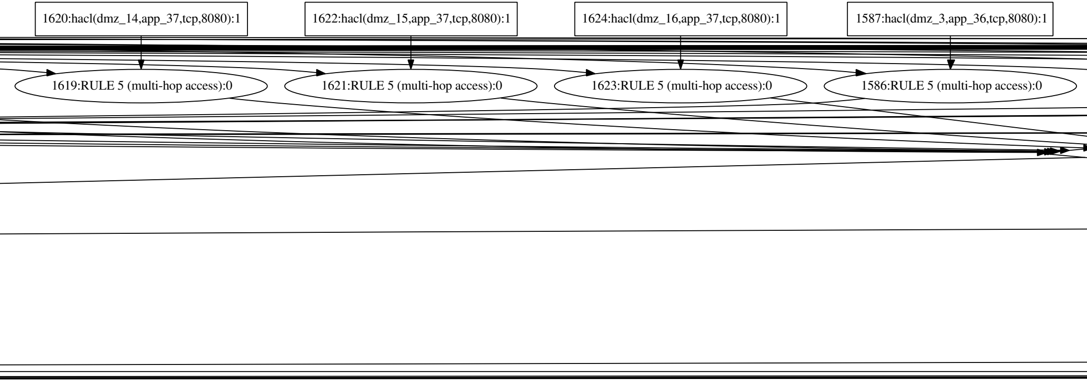

1573:hacl(dmz_14,app_36,tcp,8080):1
1575:hacl(dmz_15,app_36,tcp,8080):1
1577:hacl(dmz_16,app_36,tcp,8080):1
1540:hacl(dmz_3,app_35,tcp,8080):1
1572:RULE 5 (multi-hop access):0
1574:RULE 5 (multi-hop access):0
1576:RULE 5 (multi-hop access):0
1539:RULE 5 (multi-hop access):0

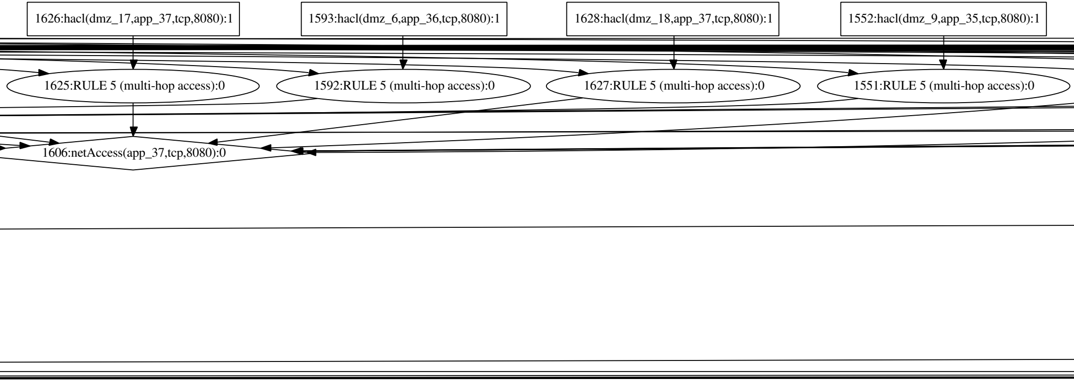

1579:hacl(dmz_17,app_36,tcp,8080):1
1546:hacl(dmz_6,app_35,tcp,8080):1
1581:hacl(dmz_18,app_36,tcp,8080):1
1505:hacl(dmz_9,app_34,tcp,8080):1
1578:RULE 5 (multi-hop access):0
1545:RULE 5 (multi-hop access):0
1580:RULE 5 (multi-hop access):0
1504:RULE 5 (multi-hop access):0
1559:netAccess(app_36,tcp,8080):0
1651:execCode(app_38,root):0
1649:RULE 5 (multi-hop access):0

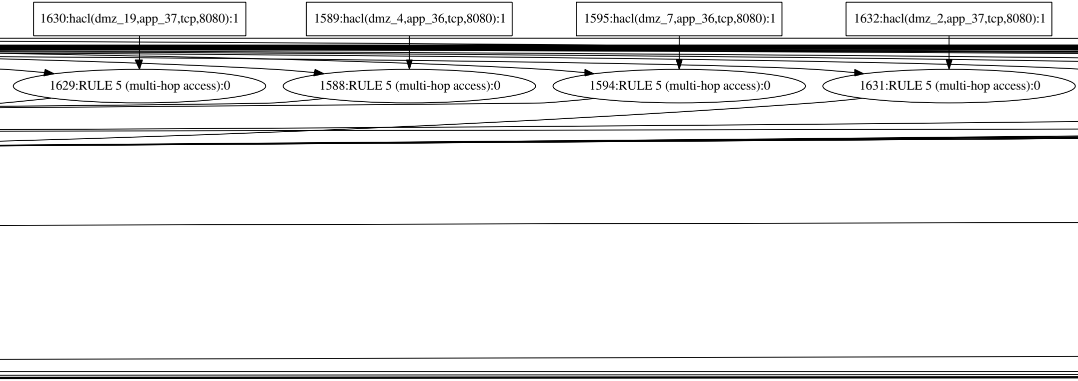

1583:hacl(dmz_19,app_36,tcp,8080):1
1542:hacl(dmz_4,app_35,tcp,8080):1
1548:hacl(dmz_7,app_35,tcp,8080):1
1585:hacl(dmz_2,app_36,tcp,8080):1
1582:RULE 5 (multi-hop access):0
1541:RULE 5 (multi-hop access):0
1547:RULE 5 (multi-hop access):0
1584:RULE 5 (multi-hop access):0
1650:hacl(app_38,db_39,tcp,5432):1

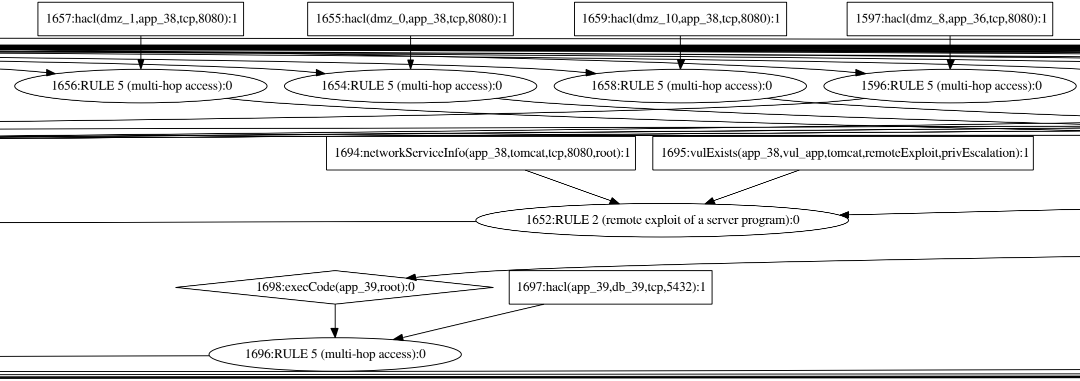

1610:hacl(dmz_1,app_37,tcp,8080):1
1608:hacl(dmz_0,app_37,tcp,8080):1
1612:hacl(dmz_10,app_37,tcp,8080):1
1550:hacl(dmz_8,app_35,tcp,8080):1
1609:RULE 5 (multi-hop access):0
1607:RULE 5 (multi-hop access):0
1611:RULE 5 (multi-hop access):0
1549:RULE 5 (multi-hop access):0
1647:networkServiceInfo(app_37,tomcat,tcp,8080,root):1
1648:vulExists(app_37,vul_app,tomcat,remoteExploit,privEscalation):1
1605:RULE 2 (remote exploit of a server program):0

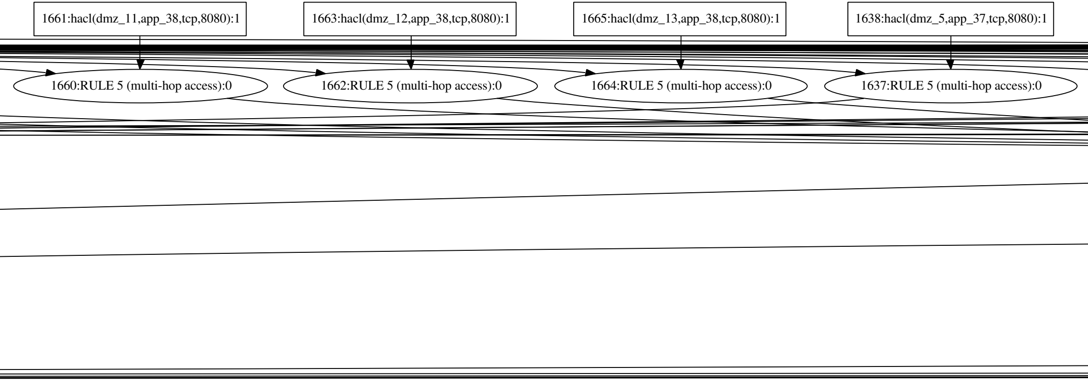

1614:hacl(dmz_11,app_37,tcp,8080):1
1616:hacl(dmz_12,app_37,tcp,8080):1
1618:hacl(dmz_13,app_37,tcp,8080):1
1591:hacl(dmz_5,app_36,tcp,8080):1
1613:RULE 5 (multi-hop access):0
1615:RULE 5 (multi-hop access):0
1617:RULE 5 (multi-hop access):0
1590:RULE 5 (multi-hop access):0

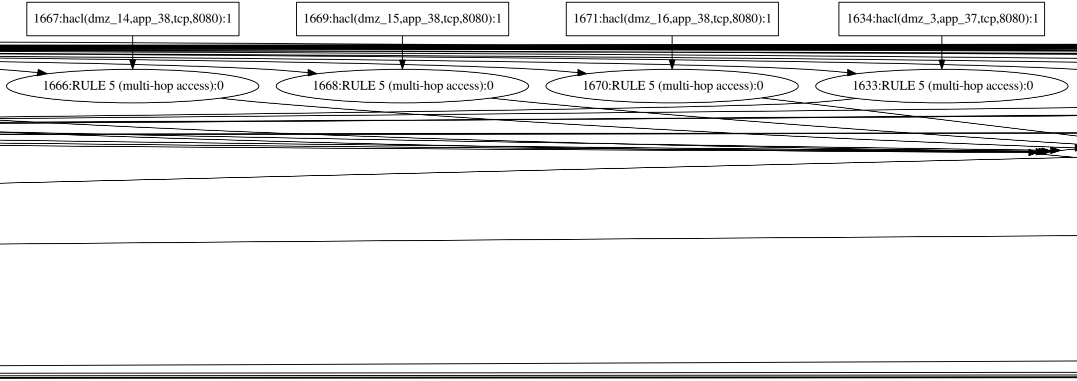

1620:hacl(dmz_14,app_37,tcp,8080):1
1622:hacl(dmz_15,app_37,tcp,8080):1
1624:hacl(dmz_16,app_37,tcp,8080):1
1587:hacl(dmz_3,app_36,tcp,8080):1
1619:RULE 5 (multi-hop access):0
1621:RULE 5 (multi-hop access):0
1623:RULE 5 (multi-hop access):0
1586:RULE 5 (multi-hop access):0

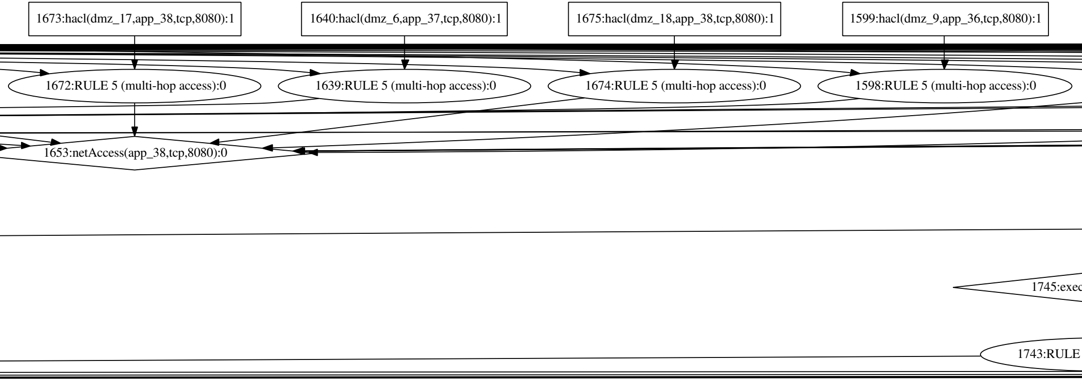

1626:hacl(dmz_17,app_37,tcp,8080):1
1593:hacl(dmz_6,app_36,tcp,8080):1
1628:hacl(dmz_18,app_37,tcp,8080):1
1552:hacl(dmz_9,app_35,tcp,8080):1
1625:RULE 5 (multi-hop access):0
1592:RULE 5 (multi-hop access):0
1627:RULE 5 (multi-hop access):0
1551:RULE 5 (multi-hop access):0
1606:netAccess(app_37,tcp,8080):0

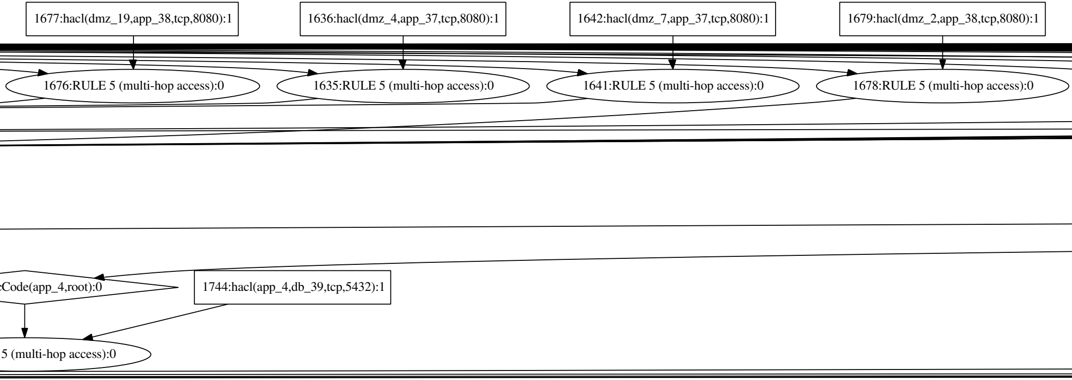

1630:hacl(dmz_19,app_37,tcp,8080):1
1589:hacl(dmz_4,app_36,tcp,8080):1
1595:hacl(dmz_7,app_36,tcp,8080):1
1632:hacl(dmz_2,app_37,tcp,8080):1
1629:RULE 5 (multi-hop access):0
1588:RULE 5 (multi-hop access):0
1594:RULE 5 (multi-hop access):0
1631:RULE 5 (multi-hop access):0

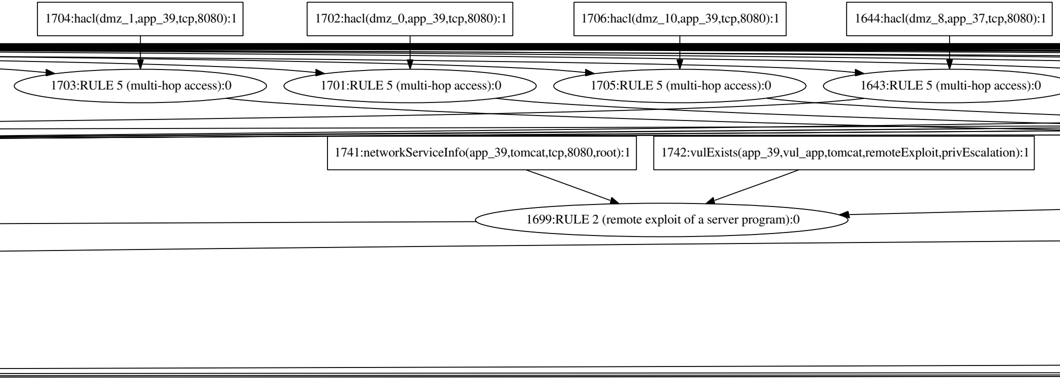

1657:hacl(dmz_1,app_38,tcp,8080):1
1655:hacl(dmz_0,app_38,tcp,8080):1
1659:hacl(dmz_10,app_38,tcp,8080):1
1597:hacl(dmz_8,app_36,tcp,8080):1
1656:RULE 5 (multi-hop access):0
1654:RULE 5 (multi-hop access):0
1658:RULE 5 (multi-hop access):0
1596:RULE 5 (multi-hop access):0
1694:networkServiceInfo(app_38,tomcat,tcp,8080,root):1
1695:vulExists(app_38,vul_app,tomcat,remoteExploit,privEscalation):1
1652:RULE 2 (remote exploit of a server program):0
1698:execCode(app_39,root):0
1697:hacl(app_39,db_39,tcp,5432):1
1696:RULE 5 (multi-hop access):0

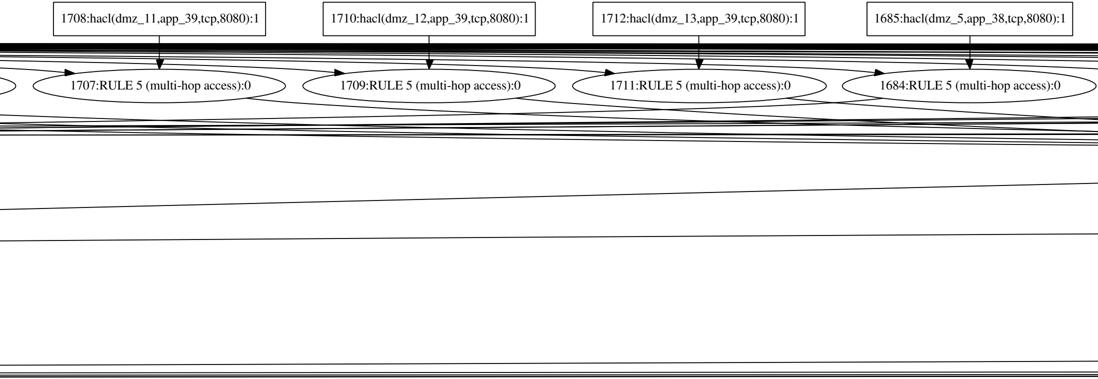

1661:hacl(dmz_11,app_38,tcp,8080):1
1663:hacl(dmz_12,app_38,tcp,8080):1
1665:hacl(dmz_13,app_38,tcp,8080):1
1638:hacl(dmz_5,app_37,tcp,8080):1
1660:RULE 5 (multi-hop access):0
1662:RULE 5 (multi-hop access):0
1664:RULE 5 (multi-hop access):0
1637:RULE 5 (multi-hop access):0

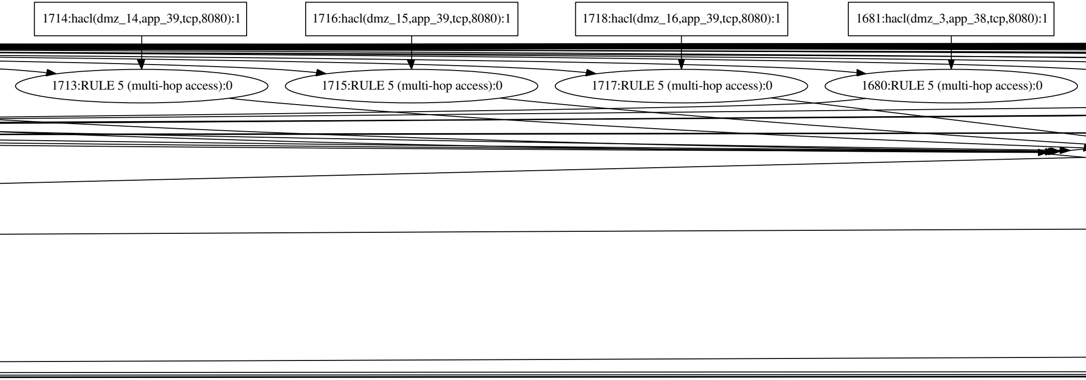

1667:hacl(dmz_14,app_38,tcp,8080):1
1669:hacl(dmz_15,app_38,tcp,8080):1
1671:hacl(dmz_16,app_38,tcp,8080):1
1634:hacl(dmz_3,app_37,tcp,8080):1
1666:RULE 5 (multi-hop access):0
1668:RULE 5 (multi-hop access):0
1670:RULE 5 (multi-hop access):0
1633:RULE 5 (multi-hop access):0

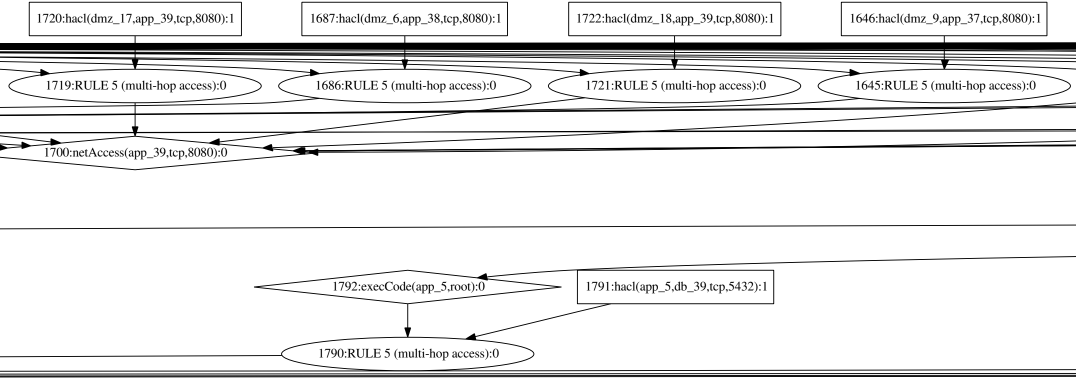

1673:hacl(dmz_17,app_38,tcp,8080):1
1640:hacl(dmz_6,app_37,tcp,8080):1
1675:hacl(dmz_18,app_38,tcp,8080):1
1599:hacl(dmz_9,app_36,tcp,8080):1
1672:RULE 5 (multi-hop access):0
1639:RULE 5 (multi-hop access):0
1674:RULE 5 (multi-hop access):0
1598:RULE 5 (multi-hop access):0
1653:netAccess(app_38,tcp,8080):0
1745:exec
1743:RULE

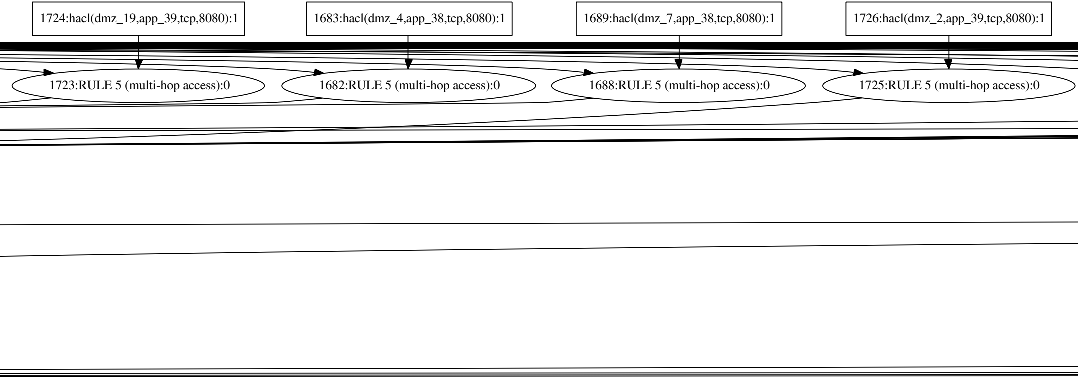

1677:hacl(dmz_19,app_38,tcp,8080):1
1636:hacl(dmz_4,app_37,tcp,8080):1
1642:hacl(dmz_7,app_37,tcp,8080):1
1679:hacl(dmz_2,app_38,tcp,8080):1
1676:RULE 5 (multi-hop access):0
1635:RULE 5 (multi-hop access):0
1641:RULE 5 (multi-hop access):0
1678:RULE 5 (multi-hop access):0
Code(app_4,root):0
1744:hacl(app_4,db_39,tcp,5432):1
5 (multi-hop access):0

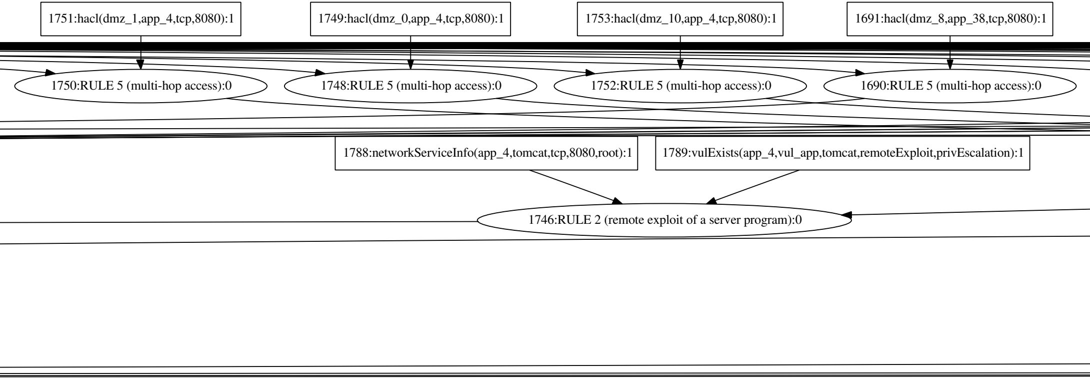

1704:hacl(dmz_1,app_39,tcp,8080):1
1702:hacl(dmz_0,app_39,tcp,8080):1
1706:hacl(dmz_10,app_39,tcp,8080):1
1644:hacl(dmz_8,app_37,tcp,8080):1
1703:RULE 5 (multi-hop access):0
1701:RULE 5 (multi-hop access):0
1705:RULE 5 (multi-hop access):0
1643:RULE 5 (multi-hop access):0
1741:networkServiceInfo(app_39,tomcat,tcp,8080,root):1
1742:vulExists(app_39,vul_app,tomcat,remoteExploit,privEscalation):1
1699:RULE 2 (remote exploit of a server program):0

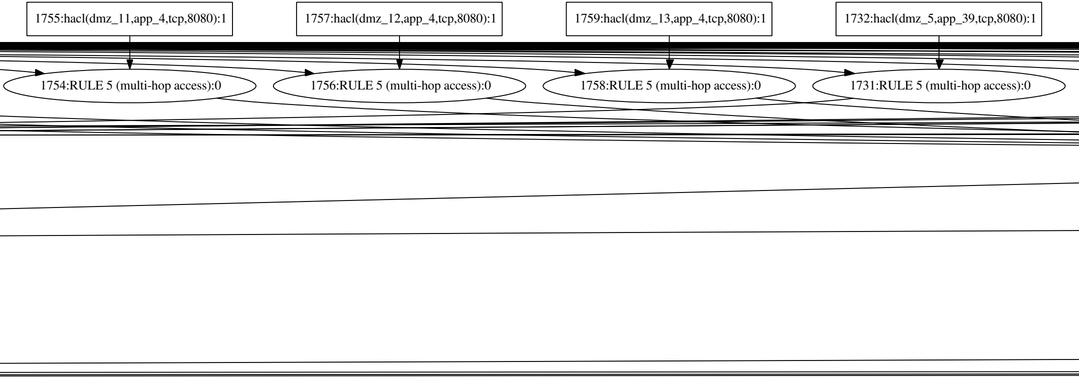

1708:hacl(dmz_11,app_39,tcp,8080):1
1710:hacl(dmz_12,app_39,tcp,8080):1
1712:hacl(dmz_13,app_39,tcp,8080):1
1685:hacl(dmz_5,app_38,tcp,8080):1
1707:RULE 5 (multi-hop access):0
1709:RULE 5 (multi-hop access):0
1711:RULE 5 (multi-hop access):0
1684:RULE 5 (multi-hop access):0

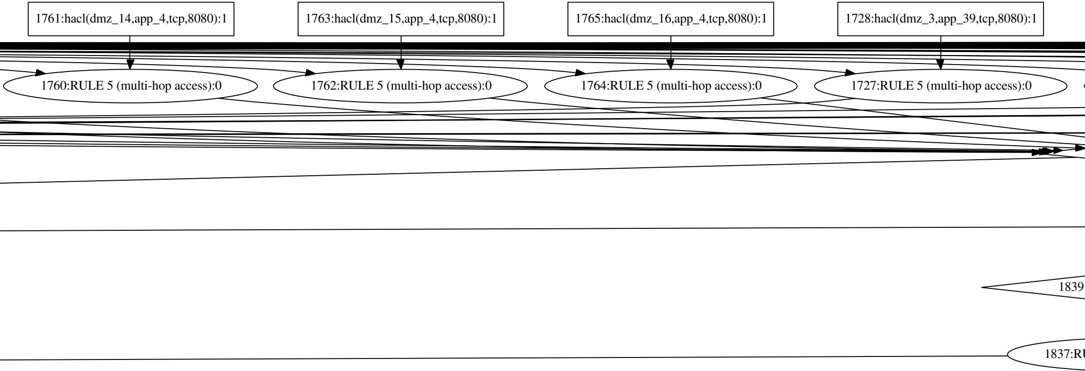

1714:hacl(dmz_14,app_39,tcp,8080):1
1716:hacl(dmz_15,app_39,tcp,8080):1
1718:hacl(dmz_16,app_39,tcp,8080):1
1681:hacl(dmz_3,app_38,tcp,8080):1
1713:RULE 5 (multi-hop access):0
1715:RULE 5 (multi-hop access):0
1717:RULE 5 (multi-hop access):0
1680:RULE 5 (multi-hop access):0

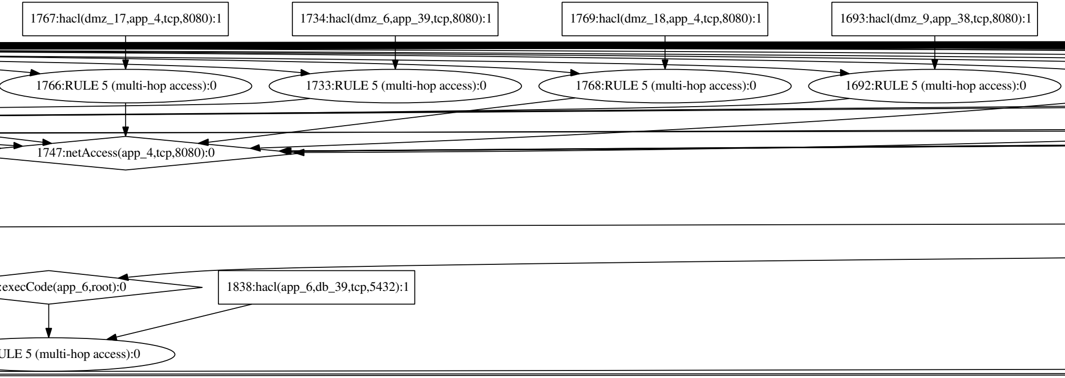

1720:hacl(dmz_17,app_39,tcp,8080):1
1687:hacl(dmz_6,app_38,tcp,8080):1
1722:hacl(dmz_18,app_39,tcp,8080):1
1646:hacl(dmz_9,app_37,tcp,8080):1
1719:RULE 5 (multi-hop access):0
1686:RULE 5 (multi-hop access):0
1721:RULE 5 (multi-hop access):0
1645:RULE 5 (multi-hop access):0
1700:netAccess(app_39,tcp,8080):0
1792:execCode(app_5,root):0
1791:hacl(app_5,db_39,tcp,5432):1
1790:RULE 5 (multi-hop access):0

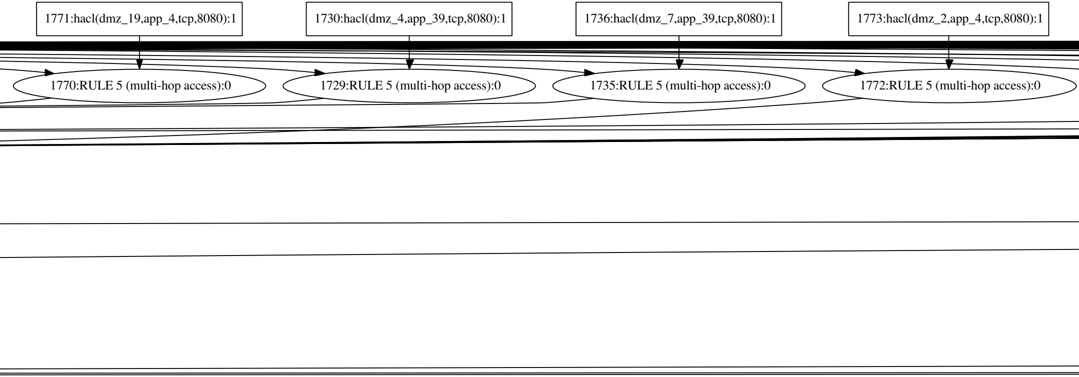

1724:hacl(dmz_19,app_39,tcp,8080):1
1683:hacl(dmz_4,app_38,tcp,8080):1
1689:hacl(dmz_7,app_38,tcp,8080):1
1726:hacl(dmz_2,app_39,tcp,8080):1
1723:RULE 5 (multi-hop access):0
1682:RULE 5 (multi-hop access):0
1688:RULE 5 (multi-hop access):0
1725:RULE 5 (multi-hop access):0

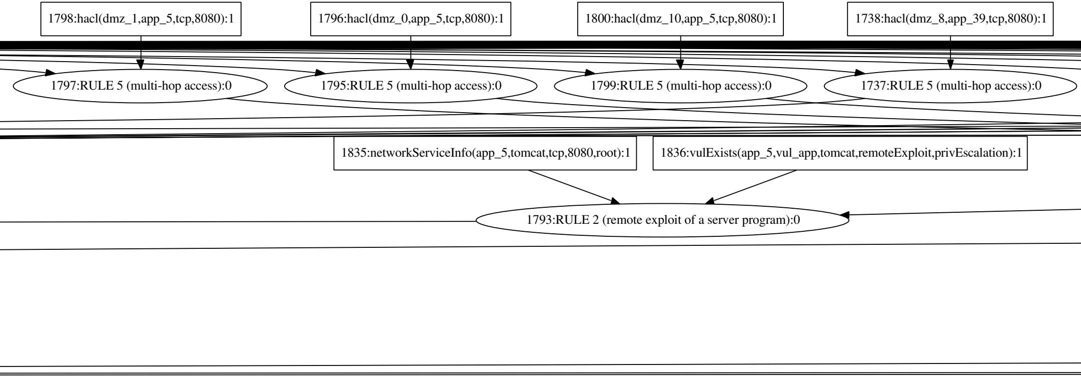

1751:hacl(dmz_1,app_4,tcp,8080):1
1749:hacl(dmz_0,app_4,tcp,8080):1
1753:hacl(dmz_10,app_4,tcp,8080):1
1691:hacl(dmz_8,app_38,tcp,8080):1
1750:RULE 5 (multi-hop access):0
1748:RULE 5 (multi-hop access):0
1752:RULE 5 (multi-hop access):0
1690:RULE 5 (multi-hop access):0
1788:networkServiceInfo(app_4,tomcat,tcp,8080,root):1
1789:vulExists(app_4,vul_app,tomcat,remoteExploit,privEscalation):1
1746:RULE 2 (remote exploit of a server program):0

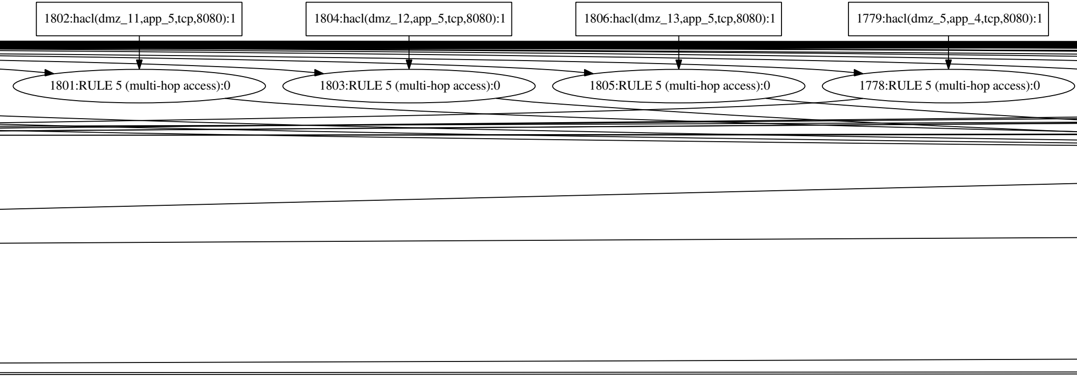

1755:hacl(dmz_11,app_4,tcp,8080):1
1757:hacl(dmz_12,app_4,tcp,8080):1
1759:hacl(dmz_13,app_4,tcp,8080):1
1732:hacl(dmz_5,app_39,tcp,8080):1
1754:RULE 5 (multi-hop access):0
1756:RULE 5 (multi-hop access):0
1758:RULE 5 (multi-hop access):0
1731:RULE 5 (multi-hop access):0

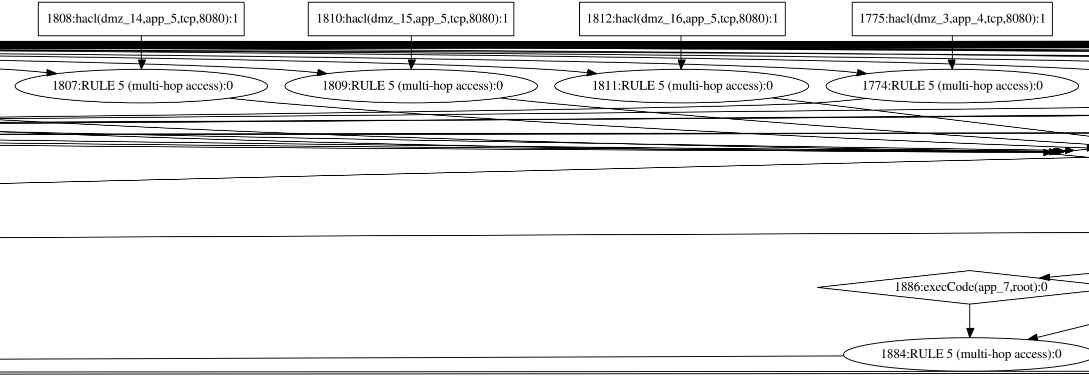

1761:hacl(dmz_14,app_4,tcp,8080):1
1763:hacl(dmz_15,app_4,tcp,8080):1
1765:hacl(dmz_16,app_4,tcp,8080):1
1728:hacl(dmz_3,app_39,tcp,8080):1
1760:RULE 5 (multi-hop access):0
1762:RULE 5 (multi-hop access):0
1764:RULE 5 (multi-hop access):0
1727:RULE 5 (multi-hop access):0
1839
1837:RU

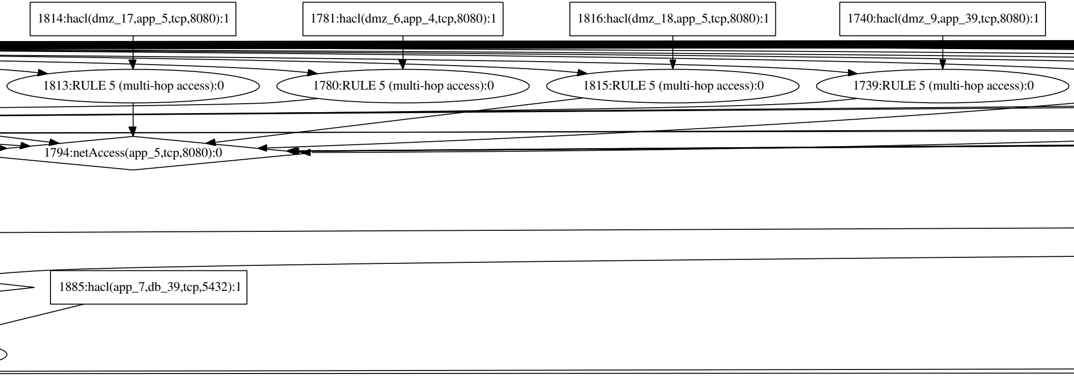

1767:hacl(dmz_17,app_4,tcp,8080):1
1734:hacl(dmz_6,app_39,tcp,8080):1
1769:hacl(dmz_18,app_4,tcp,8080):1
1693:hacl(dmz_9,app_38,tcp,8080):1
1766:RULE 5 (multi-hop access):0
1733:RULE 5 (multi-hop access):0
1768:RULE 5 (multi-hop access):0
1692:RULE 5 (multi-hop access):0
1747:netAccess(app_4,tcp,8080):0
:execCode(app_6,root):0
1838:hacl(app_6,db_39,tcp,5432):1
ULE 5 (multi-hop access):0

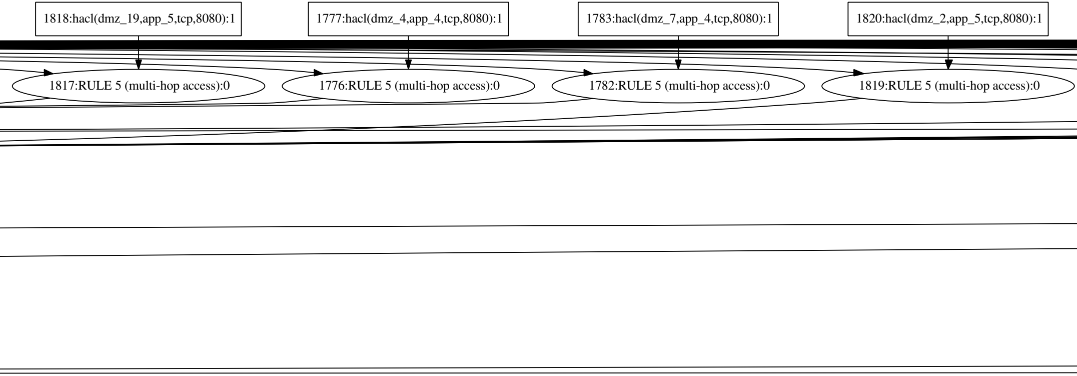

1771:hacl(dmz_19,app_4,tcp,8080):1
1730:hacl(dmz_4,app_39,tcp,8080):1
1736:hacl(dmz_7,app_39,tcp,8080):1
1773:hacl(dmz_2,app_4,tcp,8080):1
1770:RULE 5 (multi-hop access):0
1729:RULE 5 (multi-hop access):0
1735:RULE 5 (multi-hop access):0
1772:RULE 5 (multi-hop access):0

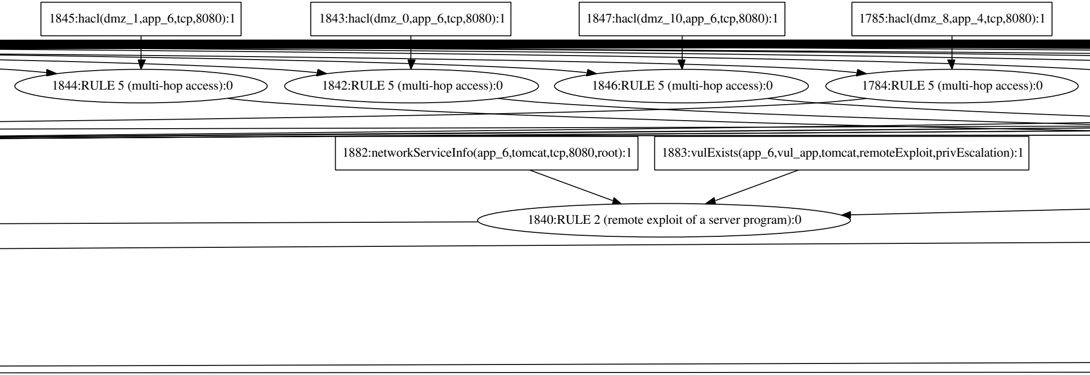

1798:hacl(dmz_1,app_5,tcp,8080):1
1796:hacl(dmz_0,app_5,tcp,8080):1
1800:hacl(dmz_10,app_5,tcp,8080):1
1738:hacl(dmz_8,app_39,tcp,8080):1
1797:RULE 5 (multi-hop access):0
1795:RULE 5 (multi-hop access):0
1799:RULE 5 (multi-hop access):0
1737:RULE 5 (multi-hop access):0
1835:networkServiceInfo(app_5,tomcat,tcp,8080,root):1
1836:vulExists(app_5,vul_app,tomcat,remoteExploit,privEscalation):1
1793:RULE 2 (remote exploit of a server program):0

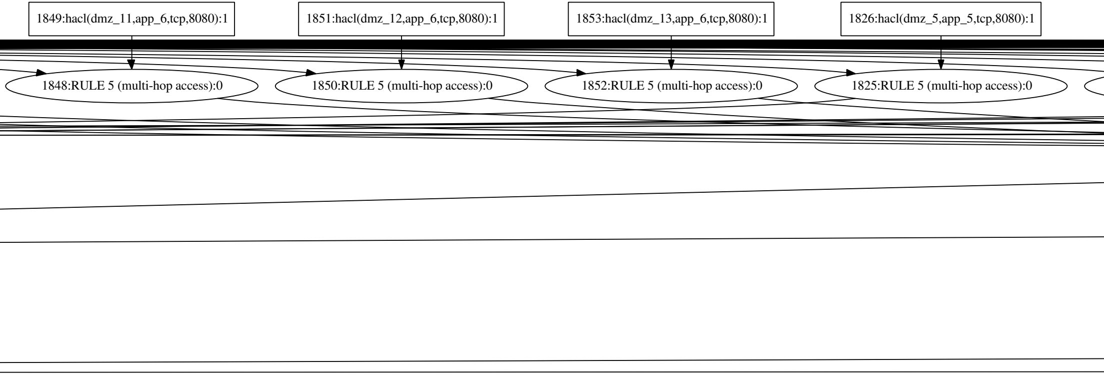

1802:hacl(dmz_11,app_5,tcp,8080):1
1804:hacl(dmz_12,app_5,tcp,8080):1
1806:hacl(dmz_13,app_5,tcp,8080):1
1779:hacl(dmz_5,app_4,tcp,8080):1
1801:RULE 5 (multi-hop access):0
1803:RULE 5 (multi-hop access):0
1805:RULE 5 (multi-hop access):0
1778:RULE 5 (multi-hop access):0

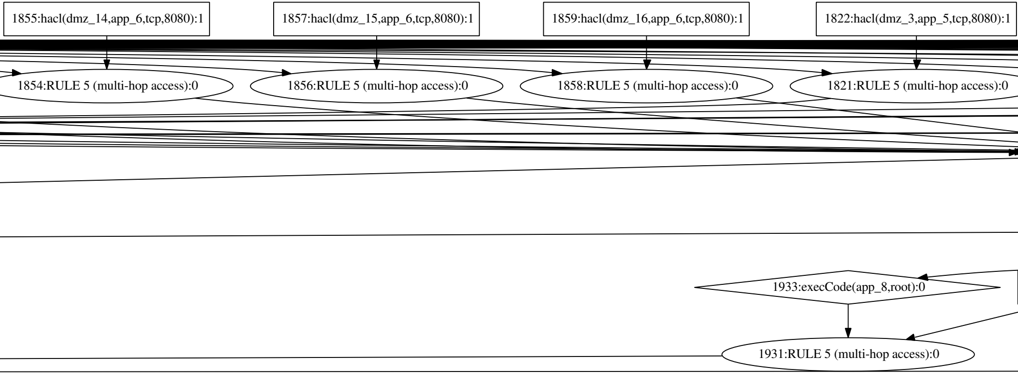

1808:hacl(dmz_14,app_5,tcp,8080):1
1810:hacl(dmz_15,app_5,tcp,8080):1
1812:hacl(dmz_16,app_5,tcp,8080):1
1775:hacl(dmz_3,app_4,tcp,8080):1
1807:RULE 5 (multi-hop access):0
1809:RULE 5 (multi-hop access):0
1811:RULE 5 (multi-hop access):0
1774:RULE 5 (multi-hop access):0
1886:execCode(app_7,root):0
1884:RULE 5 (multi-hop access):0

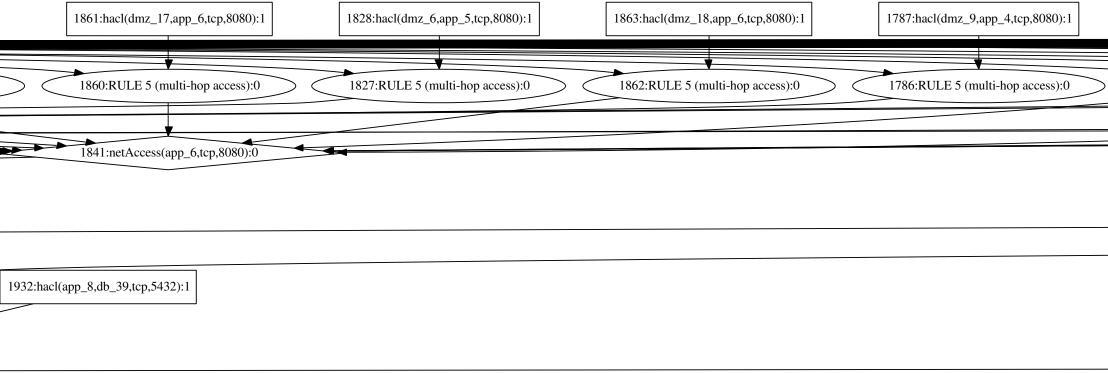

1814:hacl(dmz_17,app_5,tcp,8080):1
1781:hacl(dmz_6,app_4,tcp,8080):1
1816:hacl(dmz_18,app_5,tcp,8080):1
1740:hacl(dmz_9,app_39,tcp,8080):1
1813:RULE 5 (multi-hop access):0
1780:RULE 5 (multi-hop access):0
1815:RULE 5 (multi-hop access):0
1739:RULE 5 (multi-hop access):0
1794:netAccess(app_5,tcp,8080):0
1885:hacl(app_7,db_39,tcp,5432):1

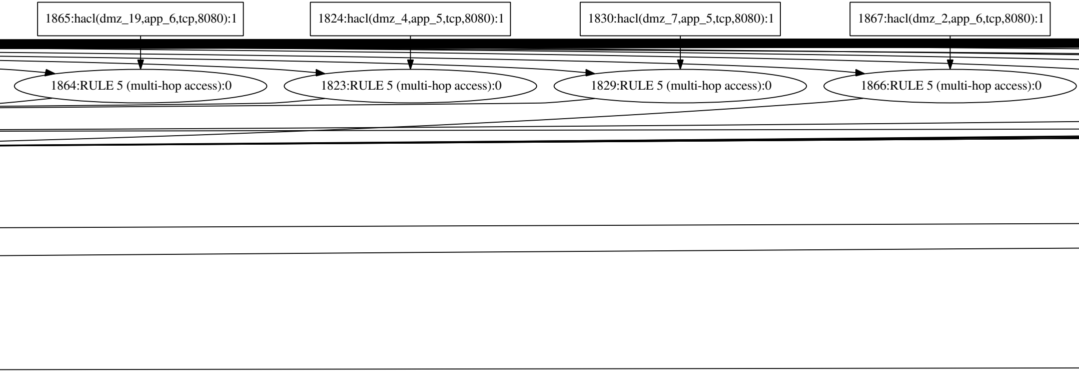

1818:hacl(dmz_19,app_5,tcp,8080):1
1777:hacl(dmz_4,app_4,tcp,8080):1
1783:hacl(dmz_7,app_4,tcp,8080):1
1820:hacl(dmz_2,app_5,tcp,8080):1
1817:RULE 5 (multi-hop access):0
1776:RULE 5 (multi-hop access):0
1782:RULE 5 (multi-hop access):0
1819:RULE 5 (multi-hop access):0

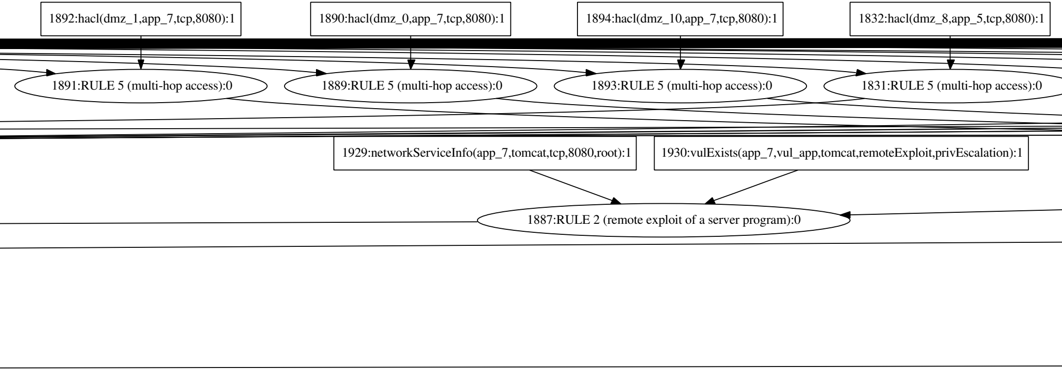

1845:hacl(dmz_1,app_6,tcp,8080):1
1843:hacl(dmz_0,app_6,tcp,8080):1
1847:hacl(dmz_10,app_6,tcp,8080):1
1785:hacl(dmz_8,app_4,tcp,8080):1
1844:RULE 5 (multi-hop access):0
1842:RULE 5 (multi-hop access):0
1846:RULE 5 (multi-hop access):0
1784:RULE 5 (multi-hop access):0
1882:networkServiceInfo(app_6,tomcat,tcp,8080,root):1
1883:vulExists(app_6,vul_app,tomcat,remoteExploit,privEscalation):1
1840:RULE 2 (remote exploit of a server program):0

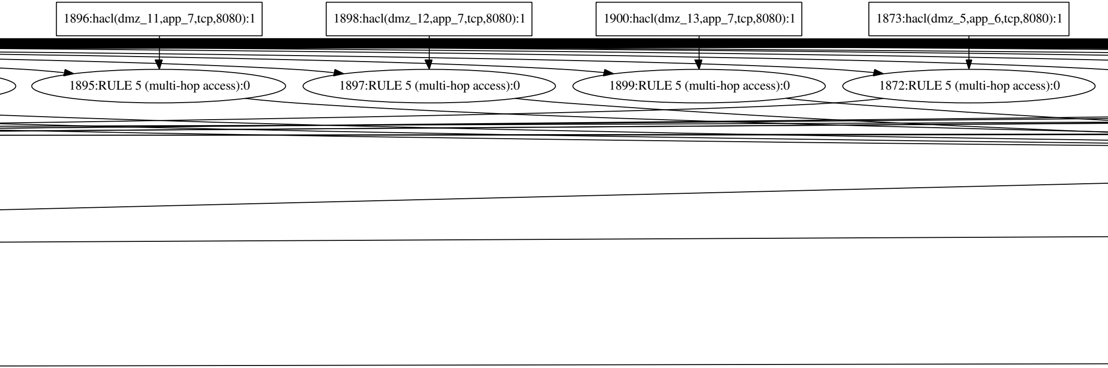

1849:hacl(dmz_11,app_6,tcp,8080):1
1851:hacl(dmz_12,app_6,tcp,8080):1
1853:hacl(dmz_13,app_6,tcp,8080):1
1826:hacl(dmz_5,app_5,tcp,8080):1
1848:RULE 5 (multi-hop access):0
1850:RULE 5 (multi-hop access):0
1852:RULE 5 (multi-hop access):0
1825:RULE 5 (multi-hop access):0

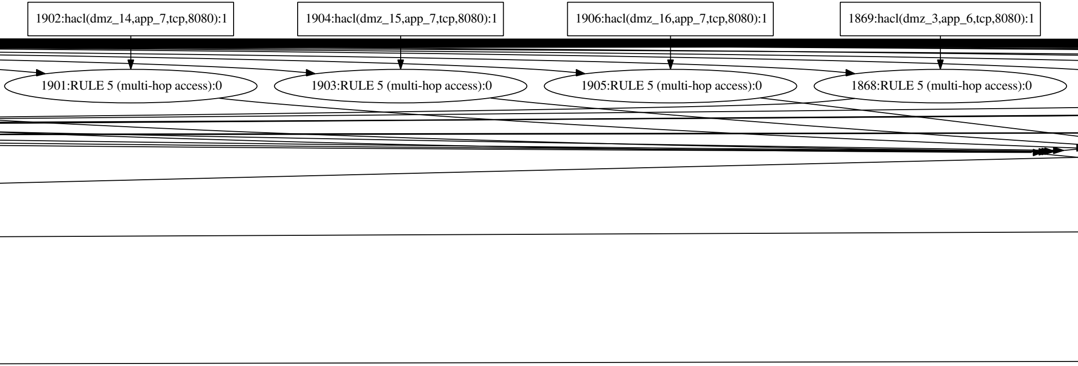

1855:hacl(dmz_14,app_6,tcp,8080):1
1857:hacl(dmz_15,app_6,tcp,8080):1
1859:hacl(dmz_16,app_6,tcp,8080):1
1822:hacl(dmz_3,app_5,tcp,8080):1
1854:RULE 5 (multi-hop access):0
1856:RULE 5 (multi-hop access):0
1858:RULE 5 (multi-hop access):0
1821:RULE 5 (multi-hop access):0
1933:execCode(app_8,root):0
1931:RULE 5 (multi-hop access):0

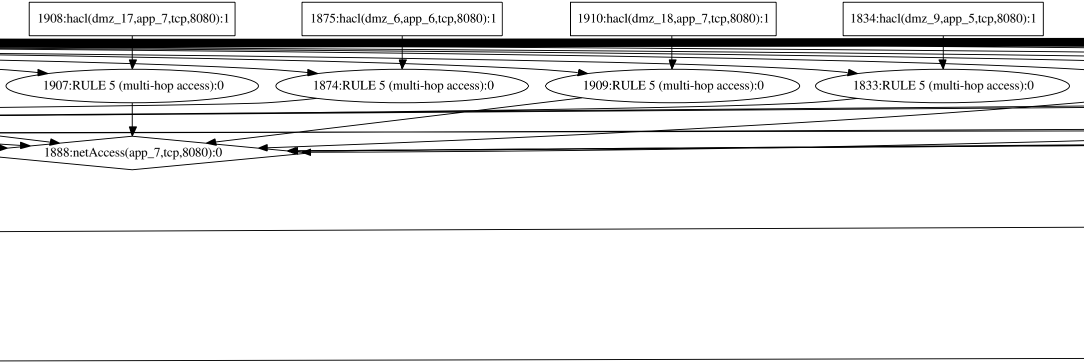

1861:hacl(dmz_17,app_6,tcp,8080):1
1828:hacl(dmz_6,app_5,tcp,8080):1
1863:hacl(dmz_18,app_6,tcp,8080):1
1787:hacl(dmz_9,app_4,tcp,8080):1
1860:RULE 5 (multi-hop access):0
1827:RULE 5 (multi-hop access):0
1862:RULE 5 (multi-hop access):0
1786:RULE 5 (multi-hop access):0
1841:netAccess(app_6,tcp,8080):0
1932:hacl(app_8,db_39,tcp,5432):1

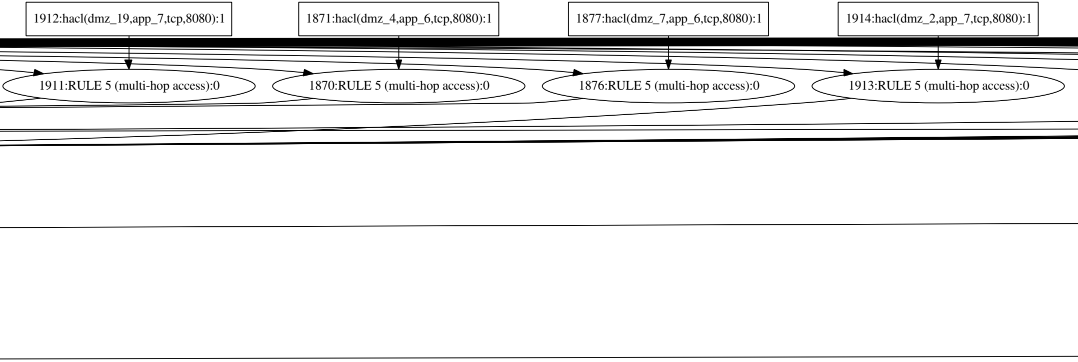

1865:hacl(dmz_19,app_6,tcp,8080):1
1824:hacl(dmz_4,app_5,tcp,8080):1
1830:hacl(dmz_7,app_5,tcp,8080):1
1867:hacl(dmz_2,app_6,tcp,8080):1
1864:RULE 5 (multi-hop access):0
1823:RULE 5 (multi-hop access):0
1829:RULE 5 (multi-hop access):0
1866:RULE 5 (multi-hop access):0

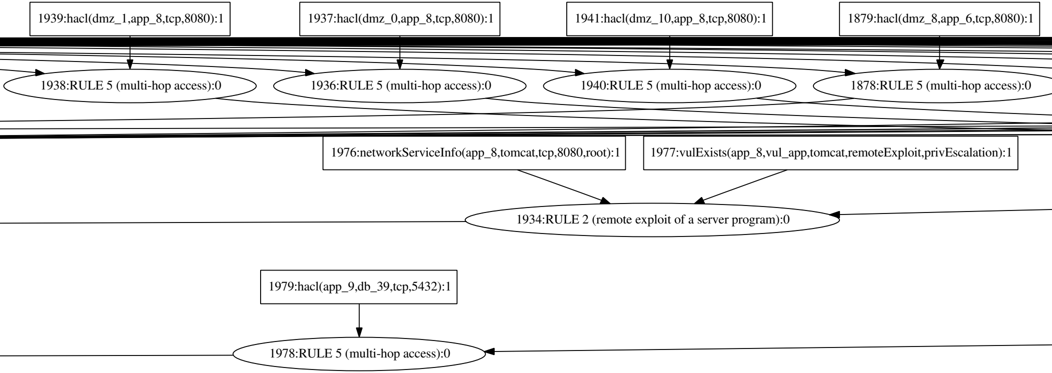

1892:hacl(dmz_1,app_7,tcp,8080):1
1890:hacl(dmz_0,app_7,tcp,8080):1
1894:hacl(dmz_10,app_7,tcp,8080):1
1832:hacl(dmz_8,app_5,tcp,8080):1
1891:RULE 5 (multi-hop access):0
1889:RULE 5 (multi-hop access):0
1893:RULE 5 (multi-hop access):0
1831:RULE 5 (multi-hop access):0
1929:networkServiceInfo(app_7,tomcat,tcp,8080,root):1
1930:vulExists(app_7,vul_app,tomcat,remoteExploit,privEscalation):1
1887:RULE 2 (remote exploit of a server program):0

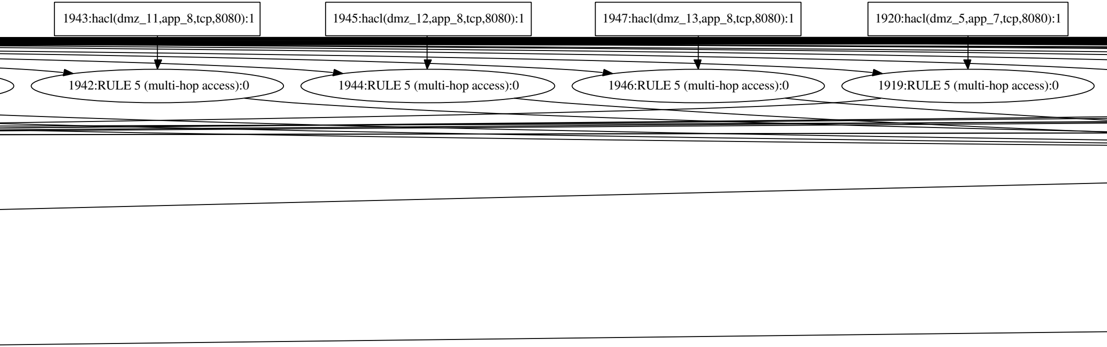

1896:hacl(dmz_11,app_7,tcp,8080):1
1898:hacl(dmz_12,app_7,tcp,8080):1
1900:hacl(dmz_13,app_7,tcp,8080):1
1873:hacl(dmz_5,app_6,tcp,8080):1
1895:RULE 5 (multi-hop access):0
1897:RULE 5 (multi-hop access):0
1899:RULE 5 (multi-hop access):0
1872:RULE 5 (multi-hop access):0

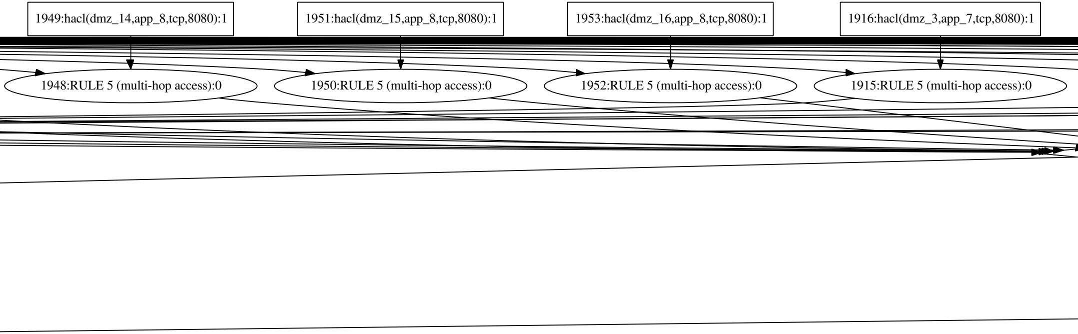

1902:hacl(dmz_14,app_7,tcp,8080):1
1904:hacl(dmz_15,app_7,tcp,8080):1
1906:hacl(dmz_16,app_7,tcp,8080):1
1869:hacl(dmz_3,app_6,tcp,8080):1
1901:RULE 5 (multi-hop access):0
1903:RULE 5 (multi-hop access):0
1905:RULE 5 (multi-hop access):0
1868:RULE 5 (multi-hop access):0

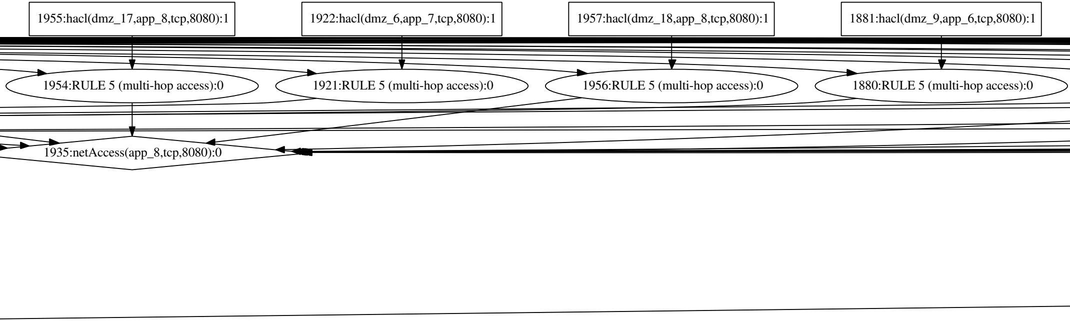

1908:hacl(dmz_17,app_7,tcp,8080):1
1875:hacl(dmz_6,app_6,tcp,8080):1
1910:hacl(dmz_18,app_7,tcp,8080):1
1834:hacl(dmz_9,app_5,tcp,8080):1
1907:RULE 5 (multi-hop access):0
1874:RULE 5 (multi-hop access):0
1909:RULE 5 (multi-hop access):0
1833:RULE 5 (multi-hop access):0
1888:netAccess(app_7,tcp,8080):0

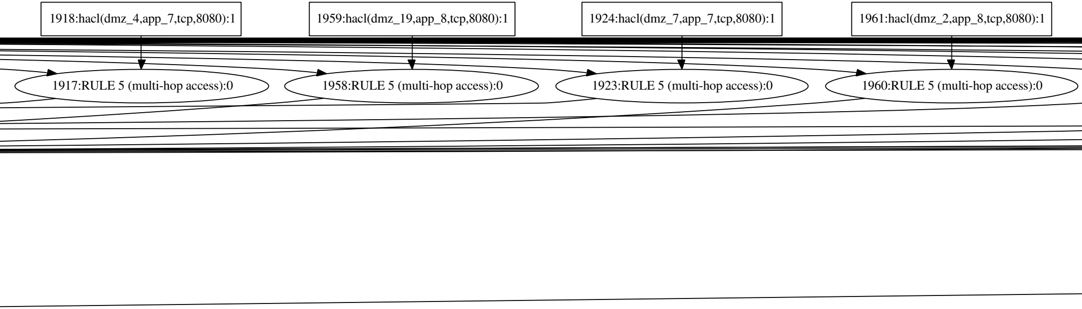

1912:hacl(dmz_19,app_7,tcp,8080):1
1871:hacl(dmz_4,app_6,tcp,8080):1
1877:hacl(dmz_7,app_6,tcp,8080):1
1914:hacl(dmz_2,app_7,tcp,8080):1
1911:RULE 5 (multi-hop access):0
1870:RULE 5 (multi-hop access):0
1876:RULE 5 (multi-hop access):0
1913:RULE 5 (multi-hop access):0

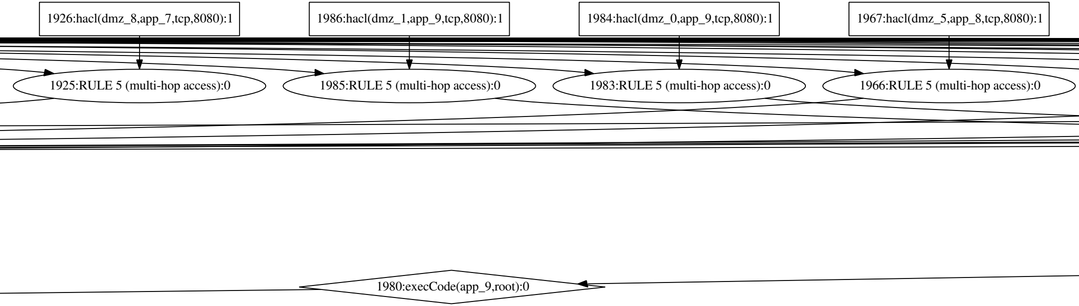

1939:hacl(dmz_1,app_8,tcp,8080):1
1937:hacl(dmz_0,app_8,tcp,8080):1
1941:hacl(dmz_10,app_8,tcp,8080):1
1879:hacl(dmz_8,app_6,tcp,8080):1
1938:RULE 5 (multi-hop access):0
1936:RULE 5 (multi-hop access):0
1940:RULE 5 (multi-hop access):0
1878:RULE 5 (multi-hop access):0
1976:networkServiceInfo(app_8,tomcat,tcp,8080,root):1
1977:vulExists(app_8,vul_app,tomcat,remoteExploit,privEscalation):1
1934:RULE 2 (remote exploit of a server program):0
1979:hacl(app_9,db_39,tcp,5432):1
1978:RULE 5 (multi-hop access):0

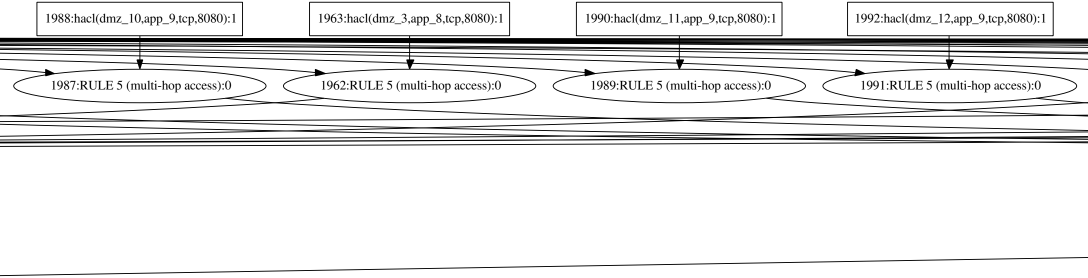

1943:hacl(dmz_11,app_8,tcp,8080):1
1945:hacl(dmz_12,app_8,tcp,8080):1
1947:hacl(dmz_13,app_8,tcp,8080):1
1920:hacl(dmz_5,app_7,tcp,8080):1
1942:RULE 5 (multi-hop access):0
1944:RULE 5 (multi-hop access):0
1946:RULE 5 (multi-hop access):0
1919:RULE 5 (multi-hop access):0

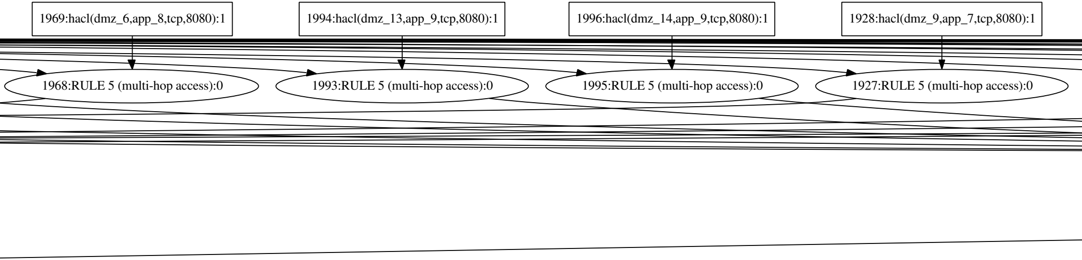

1949:hacl(dmz_14,app_8,tcp,8080):1
1951:hacl(dmz_15,app_8,tcp,8080):1
1953:hacl(dmz_16,app_8,tcp,8080):1
1916:hacl(dmz_3,app_7,tcp,8080):1
1948:RULE 5 (multi-hop access):0
1950:RULE 5 (multi-hop access):0
1952:RULE 5 (multi-hop access):0
1915:RULE 5 (multi-hop access):0

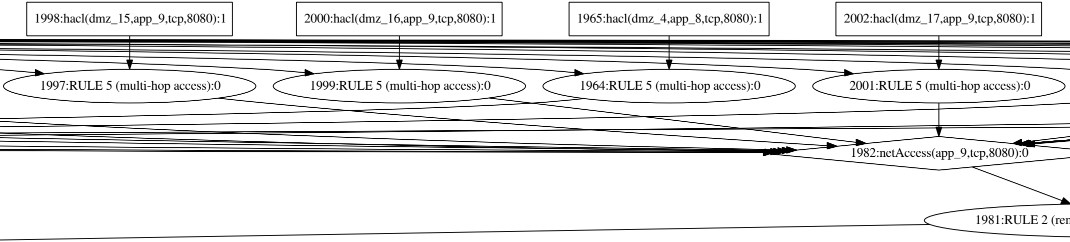

1955:hacl(dmz_17,app_8,tcp,8080):1
1922:hacl(dmz_6,app_7,tcp,8080):1
1957:hacl(dmz_18,app_8,tcp,8080):1
1881:hacl(dmz_9,app_6,tcp,8080):1
1954:RULE 5 (multi-hop access):0
1921:RULE 5 (multi-hop access):0
1956:RULE 5 (multi-hop access):0
1880:RULE 5 (multi-hop access):0
1935:netAccess(app_8,tcp,8080):0

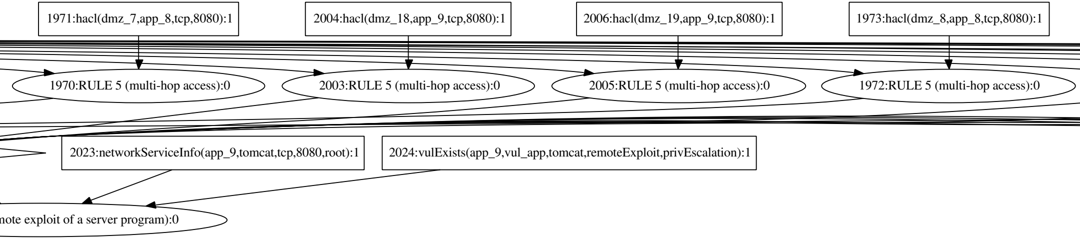

1918:hacl(dmz_4,app_7,tcp,8080):1
1959:hacl(dmz_19,app_8,tcp,8080):1
1924:hacl(dmz_7,app_7,tcp,8080):1
1961:hacl(dmz_2,app_8,tcp,8080):1
1917:RULE 5 (multi-hop access):0
1958:RULE 5 (multi-hop access):0
1923:RULE 5 (multi-hop access):0
1960:RULE 5 (multi-hop access):0

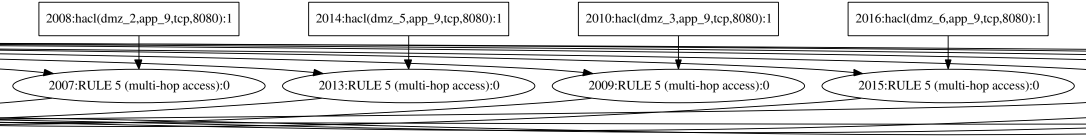

1926:hacl(dmz_8,app_7,tcp,8080):1
1986:hacl(dmz_1,app_9,tcp,8080):1
1984:hacl(dmz_0,app_9,tcp,8080):1
1967:hacl(dmz_5,app_8,tcp,8080):1
1925:RULE 5 (multi-hop access):0
1985:RULE 5 (multi-hop access):0
1983:RULE 5 (multi-hop access):0
1966:RULE 5 (multi-hop access):0
1980:execCode(app_9,root):0

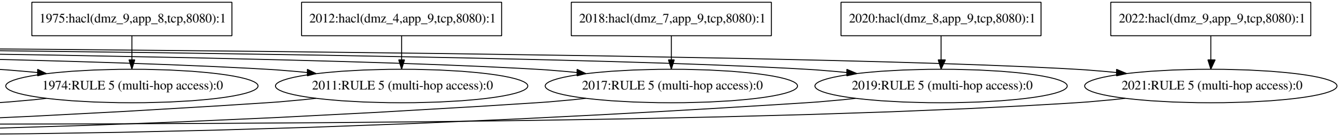

1988:hacl(dmz_10,app_9,tcp,8080):1
1963:hacl(dmz_3,app_8,tcp,8080):1
1990:hacl(dmz_11,app_9,tcp,8080):1
1992:hacl(dmz_12,app_9,tcp,8080):1
1987:RULE 5 (multi-hop access):0
1962:RULE 5 (multi-hop access):0
1989:RULE 5 (multi-hop access):0
1991:RULE 5 (multi-hop access):0


1969:hacl(dmz_6,app_8,tcp,8080):1
1994:hacl(dmz_13,app_9,tcp,8080):1
1996:hacl(dmz_14,app_9,tcp,8080):1
1928:hacl(dmz_9,app_7,tcp,8080):1
1968:RULE 5 (multi-hop access):0
1993:RULE 5 (multi-hop access):0
1995:RULE 5 (multi-hop access):0
1927:RULE 5 (multi-hop access):0


1998:hacl(dmz_15,app_9,tcp,8080):1
2000:hacl(dmz_16,app_9,tcp,8080):1
1965:hacl(dmz_4,app_8,tcp,8080):1
2002:hacl(dmz_17,app_9,tcp,8080):1
1997:RULE 5 (multi-hop access):0
1999:RULE 5 (multi-hop access):0
1964:RULE 5 (multi-hop access):0
2001:RULE 5 (multi-hop access):0
1982:netAccess(app_9,tcp,8080):0
1981:RULE 2 (rem


1971:hacl(dmz_7,app_8,tcp,8080):1
2004:hacl(dmz_18,app_9,tcp,8080):1
2006:hacl(dmz_19,app_9,tcp,8080):1
1973:hacl(dmz_8,app_8,tcp,8080):1
1970:RULE 5 (multi-hop access):0
2003:RULE 5 (multi-hop access):0
2005:RULE 5 (multi-hop access):0
1972:RULE 5 (multi-hop access):0
2023:networkServiceInfo(app_9,tomcat,tcp,8080,root):1
2024:vulExists(app_9,vul_app,tomcat,remoteExploit,privEscalation):1
note exploit of a server program):0


2008:hacl(dmz_2,app_9,tcp,8080):1
2014:hacl(dmz_5,app_9,tcp,8080):1
2010:hacl(dmz_3,app_9,tcp,8080):1
2016:hacl(dmz_6,app_9,tcp,8080):1
2007:RULE 5 (multi-hop access):0
2013:RULE 5 (multi-hop access):0
2009:RULE 5 (multi-hop access):0
2015:RULE 5 (multi-hop access):0


1975:hacl(dmz_9,app_8,tcp,8080):1
2012:hacl(dmz_4,app_9,tcp,8080):1
2018:hacl(dmz_7,app_9,tcp,8080):1
2020:hacl(dmz_8,app_9,tcp,8080):1
2022:hacl(dmz_9,app_9,tcp,8080):1
1974:RULE 5 (multi-hop access):0
2011:RULE 5 (multi-hop access):0
2017:RULE 5 (multi-hop access):0
2019:RULE 5 (multi-hop access):0
2021:RULE 5 (multi-hop access):0